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# Exercises

## Chapter 2 Exercises

### Exercises for Section 2.1

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

*Provide a reasonable description of the sample space for each of the random experiments in Exercises 2.1.1 to 2.1.11. There can be more than one acceptable interpretation of each experiment. Describe any assumptions you make.*

**2.1.1** Each of four transmitted bits is classified as either in error or not in error.

**2.1.2** The number of hits (views) is recorded at a high-volume Web site in a day.

**2.1.3** In the final inspection of electronic power supplies, either units pass or three types of nonconformities might occur: functional, minor, or cosmetic. Three units are inspected.

**2.1.4** An ammeter that displays three digits is used to measure current in milliamperes.

**2.1.5** WP The following two questions appear on an employee survey questionnaire. Each answer is chosen from the five-point scale 1 (never), 2, 3, 4, 5 (always).

Is the corporation willing to listen to and fairly evaluate new ideas?

How often are my coworkers important in my overall job performance?

**2.1.6** The time until a service transaction is requested of a computer to the nearest millisecond.

**2.1.7** SS The pH reading of a water sample to the nearest tenth of a unit.

**2.1.8** The voids in a ferrite slab are classified as small, medium, or large. The number of voids in each category is measured by an optical inspection of a sample.

**2.1.9** A sampled injection-molded part could have been produced in either one of two presses and in any one of the eight cavities in each press.

**2.1.10** An order for an automobile can specify either an automatic or a standard transmission, either with or without air conditioning, and with any one of the four colors red, blue, black, or white. Describe the set of possible orders for this experiment.

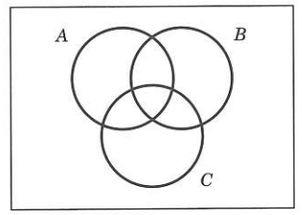
**2.1.11** WP Calls are repeatedly placed to a busy phone line until a connection is achieved.

**2.1.12** Three attempts are made to read data in a magnetic storage device before an error recovery procedure is used. The error recovery procedure attempts three corrections before an "abort" message is sent to the operator. Let

$s$ denote the success of a read operation
$f$ denote the failure of a read operation
$S$ denote the success of an error recovery procedure
$F$ denote the failure of an error recovery procedure
$A$ denote an abort message sent to the operator

Describe the sample space of this experiment with a tree diagram.

**2.1.13** SS VS Three events are shown on the Venn diagram in the following figure:




Reproduce the figure and shade the region that corresponds to each of the following events.

**a.** $A'$ **b.** $A \cap B$ **c.** $(A \cap B) \cup C$
**d.** $(B \cup C)'$ **e.** $(A \cap B)' \cup C$

**2.1.14** In an injection-molding operation, the length and width, denoted as $X$ and $Y$, respectively, of each molded part are evaluated. Let

$A$ denote the event of $48 < X < 52$ centimeters
$B$ denote the event of $9 < Y < 11$ centimeters

Construct a Venn diagram that includes these events. Shade the areas that represent the following:

**a.** $A$ **b.** $A \cap B$
**c.** $A' \cup B$ **d.** $A \cap B$
**e.** If these events were mutually exclusive, how successful would this production operation be? Would the process produce parts with $X = 50$ centimeters and $Y = 10$ centimeters?

**2.1.15** WP A digital scale that provides weights to the nearest gram is used.

**a.** What is the sample space for this experiment?

Let $A$ denote the event that a weight exceeds 11 grams, let $B$ denote the event that a weight is less than or equal to 15 grams,

and let $C$ denote the event that a weight is greater than or equal to 8 grams and less than 12 grams.
Describe the following events.

**b.** $A \cup B$ **c.** $A \cap B$
**d.** $A'$ **e.** $A \cup B \cup C$
**f.** $(A \cup C)'$ **g.** $A \cap B \cap C$
**h.** $B' \cap C$ **i.** $A \cup (B \cap C)$

**2.1.16** In light-dependent photosynthesis, light quality refers to the wavelengths of light that are important. The wavelength of a sample of photosynthetically active radiations (PAR) is measured to the nearest nanometer. The red range is 675–700 nm and the blue range is 450–500 nm. Let $A$ denote the event that PAR occurs in the red range, and let $B$ denote the event that PAR occurs in the blue range. Describe the sample space and indicate each of the following events:

**a.** $A$ **b.** $B$ **c.** $A \cap B$ **d.** $A \cup B$

**2.1.17** WP GO Tutorial Four bits are transmitted over a digital communications channel. Each bit is either distorted or received without distortion. Let $A_i$ denote the event that the $i$th bit is distorted, $i = 1, \ldots, 4$.

**a.** Describe the sample space for this experiment.
**b.** Are the $A_i$'s mutually exclusive?

Describe the outcomes in each of the following events:

**c.** $A_1$ **d.** $A_1'$
**e.** $A_1 \cap A_2 \cap A_3 \cap A_4$ **f.** $(A_1 \cap A_2) \cup (A_3 \cap A_4)$

**2.1.18** WP Disks of polycarbonate plastic from a supplier are analyzed for scratch and shock resistance. The results from 100 disks are summarized here:

| | | Shock Resistance: High | Shock Resistance: Low |
|---|---|---|---|
| Scratch Resistance | High | 70 | 9 |
| | Low | 16 | 5 |

Let $A$ denote the event that a disk has high shock resistance, and let $B$ denote the event that a disk has high scratch resistance. Determine the number of disks in $A \cap B$, $A'$, and $A \cup B$.

**2.1.19** SS In control replication, cells are replicated over a period of two days. Two control mechanisms have been identified—one positive and one negative. Suppose that a replication is observed in three cells. Let $A$ denote the event that all cells are identified as positive, and let $B$ denote the event that all cells are negative. Describe the sample space graphically and display each of the following events:

**a.** $A$ **b.** $B$ **c.** $A \cap B$ **d.** $A \cup B$

**2.1.20** WP Samples of emissions from three suppliers are classified for conformance to air-quality specifications. The results from 100 samples are summarized as follows:

| | | Conforms: Yes | Conforms: No |
|---|---|---|---|
| Supplier | 1 | 22 | 8 |
| | 2 | 25 | 5 |
| | 3 | 30 | 10 |

Let $A$ denote the event that a sample is from supplier 1, and let $B$ denote the event that a sample conforms to specifications. Determine the number of samples in $A' \cap B$, $B'$, and $A \cup B$.

**2.1.21** WP The rise time of a reactor is measured in minutes (and fractions of minutes). Let the sample space be positive, real numbers. Define the events $A$ and $B$ as follows: $A = \{x \mid x < 72.5\}$ and $B = \{x \mid x > 52.5\}$.
Describe each of the following events:

**a.** $A'$ **b.** $B'$
**c.** $A \cap B$ **d.** $A \cup B$

**2.1.22** The following table summarizes 204 endothermic reactions involving sodium bicarbonate.

| Final Temperature Conditions | Heat Absorbed (cal): Below Target | Heat Absorbed (cal): Above Target |
|---|---|---|
| 266 K | 12 | 40 |
| 271 K | 44 | 16 |
| 274 K | 56 | 36 |

Let $A$ denote the event that a reaction's final temperature is 271 K or less. Let $B$ denote the event that the heat absorbed is below target. Determine the number of reactions in each of the following events.

**a.** $A \cap B$ **b.** $A'$ **c.** $A \cup B$
**d.** $A \cup B'$ **e.** $A' \cap B'$

**2.1.23** SS A Web ad can be designed from four different colors, three font types, five font sizes, three images, and five text phrases. How many different designs are possible?

**2.1.24** Consider the hospital emergency department data in Example 2.6. Let $A$ denote the event that a visit is to hospital 1, and let $B$ denote the event that a visit results in admittance to any hospital. Determine the number of persons in each of the following events.

**a.** $A \cap B$ **b.** $A'$ **c.** $A \cup B$
**d.** $A \cup B'$ **e.** $A' \cap B'$

**2.1.25** The article "Term Efficacy of Ribavirin Plus Interferon Alfa in the Treatment of Chronic Hepatitis C" [*Gastroenterology* (1996, Vol. 111(5), pp. 1307–1312)], considered the effect of two treatments and a control for treatment of hepatitis C. The following table provides the total patients in each group and the number that showed a complete (positive) response after 24 weeks of treatment.

| | Complete Response | Total |
|---|---|---|
| Ribavirin plus interferon alfa | 16 | 21 |
| Interferon alfa | 6 | 19 |
| Untreated controls | 0 | 20 |

Let $A$ denote the event that the patient was treated with ribavirin plus interferon alfa, and let $B$ denote the event that the response was complete. Determine the number of patients in each of the following events.

**a.** $A$ **b.** $A \cap B$ **c.** $A \cup B$ **d.** $A' \cap B'$



**2.1.26** A computer system uses passwords that contain exactly eight characters, and each character is 1 of the 26 lowercase letters ($a$–$z$) or 26 uppercase letters ($A$–$Z$) or 10 integers (0–9). Let $\Omega$ denote the set of all possible passwords, and let $A$ and $B$ denote the events that consist of passwords with only letters or only integers, respectively. Determine the number of passwords in each of the following events.

**a.** $\Omega$ **b.** $A$ **c.** $A' \cap B'$
**d.** Passwords that contain at least 1 integer
**e.** Passwords that contain exactly 1 integer

## Exercises for Section 2.2

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
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**2.2.1** WP A sample of two printed circuit boards is selected without replacement from a batch. Describe the (ordered) sample space for each of the following batches:

**a.** The batch contains 90 boards that are not defective, 8 boards with minor defects, and 2 boards with major defects.
**b.** The batch contains 90 boards that are not defective, 8 boards with minor defects, and 1 board with major defects.

**2.2.2** A sample of two items is selected without replacement from a batch. Describe the (ordered) sample space for each of the following batches:

**a.** The batch contains the items $\{a, b, c, d\}$.
**b.** The batch contains the items $\{a, b, c, d, e, f, g\}$.
**c.** The batch contains 4 defective items and 20 good items.
**d.** The batch contains 1 defective item and 20 good items.

**2.2.3** WP SS A wireless garage door opener has a code determined by the up or down setting of 12 switches. How many outcomes are in the sample space of possible codes?

**2.2.4** WP In a manufacturing operation, a part is produced by machining, polishing, and painting. If there are three machine tools, four polishing tools, and three painting tools, how many different routings (consisting of machining, followed by polishing, and followed by painting) for a part are possible?

**2.2.5** WP New designs for a wastewater treatment tank have proposed three possible shapes, four possible sizes, three locations for input valves, and four locations for output valves. How many different product designs are possible?

**2.2.6** WP A manufacturing process consists of 10 operations that can be completed in any order. How many different production sequences are possible?

**2.2.7** WP VS A batch of 140 semiconductor chips is inspected by choosing a sample of 5 chips without replacement. Assume 10 of the chips do not conform to customer requirements.

**a.** How many different samples are possible?
**b.** How many samples contain exactly one nonconforming chip?
**c.** How many samples contain at least one nonconforming chip?

**2.2.8** WP In a sheet metal operation, three notches and four bends are required. If the operations can be done in any order, how many different ways of completing the manufacturing are possible?

**2.2.9** In the laboratory analysis of samples from a chemical process, five samples from the process are analyzed daily. In addition, a control sample is analyzed twice each day to check the calibration of the laboratory instruments.

**a.** How many different sequences of process and control samples are possible each day? Assume that the five process samples are considered identical and that the two control samples are considered identical.
**b.** How many different sequences of process and control samples are possible if we consider the five process samples to be different and the two control samples to be identical?
**c.** For the same situation as part (b), how many sequences are possible if the first test of each day must be a control sample?

**2.2.10** In the layout of a printed circuit board for an electronic product, 12 different locations can accommodate chips.

**a.** If five different types of chips are to be placed on the board, how many different layouts are possible?
**b.** If the five chips that are placed on the board are of the same type, how many different layouts are possible?

**2.2.11** SS Consider the design of a communication system.

**a.** How many three-digit phone prefixes that are used to represent a particular geographic area (such as an area code) can be created from the digits 0 through 9?
**b.** As in part (a), how many three-digit phone prefixes are possible that do not start with 0 or 1, but contain 0 or 1 as the middle digit?
**c.** How many three-digit phone prefixes are possible in which no digit appears more than once in each prefix?

**2.2.12** In the design of an electromechanical product, 12 components are to be stacked into a cylindrical casing in a manner that minimizes the impact of shocks. One end of the casing is designated as the bottom and the other end is the top.

**a.** If all components are different, how many different designs are possible?
**b.** If 7 components are identical to one another, but the others are different, how many different designs are possible?
**c.** If 3 components are of one type and identical to one another, and 4 components are of another type and identical to one another, but the others are different, how many different designs are possible?

**2.2.13** A bin of 50 parts contains 5 that are defective. A sample of 10 parts is selected without replacement. How many samples contain at least 4 defective parts?

**2.2.14** WP Plastic parts produced by an injection-molding operation are checked for conformance to specifications. Each tool contains 12 cavities in which parts are produced, and these parts

fall into a conveyor when the press opens. An inspector selects 3 parts without replacement from among the 12 at random. Two cavities are affected by a temperature malfunction that results in parts that do not conform to specifications.

**a.** How many samples contain exactly one nonconforming part?

**b.** How many samples contain at least one nonconforming part?

**2.2.15** Similar to the hospital schedule in Example 2.9, suppose that an operating room needs to handle three knee, four hip, and five shoulder surgeries.

**a.** How many different sequences are possible?

**b.** How many different sequences have all hip, knee, and shoulder surgeries scheduled consecutively?

**c.** How many different schedules begin and end with a knee surgery?

## Exercises for Section 2.3

WP Problem available in WileyPLUS at instructor's discretion.
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SS Student solution available in WileyPLUS.
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**2.3.1** WP The sample space of a random experiment is $\{a, b, c, d, e\}$ with probabilities 0.1, 0.1, 0.2, 0.4, and 0.2, respectively. Let $A$ denote the event $\{a, b, c\}$, and let $B$ denote the event $\{c, d, e\}$. Determine the following:

**a.** $P(A)$ **b.** $P(B)$ **c.** $P(A')$
**d.** $P(A \cup B)$ **e.** $P(A \cap B)$

**2.3.2** A part selected for testing is equally likely to have been produced on any one of six cutting tools.

**a.** What is the sample space?

**b.** What is the probability that the part is from tool 1?

**c.** What is the probability that the part is from tool 3 or tool 5?

**d.** What is the probability that the part is not from tool 4?

**2.3.3** WP SS An injection-molded part is equally likely to be obtained from any one of the eight cavities on a mold.

**a.** What is the sample space?

**b.** What is the probability that a part is from cavity 1 or 2?

**c.** What is the probability that a part is from neither cavity 3 nor 4?

**2.3.4** WP A credit card contains 16 digits between 0 and 9. However, only 100 million numbers are valid. If a number is entered randomly, what is the probability that it is a valid number?

**2.3.5** WP In a NiCd battery, a fully charged cell is composed of nickelic hydroxide. Nickel is an element that has multiple oxidation states and that is usually found in the following states:

| Nickel Charge | Proportions Found |
|---|---|
| 0 | 0.17 |
| +2 | 0.35 |
| +3 | 0.33 |
| +4 | 0.15 |

**a.** What is the probability that a cell has at least one of the positive nickel-charged options?

**b.** What is the probability that a cell is *not* composed of a positive nickel charge greater than +3?

**2.3.6** A message can follow different paths through servers on a network. The sender's message can go to one of five servers for the first step; each of them can send to five servers at the second step; each of those can send to four servers at the third step; and then the message goes to the recipient's server.

**a.** How many paths are possible?

**b.** If all paths are equally likely, what is the probability that a message passes through the first of four servers at the third step?

**2.3.7** WP SS Suppose your vehicle is licensed in a state that issues license plates that consist of three digits (between 0 and 9) followed by three letters (between $A$ and $Z$). If a license number is selected randomly, what is the probability that yours is the one selected?

**2.3.8** WP Disks of polycarbonate plastic from a supplier are analyzed for scratch and shock resistance. The results from 100 disks are summarized as follows:

| | | Shock Resistance | |
|---|---|---|---|
| | | High | Low |
| Scratch Resistance | High | 70 | 9 |
| | Low | 16 | 5 |

Let $A$ denote the event that a disk has high shock resistance, and let $B$ denote the event that a disk has high scratch resistance. If a disk is selected at random, determine the following probabilities:

**a.** $P(A)$ **b.** $P(B)$ **c.** $P(A')$
**d.** $P(A \cap B)$ **e.** $P(A \cup B)$ **f.** $P(A' \cup B)$

**2.3.9** Magnesium alkyls are used as homogenous catalysts in the production of linear low-density polyethylene (LLDPE), which requires a finer magnesium powder to sustain a reaction. Redox reaction experiments using four different amounts of magnesium powder are performed. Each result may or may not be further reduced in a second step using three different magnesium powder amounts. Each of these results may or may not be further reduced in a third step using three different amounts of magnesium powder.

**a.** How many experiments are possible?

**b.** If all outcomes are equally likely, what is the probability that the best result is obtained from an experiment that uses all three steps?



**c.** Does the result in part (b) change if five or six or seven different amounts are used in the first step? Explain.

**2.3.10** WP An article in the *Journal of Database Management* ["Experimental Study of a Self-Tuning Algorithm for DBMS Buffer Pools" (2005, Vol. 16, pp. 1–20)] provided the workload used in the TPC-C OLTP (Transaction Processing Performance Council's Version C On-Line Transaction Processing) benchmark, which simulates a typical order entry application.

**Average Frequencies and Operations in TPC-C**

| Transaction | Frequency | Selects | Updates |
|---|---|---|---|
| New order | 43 | 23 | 11 |
| Payment | 44 | 4.2 | 3 |
| Order status | 4 | 11.4 | 0 |
| Delivery | 5 | 130 | 120 |
| Stock level | 4 | 0 | 0 |

The frequency of each type of transaction (in the second column) can be used as the percentage of each type of transaction. The average number of *selects* operations required for each type of transaction is shown. Let $A$ denote the event of transactions with an average number of *selects* operations of 12 or fewer. Let $B$ denote the event of transactions with an average number of *updates* operations of 12 or fewer. Calculate the following probabilities.

**a.** $P(A)$ **b.** $P(B)$ **c.** $P(A \cap B)$
**d.** $P(A \cap B')$ **e.** $P(A \cup B)$

**2.3.11** VS Samples of emissions from three suppliers are classified for conformance to air-quality specifications. The results from 100 samples are summarized as follows:

| | | Conforms | |
|---|---|---|---|
| | | Yes | No |
| Supplier | 1 | 22 | 8 |
| | 2 | 25 | 5 |
| | 3 | 30 | 10 |

Let $A$ denote the event that a sample is from supplier 1, and let $B$ denote the event that a sample conforms to specifications. If a sample is selected at random, determine the following probabilities:

**a.** $P(A)$ **b.** $P(B)$ **c.** $P(A')$
**d.** $P(A \cap B)$ **e.** $P(A \cup B)$ **f.** $P(A' \cap B)$

**2.3.12** WP VS Consider the hospital emergency room data in Example 2.6. Let $A$ denote the event that a visit is to hospital 4, and let $B$ denote the event that a visit results in LWBS (at any hospital). Determine the following probabilities.

**a.** $P(A \cap B)$ **b.** $P(A')$ **c.** $P(A \cup B)$
**d.** $P(A \cup B')$ **e.** $P(A' \cap B')$

**2.3.13** Use the axioms of probability to show the following:

**a.** For any event $E$, $P(E') = 1 - P(E)$.

**b.** $P(\emptyset) = 0$

**c.** If $A$ is contained in $B$, then $P(A) \leq P(B)$.

**2.3.14** Suppose that a patient is selected randomly from those described in Exercise 2.1.25. Let $A$ denote the event that the patient is in the group treated with interferon alfa, and let $B$ denote the event that the patient has a complete response. Determine the following probabilities.

**a.** $P(A)$ **b.** $P(B)$ **c.** $P(A \cap B)$
**d.** $P(A \cup B)$ **e.** $P(A' \cup B)$

**2.3.15** SS A Web ad can be designed from four different colors, three font types, five font sizes, three images, and five text phrases. A specific design is randomly generated by the Web server when you visit the site. If you visit the site five times, what is the probability that you will not see the same design?

**2.3.16** Similar to the hospital schedule in Example 2.9, suppose that an operating room needs to schedule three knee, four hip, and five shoulder surgeries. Assume that all schedules are equally likely. Determine the probability for each of the following:

**a.** All hip surgeries are completed before another type of surgery.

**b.** The schedule begins with a hip surgery.

**c.** The first and last surgeries are hip surgeries.

**d.** The first two surgeries are hip surgeries.

**2.3.17** A computer system uses passwords that contain exactly eight characters, and each character is one of 26 lowercase letters ($a$–$z$) or 26 uppercase letters ($A$–$Z$) or 10 integers (0–9). Let $\Omega$ denote the set of all possible passwords, and let $A$ and $B$ denote the events that consist of passwords with only letters or only integers, respectively. Suppose that all passwords in $\Omega$ are equally likely. Determine the probability of each of the following:

**a.** $A$ **b.** $B$

**c.** A password contains at least 1 integer.

**d.** A password contains exactly 2 integers.

## Exercises for Section 2.4

WP Problem available in WileyPLUS at instructor's discretion.
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SS Student solution available in WileyPLUS.
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**2.4.1** WP If $A$, $B$, and $C$ are mutually exclusive events with $P(A) = 0.2$, $P(B) = 0.3$, and $P(C) = 0.4$, determine the following probabilities:

**a.** $P(A \cup B \cup C)$ **b.** $P(A \cap B \cap C)$
**c.** $P(A \cap B)$ **d.** $P[(A \cup B) \cap C]$
**e.** $P(A' \cap B' \cap C')$

**2.4.2** WP If $P(A) = 0.3$, $P(B) = 0.2$, and $P(A \cap B) = 0.1$, determine the following probabilities:

**a.** $P(A')$ **b.** $P(A \cup B)$ **c.** $P(A' \cap B)$
**d.** $P(A \cap B')$ **e.** $P[(A \cup B)']$ **f.** $P(A' \cup B)$

**2.4.3** WP A manufacturer of front lights for automobiles tests lamps under a high-humidity, high-temperature environment using intensity and useful life as the responses of interest. The following table shows the performance of 130 lamps:

| | | Useful life | |
|---|---|---|---|
| | | Satisfactory | Unsatisfactory |
| Intensity | Satisfactory | 117 | 3 |
| | Unsatisfactory | 8 | 2 |

**a.** Find the probability that a randomly selected lamp will yield unsatisfactory results under any criteria.
**b.** The customers for these lamps demand 95% satisfactory results. Can the lamp manufacturer meet this demand?

**2.4.4** WP In the article "ACL Reconstruction Using Bone-Patellar Tendon-Bone Press-Fit Fixation: 10-Year Clinical Results" in *Knee Surgery, Sports Traumatology, Arthroscopy* (2005, Vol. 13, pp. 248–255), the following causes for knee injuries were considered:

| Activity | Percentage of Knee Injuries |
|---|---|
| Contact sport | 46% |
| Noncontact sport | 44% |
| Activity of daily living | 9% |
| Riding motorcycle | 1% |

**a.** What is the probability that a knee injury resulted from a sport (contact or noncontact)?
**b.** What is the probability that a knee injury resulted from an activity other than a sport?

**2.4.5** Consider the hospital emergency room data in Example 2.6. Let $A$ denote the event that a visit is to hospital 4, and let $B$ denote the event that a visit results in LWBS (at any hospital). Use the addition rules to calculate the following probabilities.
**a.** $P(A \cup B)$ **b.** $P(A \cup B')$ **c.** $P(A' \cup B')$

**2.4.6** WP Strands of copper wire from a manufacturer are analyzed for strength and conductivity. The results from 100 strands are as follows:

| | Strength | |
|---|---|---|
| | High | Low |
| High conductivity | 74 | 8 |
| Low conductivity | 15 | 3 |

**a.** If a strand is randomly selected, what is the probability that its conductivity is high and its strength is high?
**b.** If a strand is randomly selected, what is the probability that its conductivity is low or its strength is low?
**c.** Consider the event that a strand has low conductivity and the event that the strand has low strength. Are these two events mutually exclusive?

**2.4.7** WP SS A computer system uses passwords that are six characters, and each character is one of the 26 letters ($a$–$z$) or 10 integers (0–9). Uppercase letters are not used. Let $A$ denote the event that a password begins with a vowel (either $a$, $e$, $i$, $o$, or $u$), and let $B$ denote the event that a password ends with an even number (either 0, 2, 4, 6, or 8). Suppose a hacker selects a password at random. Determine the following probabilities:
**a.** $P(A)$ **b.** $P(B)$
**c.** $P(A \cap B)$ **d.** $P(A \cup B)$

**2.4.8** A Web ad can be designed from four different colors, three font types, five font sizes, three images, and five text phrases. A specific design is randomly generated by the Web server when you visit the site. Let $A$ denote the event that the design color is red, and let $B$ denote the event that the font size is not the smallest one. Use the addition rules to calculate the following probabilities.
**a.** $P(A \cup B)$ **b.** $P(A \cup B')$ **c.** $P(A' \cup B')$

**2.4.9** A computer system uses passwords that contain exactly eight characters, and each character is one of the 26 lowercase letters ($a$–$z$) or 26 uppercase letters ($A$–$Z$) or 10 integers (0–9). Assume all passwords are equally likely. Let $A$ and $B$ denote the events that consist of passwords with only letters or only integers, respectively. Determine the following probabilities:
**a.** $P(A \cup B)$ **b.** $P(A' \cup B)$
**c.** $P$ (Password contains exactly 1 or 2 integers)

**2.4.10** Consider the three patient groups in Exercise 2.1.25. Let $A$ denote the event that the patient was treated with ribavirin plus interferon alfa, and let $B$ denote the event that the response was complete. Determine the following probabilities:
**a.** $P(A \cup B)$ **b.** $P(A' \cup B)$ **c.** $P(A \cup B')$

**2.4.11** The article "Clinical and Radiographic Outcomes of Four Different Treatment Strategies in Patients with Early Rheumatoid Arthritis" [*Arthritis & Rheumatism* (2005, Vol. 52, pp. 3381–3390)] considered four treatment groups. The groups consisted of patients with different drug therapies (such as prednisone and infliximab): sequential monotherapy (group 1), step-up combination therapy (group 2), initial combination therapy (group 3), or initial combination therapy with infliximab (group 4). Radiographs of hands and feet were used to evaluate disease progression. The number of patients without progression of joint damage was 76 of 114 patients (67%), 82 of 112 patients (73%), 104 of 120 patients (87%), and 113 of 121 patients (93%) in groups 1–4, respectively. Suppose that a patient is selected randomly. Let $A$ denote the event that the patient is in group 1, and let $B$ denote the event that there is no progression. Determine the following probabilities:
**a.** $P(A \cup B)$ **b.** $P(A' \cup B')$ **c.** $P(A \cup B')$



## Exercises for Section 2.5

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**2.5.1** WP SS VS The analysis of results from a leaf transmutation experiment (turning a leaf into a petal) is summarized by type of transformation completed:

| | | Total Textural Transformation | |
|---|---|---|---|
| | | Yes | No |
| Total color transformation | Yes | 243 | 26 |
| | No | 13 | 18 |

**a.** If a leaf completes the color transformation, what is the probability that it will complete the textural transformation?
**b.** If a leaf does not complete the textural transformation, what is the probability it will complete the color transformation?

**2.5.2** WP Samples of skin experiencing desquamation are analyzed for both moisture and melanin content. The results from 100 skin samples are as follows:

| | | Melanin Content | |
|---|---|---|---|
| | | High | Low |
| Moisture content | High | 13 | 7 |
| | Low | 48 | 32 |

Let $A$ denote the event that a sample has low melanin content, and let $B$ denote the event that a sample has high moisture content. Determine the following probabilities:
**a.** $P(A)$ **b.** $P(B)$ **c.** $P(A \mid B)$ **d.** $P(B \mid A)$

**2.5.3** WP The following table summarizes the number of deceased beetles under autolysis (the destruction of a cell after its death by the action of its own enzymes) and putrefaction (decomposition of organic matter, especially protein, by microorganisms, resulting in production of foul-smelling matter):

| | | Autolysis | |
|---|---|---|---|
| | | High | Low |
| Putrefaction | High | 14 | 59 |
| | Low | 18 | 9 |

**a.** If the autolysis of a sample is high, what is the probability that the putrefaction is low?
**b.** If the putrefaction of a sample is high, what is the probability that the autolysis is high?
**c.** If the putrefaction of a sample is low, what is the probability that the autolysis is low?

**2.5.4** WP A maintenance firm has gathered the following information regarding the failure mechanisms for air conditioning systems:

| | | Evidence of Gas Leaks | |
|---|---|---|---|
| | | Yes | No |
| Evidence of electrical failure | Yes | 55 | 17 |
| | No | 32 | 3 |

The units without evidence of gas leaks or electrical failure showed other types of failure. If this is a representative sample of AC failure, find the probability
**a.** That failure involves a gas leak
**b.** That there is evidence of electrical failure given that there was a gas leak
**c.** That there is evidence of a gas leak given that there is evidence of electrical failure

**2.5.5** Consider the endothermic reactions in Exercise 2.1.22. Let $A$ denote the event that a reaction's final temperature is 271 K or less. Let $B$ denote the event that the heat absorbed is above target. Determine the following probabilities.
**a.** $P(A \mid B)$ **b.** $P(A' \mid B)$
**c.** $P(A \mid B')$ **d.** $P(B \mid A)$

**2.5.6** WP A batch of 500 containers for frozen orange juice contains 5 that are defective. Three are selected, at random, without replacement from the batch.
**a.** What is the probability that the second one selected is defective given that the first one was defective?
**b.** What is the probability that the first two selected are defective?
**c.** What is the probability that the first two selected are both acceptable?
**d.** What is the probability that the third one selected is defective given that the first and second ones selected were defective?
**e.** What is the probability that the third one selected is defective given that the first one selected was defective and the second one selected was okay?
**f.** What is the probability that all three selected ones are defective?

**2.5.7** Suppose $A$ and $B$ are mutually exclusive events. Construct a Venn diagram that contains the three events $A$, $B$, and $C$ such that $P(A \mid C) = 1$ and $P(B \mid C) = 0$.

**2.5.8** An article in *The Canadian Entomologist* (Harcourt et al., 1977, Vol. 109, pp. 1521–1534) reported on the life of the alfalfa weevil from eggs to adulthood. The following table shows the number of larvae that survived at each stage of development from eggs to adults.

| Eggs | Early Larvae | Late Larvae | Pre-pupae | Late Pupae | Adults |
|---|---|---|---|---|---|
| 421 | 412 | 306 | 45 | 35 | 31 |

**a.** What is the probability an egg survives to adulthood?
**b.** What is the probability of survival to adulthood given survival to the late larvae stage?
**c.** What stage has the lowest probability of survival to the next stage?

**2.5.9** WP SS VS Consider the hospital emergency room data in Example 2.6. Let $A$ denote the event that a visit is to hospital 4, and let $B$ denote the event that a visit results in LWBS (at any hospital). Determine the following probabilities.

**a.** $P(A \mid B)$ **b.** $P(A' \mid B)$
**c.** $P(A \mid B')$ **d.** $P(B \mid A)$

**2.5.10** A computer system uses passwords that contain exactly eight characters, and each character is one of the 26 lowercase letters ($a$–$z$) or 26 uppercase letters ($A$–$Z$) or 10 integers (0–9). Let $\Omega$ denote the set of all possible passwords. Suppose that all passwords in $\Omega$ are equally likely. Determine the probability for each of the following:

**a.** Password contains all lowercase letters given that it contains only letters
**b.** Password contains at least 1 uppercase letter given that it contains only letters
**c.** Password contains only even numbers given that it contains all numbers

**2.5.11** Suppose that a patient is selected randomly from those described in Exercise 2.4.11. Let $A$ denote the event that the patient is in group 1, and let $B$ denote the event that there is no progression. Determine the following probabilities:

**a.** $P(B \mid A)$ **b.** $P(A \mid B)$
**c.** $P(A \mid B')$ **d.** $P(A' \mid B)$

## Exercises for Section 2.6

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**2.6.1** WP Suppose that $P(A \mid B) = 0.4$ and $P(B) = 0.5$. Determine the following:

**a.** $P(A \cap B)$
**b.** $P(A' \cap B)$

**2.6.2** WP Suppose that $P(A \mid B) = 0.2$, $P(A \mid B') = 0.3$, and $P(B) = 0.8$. What is $P(A)$?

**2.6.3** WP The probability is 1% that an electrical connector that is kept dry fails during the warranty period. If the connector is ever wet, the probability of a failure during the warranty period is 5%. If 90% of the connectors are kept dry and 10% are wet, what proportion of connectors fail during the warranty period?

**2.6.4** WP Heart failures are due to either natural occurrences (87%) or outside factors (13%). Outside factors are related to induced substances (73%) or foreign objects (27%). Natural occurrences are caused by arterial blockage (56%), disease (27%), and infection (e.g., staph infection) (17%).

**a.** Determine the probability that a failure is due to an induced substance.
**b.** Determine the probability that a failure is due to disease or infection.

**2.6.5** WP SS The edge roughness of slit paper products increases as knife blades wear. Only 1% of products slit with new blades have rough edges, 3% of products slit with blades of average sharpness exhibit roughness, and 5% of products slit with worn blades exhibit roughness. If 25% of the blades in manufacturing are new, 60% are of average sharpness, and 15% are worn, what is the proportion of products that exhibit edge roughness?

**2.6.6** WP A lot of 100 semiconductor chips contains 20 that are defective.

**a.** Two are selected, at random, without replacement, from the lot. Determine the probability that the second chip selected is defective.
**b.** Three are selected, at random, without replacement, from the lot. Determine the probability that all are defective.

**2.6.7** WP Computer keyboard failures are due to faulty electrical connects (12%) or mechanical defects (88%). Mechanical defects are related to loose keys (27%) or improper assembly (73%). Electrical connect defects are caused by defective wires (35%), improper connections (13%), or poorly welded wires (52%).

**a.** Find the probability that a failure is due to loose keys.
**b.** Find the probability that a failure is due to improperly connected or poorly welded wires.

**2.6.8** An article in the *British Medical Journal* ["Comparison of Treatment of Renal Calculi by Operative Surgery, Percutaneous Nephrolithotomy, and Extracorporeal Shock Wave Lithotripsy" (1986, Vol. 82, pp. 879–892)] provided the following discussion of success rates in kidney stone removals. Open surgery had a success rate of 78% (273/350) and a newer method, percutaneous nephrolithotomy (PN), had a success rate of 83% (289/350). This newer method looked better, but the results changed when stone diameter was considered. For stones with diameters less than 2 centimeters, 93% (81/87) of cases of open surgery were successful compared with only 83% (234/270) of cases of PN. For stones greater than or equal to 2 centimeters, the success rates were 73% (192/263) and 69% (55/80) for open surgery and PN, respectively. Open surgery is better for both stone sizes, but less successful in total. In 1951, E. H. Simpson pointed out this apparent contradiction (known as **Simpson's paradox**), and the hazard still persists today. Explain how open surgery can be better for both stone sizes but worse in total.



**2.6.9** SS Consider the hospital emergency room data in Example 2.6. Let $A$ denote the event that a visit is to hospital 4 and let $B$ denote the event that a visit results in LWBS (at any hospital). Determine the following probabilities.

**a.** $P(A \cap B)$ **b.** $P(A \cup B)$ **c.** $P'(A' \cup B')$
**d.** Use the total probability rule to determine $P(A)$

**2.6.10** Suppose that a patient is selected randomly from those described in Exercise 2.4.11. Let $A$ denote the event that the patient is in group 1, and let $B$ denote the event for which there is no progression. Determine the following probabilities:

**a.** $P(A \cap B)$ **b.** $P(B)$ **c.** $P(A' \cap B)$
**d.** $P(A \cup B)$ **e.** $P(A' \cup B)$

**2.6.11** A Web ad can be designed from four different colors, three font types, five font sizes, three images, and five text phrases. A specific design is randomly generated by the Web server when you visit the site. Determine the probability that the ad color is red and the font size is not the smallest one.

**2.6.12** Similar to the hospital schedule in Example 2.9, suppose that an operating room needs to schedule three knee, four hip, and five shoulder surgeries. Assume that all schedules are equally likely. Determine the following probabilities:

**a.** All hip surgeries are completed first given that all knee surgeries are last.
**b.** The schedule begins with a hip surgery given that all knee surgeries are last.
**c.** The first and last surgeries are hip surgeries given that knee surgeries are scheduled in time periods 2 through 4.
**d.** The first two surgeries are hip surgeries given that all knee surgeries are last.

## Exercises for Section 2.7

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**2.7.1** WP If $P(A \mid B) = 0.3$, $P(B) = 0.8$, and $P(A) = 0.3$, are the events $B$ and the complement of $A$ independent?

**2.7.2** WP If $P(A) = 0.2$, $P(B) = 0.2$, and $A$ and $B$ are mutually exclusive, are they independent?

**2.7.3** WP A batch of 500 containers of frozen orange juice contains 5 that are defective. Two are selected, at random, without replacement, from the batch. Let $A$ and $B$ denote the events that the first and second containers selected are defective, respectively.

**a.** Are $A$ and $B$ independent events?
**b.** If the sampling were done with replacement, would $A$ and $B$ be independent?

**2.7.4** WP Disks of polycarbonate plastic from a supplier are analyzed for scratch and shock resistance. The results from 100 disks are summarized as follows:

| | | Shock Resistance | |
|---|---|---|---|
| | | High | Low |
| Scratch Resistance | High | 70 | 9 |
| | Low | 16 | 5 |

Let $A$ denote the event that a disk has high shock resistance, and let $B$ denote the event that a disk has high scratch resistance. Are events $A$ and $B$ independent?

**2.7.5** WP SS Redundant array of inexpensive disks (RAID) is a technology that uses multiple hard drives to increase the speed of data transfer and provide instant data backup. Suppose that the probability of any hard drive failing in a day is 0.001 and the drive failures are independent.

**a.** A RAID 0 scheme uses two hard drives, each containing a mirror image of the other. What is the probability of data loss? Assume that data loss occurs if both drives fail within the same day.
**b.** A RAID 1 scheme splits the data over two hard drives. What is the probability of data loss? Assume that data loss occurs if at least one drive fails within the same day.

**2.7.6** WP A test of a printed circuit board uses a random test pattern with an array of 10 bits and each is equally likely to be 0 or 1. Assume the bits are independent.

**a.** What is the probability that all bits are 1s?
**b.** What is the probability that all bits are 0s?
**c.** What is the probability that exactly 5 bits are 1s and 5 bits are 0s?

**2.7.7** WP The probability that a lab specimen contains high levels of contamination is 0.10. Five samples are checked, and the samples are independent.

**a.** What is the probability that none contain high levels of contamination?
**b.** What is the probability that exactly one contains high levels of contamination?
**c.** What is the probability that at least one contains high levels of contamination?

**2.7.8** WP A player of a video game is confronted with a series of four opponents and an 80% probability of defeating each opponent. Assume that the results from opponents are independent (and that when the player is defeated by an opponent the game ends).

**a.** What is the probability that a player defeats all four opponents in a game?
**b.** What is the probability that a player defeats at least two opponents in a game?
**c.** If the game is played three times, what is the probability that the player defeats all four opponents at least once?

**2.7.9** WP Eight cavities in an injection-molding tool produce plastic connectors that fall into a common stream. A sample is chosen every several minutes. Assume that the samples are independent.

**a.** What is the probability that five successive samples were all produced in cavity 1 of the mold?

**b.** What is the probability that five successive samples were all produced in the same cavity of the mold?

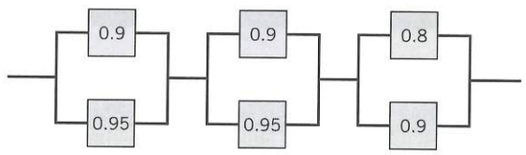
**c.** What is the probability that four out of five successive samples were produced in cavity 1 of the mold?

**2.7.10** A credit card contains 16 digits. It also contains the month and year of expiration. Suppose there are 1 million credit card holders with unique card numbers. A hacker randomly selects a 16-digit credit card number.

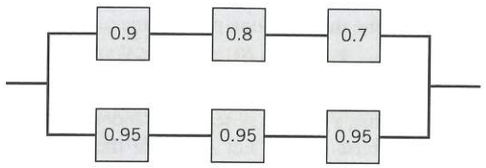
**a.** What is the probability that it belongs to a user?

**b.** Suppose a hacker has a 25% chance of correctly guessing the year your card expires and randomly selects 1 of the 12 months. What is the probability that the hacker correctly selects the month and year of expiration?

**2.7.11** SS The following circuit operates if and only if there is a path of functional devices from left to right. The probability that each device functions is as shown. Assume that the probability that a device is functional does not depend on whether or not other devices are functional. What is the probability that the circuit operates?




**2.7.12** VS The following circuit operates if and only if there is a path of functional devices from left to right. The probability that each device functions is as shown. Assume that the probability that a device is functional does not depend on whether or not other devices are functional. What is the probability that the circuit operates?




**2.7.13** Consider the hospital emergency room data in Example 2.6. Let $A$ denote the event that a visit is to hospital 4, and let $B$ denote the event that a visit results in LWBS (at any hospital). Are these events independent?

**2.7.14** A Web ad can be designed from four different colors, three font types, five font sizes, three images, and five text phrases. A specific design is randomly generated by the Web server when you visit the site. Let $A$ denote the event that the design color is red, and let $B$ denote the event that the font size is not the smallest one. Are $A$ and $B$ independent events? Explain why or why not.

**2.7.15** An integrated circuit contains 10 million logic gates (each can be a logical AND or OR circuit). Assume the probability of a gate failure is $p$ and that the failures are independent. The integrated circuit fails to function if any gate fails. Determine the value for $p$ so that the probability that the integrated circuit functions is 0.95.

**2.7.16** Table 2.1 provides data on wafers categorized by location and contamination levels. Let $A$ denote the event that contamination is *low*, and let $B$ denote the event that the location is *center*. Are $A$ and $B$ independent? Why or why not?

## Exercises for Section 2.8

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**2.8.1** WP SS VS Customers are used to evaluate preliminary product designs. In the past, 95% of highly successful products received good reviews, 60% of moderately successful products received good reviews, and 10% of poor products received good reviews. In addition, 40% of products have been highly successful, 35% have been moderately successful, and 25% have been poor products.

**a.** What is the probability that a product attains a good review?

**b.** If a new design attains a good review, what is the probability that it will be a highly successful product?

**c.** If a product does not attain a good review, what is the probability that it will be a highly successful product?

**2.8.2** WP Suppose that $P(A \mid B) = 0.7$, $P(A) = 0.5$, and $P(B) = 0.2$. Determine $P(B \mid A)$.

**2.8.3** WP A new analytical method to detect pollutants in water is being tested. This new method of chemical analysis is important because, if adopted, it could be used to detect three different contaminants—organic pollutants, volatile solvents, and chlorinated compounds—instead of having to use a single test for each pollutant. The makers of the test claim that it can detect high levels of organic pollutants with 99.7% accuracy, volatile solvents with 99.95% accuracy, and chlorinated compounds with 89.7% accuracy. If a pollutant is not present, the test does not signal. Samples are prepared for the calibration of the test and 60% of them are contaminated with organic pollutants, 27% with volatile solvents, and 13% with traces of chlorinated compounds. A test sample is selected randomly.

**a.** What is the probability that the test will signal?

**b.** If the test signals, what is the probability that chlorinated compounds are present?

**2.8.4** WP Software to detect fraud in consumer phone cards tracks the number of metropolitan areas where calls originate each day. It is found that 1% of the legitimate users originate calls from two or more metropolitan areas in a single day. However, 30% of fraudulent users originate calls from two or more metropolitan areas in a single day. The proportion of fraudulent users is 0.01%. If the same user originates calls from two or more metropolitan areas in a single day, what is the probability that the user is fraudulent?



**2.8.5** Consider the hospital emergency room data in Example 2.6. Use Bayes' theorem to calculate the probability that a person visits hospital 4 given they are LWBS.

**2.8.6** Suppose that a patient is selected randomly from those described in Exercise 2.4.11. Let $A$ denote the event that the patient is in group 1, and let $B$ denote the event that there is no progression. Determine the following probabilities:

**a.** $P(B)$ **b.** $P(B \mid A)$ **c.** $P(A \mid B)$

**2.8.7** SS Two Web colors are used for a site advertisement. If a site visitor arrives from an affiliate, the probabilities of the blue or green colors being used in the advertisement are 0.8 and 0.2, respectively. If the site visitor arrives from a search site, the probabilities of blue and green colors in the advertisement are 0.4 and 0.6, respectively. The proportions of visitors from affiliates and search sites are 0.3 and 0.7, respectively. What is the probability that a visitor is from a search site given that the blue ad was viewed?

**2.8.8** A recreational equipment supplier finds that among orders that include tents, 40% also include sleeping mats. Only 5% of orders that do not include tents do include sleeping mats. Also, 20% of orders include tents. Determine the following probabilities:

**a.** The order includes sleeping mats.

**b.** The order includes a tent given it includes sleeping mats.

**2.8.9** An e-mail filter is planned to separate valid e-mails from spam. The word *free* occurs in 60% of the spam messages and only 4% of the valid messages. Also, 20% of the messages are spam. Determine the following probabilities:

**a.** The message contains *free*.

**b.** The message is spam given that it contains *free*.

**c.** The message is valid given that it does not contain *free*.

## Exercises for Section 2.9

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**2.9.1** SS Decide whether a discrete or continuous random variable is the best model for each of the following variables:

**a.** The number of cracks exceeding one-half inch in 10 miles of an interstate highway.

**b.** The weight of an injection-molded plastic part.

**c.** The number of molecules in a sample of gas.

**d.** The concentration of output from a reactor.

**e.** The current in an electronic circuit.

**2.9.2** Decide whether a discrete or continuous random variable is the best model for each of the following variables:

**a.** The time until a projectile returns to earth.

**b.** The number of times a transistor in a computer memory changes state in one operation.

**c.** The volume of gasoline that is lost to evaporation during the filling of a gas tank.

**d.** The outside diameter of a machined shaft.

**2.9.3** WP Decide whether a discrete or continuous random variable is the best model for each of the following variables:

**a.** The time for a computer algorithm to assign an image to a category.

**b.** The number of bytes used to store a file in a computer.

**c.** The ozone concentration in micrograms per cubic meter.

**d.** The ejection fraction (volumetric fraction of blood pumped from a heart ventricle with each beat).

**e.** The fluid flow rate in liters per minute.

## Supplemental Exercises for Chapter 2

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**2.S4** WP Samples of laboratory glass are in small, light packaging or heavy, large packaging. Suppose that 2% and 1%, respectively, of the sample shipped in small and large packages, respectively, break during transit. If 60% of the samples are shipped in large packages and 40% are shipped in small packages, what proportion of samples break during shipment?

**2.S5** WP SS Samples of a cast aluminum part are classified on the basis of surface finish (in microinches) and edge finish. The results of 100 parts are summarized as follows:

| | | Edge Finish | |
|---|---|---|---|
| | | Excellent | Good |
| Surface finish | Excellent | 80 | 2 |
| | Good | 10 | 8 |

Let $A$ denote the event that a sample has excellent surface finish, and let $B$ denote the event that a sample has excellent edge finish. If a part is selected at random, determine the following probabilities:

**a.** $P(A)$ **b.** $P(B)$ **c.** $P(A')$

**d.** $P(A \cap B)$ **e.** $P(A \cup B)$ **f.** $P(A' \cup B)$

**2.S6** WP A lot contains 15 castings from a local supplier and 25 castings from a supplier in the next state. Two castings are

**525** A machine tool is idle 15% of the time. You request immediate use of the tool on five different occasions during the year. Assume that your requests represent independent events.

**a.** What is the probability that the tool is idle at the time of all of your requests?

**b.** What is the probability that the machine is idle at the time of exactly four of your requests?

**c.** What is the probability that the tool is idle at the time of at least three of your requests?

**2-26** A company that tracks the use of its Web site determined that the more pages a visitor views, the more likely the visitor is to request more information (RMI). Use the following table to answer the questions:

| Number of pages viewed: | 1 | 2 | 3 | 4 or more |
|---|---|---|---|---|
| Percentage of visitors: | 40 | 30 | 20 | 10 |
| Percentage who RMI | 10 | 10 | 20 | 40 |

**a.** What is the probability that a visitor to the Web site RMIs?

**b.** If a visitor RMIs, what is the probability that the visitor viewed four or more pages?

**2-27** SS An article in *Genome Research* ["An Assessment of the Prediction Accuracy in Large DNA Sequences" (2000, Vol. pp. 1631–1642)] considered the accuracy of commercial software to predict nucleotides in gene sequences. The following table shows the number of sequences for which the programs produced predictions and the number of nucleotides correctly predicted computed globally from the total number of prediction successes failures on all sequences).

| | Number of Sequences | Proportion |
|---|---|---|
| GenScan | 177 | 0.93 |
| Fgenesx default | 175 | 0.91 |
| Fgenesx topcomboN | 174 | 0.97 |
| Fgenesx 2 stages | 175 | 0.90 |
| GeneWise | 177 | 0.98 |
| Procrustes | 177 | 0.93 |

Assume the prediction successes and failures are independent among the programs.

**a.** What is the probability that all programs predict a nucleotide correctly?

**b.** What is the probability that all programs predict a nucleotide incorrectly?

**c.** What is the probability that at least one Blastx program predicts a nucleotide correctly?

**2.S28** WP A batch contains 36 bacteria cells. Assume that 12 of the cells are not capable of cellular replication. Of the cells, 6 are selected at random, without replacement, to be checked for replication.

**a.** What is the probability that all 6 of the selected cells are able to replicate?

**b.** What is the probability that at least 1 of the selected cells is not capable of replication?

**2.S29** A computer system uses passwords that are exactly seven characters, and each character is one of the 26 letters ($a$–$z$) or 10 integers (0–9). Uppercase letters are not used.

**a.** How many passwords are possible?

**b.** If a password consists of exactly 6 letters and 1 number, how many passwords are possible?

**c.** If a password consists of 5 letters followed by 2 numbers, how many passwords are possible?

**2.S30** Consider the treatments in Exercise 2.1.25. Suppose a patient is selected randomly. Let $A$ denote the event that the patient is treated with ribavirin plus interferon alfa or interferon alfa, and let $B$ denote the event that the response is complete. Determine the following probabilities.

**a.** $P(A \mid B)$ **b.** $P(B \mid A)$ **c.** $P(A \cap B)$ **d.** $P(A \cup B)$

**2.S31** Consider the patient groups in Exercise 2.4.11. Suppose a patient is selected randomly. Let $A$ denote the event that the patient is in group 1 or 2, and let $B$ denote the event that there is no progression. Determine the following probabilities:

**a.** $P(A \mid B)$ **b.** $P(B \mid A)$ **c.** $P(A \cap B)$ **d.** $P(A \cup B)$

# Chapter 3 Exercises

## Exercises for Section 3.1

Problem available in WileyPLUS at instructor's discretion.
GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
Student solution available in WileyPLUS.
Student video solution available in WileyPLUS.

*For each of the following exercises, determine the range (possible values) of the random variable.*

**3.1.1** WP The random variable is the number of nonconforming solder connections on a printed circuit board with 1000 connections.

**3.1.2** WP A batch of 500 machined parts contains 10 that do not conform to customer requirements. The random variable is the number of parts in a sample of five parts that do not conform to customer requirements.

**3.1.3** WP A batch of 500 machined parts contains 10 that do not conform to customer requirements. Parts are selected successively, without replacement, until a nonconforming part is obtained. The random variable is the number of parts selected.

**3.1.4** The random variable is the number of computer clock cycles required to complete a selected arithmetic calculation.



**3.1.5** The random variable is the number of surface flaws in a large coil of galvanized steel.

**3.1.6** WP A group of 10,000 people are tested for a gene called *Ifi202* that has been found to increase the risk for lupus. The random variable is the number of people who carry the gene.

**3.1.7** WP SS The number of mutations in a nucleotide sequence of length 40,000 in a DNA strand after exposure to radiation is measured. Each nucleotide may be mutated.

**3.1.8** A healthcare provider schedules 30 minutes for each patient's visit, but some visits require extra time. The random variable is the number of patients treated in an eight-hour day.

**3.1.9** WP The sample space of a random experiment is $\{a, b, c, d, e, f\}$, and each outcome is equally likely. A random variable is defined as follows:

| outcome | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|---|---|---|---|---|---|---|
| $x$ | 0 | 0 | 1.5 | 1.5 | 2 | 3 |

Determine the probability mass function of $X$. Use the probability mass function to determine the following probabilities:

**a.** $P(X = 1.5)$ **b.** $P(0.5 < X < 2.7)$
**c.** $P(X > 3)$ **d.** $P(0 \le X < 2)$
**e.** $P(X = 0 \text{ or } X = 2)$

*For Exercises 3.1.10 to 3.1.12, verify that the following functions are probability mass functions, and determine the requested probabilities.*

**3.1.10** WP $f(x) = (8/7)(1/2)^x$, $x = 1, 2, 3$

**a.** $P(X \le 1)$ **b.** $P(X > 1)$
**c.** $P(2 < X < 6)$ **d.** $P(X \le 1 \text{ or } X > 1)$

**3.1.11** WP SS VS $f(x) = \dfrac{2x+1}{25}$, $x = 0, 1, 2, 3, 4$

**a.** $P(X = 4)$ **b.** $P(X \le 1)$
**c.** $P(2 \le X < 4)$ **d.** $P(X > -10)$

**3.1.12** Consider the hospital patients in Example 2.6. Two patients are selected randomly, with replacement, from the total patients at Hospital 1. What is the probability mass function of the number of patients in the sample who are admitted?

**3.1.13**

| $x$ | 1.25 | 1.5 | 1.75 | 2 | 2.25 |
|---|---|---|---|---|---|
| $f(x)$ | 0.2 | 0.4 | 0.1 | 0.2 | 0.1 |

**a.** $P(X \ge 2)$ **b.** $P(X < 1.65)$
**c.** $P(X = 1.5)$ **d.** $P(X < 1.3 \text{ or } X > 2.1)$

**3.1.14** WP An article in *Knee Surgery, Sports Traumatology, Arthroscopy* ["Arthroscopic Meniscal Repair with an Absorbable Screw: Results and Surgical Technique" (2005, Vol. 13, pp. 273–279)] cited a success rate of more than 90% for meniscal tears with a rim width under 3 mm, but only a 67% success rate for tears of 3–6 mm. If you are unlucky enough to suffer a meniscal tear of under 3 mm on your left knee and one of width 3–6 mm on your right knee, what is the probability mass function of the number of successful surgeries? Assume that the surgeries are independent.

**3.1.15** WP In a semiconductor manufacturing process, three wafers from a lot are tested. Each wafer is classified as *pass* or *fail*. Assume that the probability that a wafer passes the test is 0.8 and that wafers are independent. Determine the probability mass function of the number of wafers from a lot that pass the test.

**3.1.16** WP Marketing estimates that a new instrument for the analysis of soil samples will be very successful, moderately successful, or unsuccessful with probabilities 0.3, 0.6, and 0.1, respectively. The yearly revenue associated with a very successful, moderately successful, or unsuccessful product is \$10 million, \$5 million, and \$1 million, respectively. Let the random variable $X$ denote the yearly revenue of the product. Determine the probability mass function of $X$.

**3.1.17** WP SS An assembly consists of three mechanical components. Suppose that the probabilities that the first, second, and third components meet specifications are 0.95, 0.98, and 0.99, respectively. Assume that the components are independent. Determine the probability mass function of the number of components in the assembly that meet specifications.

**3.1.18** The distribution of the time until a Web site changes is important to Web crawlers that search engines use to maintain current information about Web sites. The distribution of the time until change (in days) of a Web site is approximated in the following table.

| Days until Changes | Probability |
|---|---|
| 1.5 | 0.05 |
| 3.0 | 0.25 |
| 4.5 | 0.35 |
| 5.0 | 0.20 |
| 7.0 | 0.15 |

Calculate the probability mass function of the days until change.

**3.1.19** WP Actual lengths of stay at a hospital's emergency department in 2009 are shown in the following table (rounded to the nearest hour). Length of stay is the total of wait and service times. Some longer stays are also approximated as 15 hours in this table.

| Hours | Count | Percent |
|---|---|---|
| 1 | 19 | 3.80 |
| 2 | 51 | 10.20 |
| 3 | 86 | 17.20 |
| 4 | 102 | 20.40 |
| 5 | 87 | 17.40 |
| 6 | 62 | 12.40 |
| 7 | 40 | 8.00 |
| 8 | 18 | 3.60 |
| 9 | 14 | 2.80 |
| 10 | 11 | 2.20 |
| 15 | 10 | 2.00 |

Calculate the probability mass function of the wait time for service.

## Exercises for Section 3.2

P Problem available in WileyPLUS at instructor's discretion.
P GO Tutorial Tutoring problem available in WileyPLUS at tructor's discretion
S Student solution available in WileyPLUS.
S Student video solution available in WileyPLUS.

.1 Determine the cumulative distribution function for the dom variable in Exercise 3.1.9.
.2 Determine the cumulative distribution function of the ran- n variable in Exercise 3.1.10.
.3 Determine the cumulative distribution function for the dom variable in Exercise 3.1.11.
.4 WP Determine the cumulative distribution function for random variable in Exercise 3.1.12; also determine the owing probabilities:
**a.** $P(X < 1.5)$ **b.** $P(X \le 3)$
**c.** $P(X > 2)$ **d.** $P(1 < X \le 2)$
.5 WP SS Determine the cumulative distribution function the random variable in Exercise 3.1.15.
6 WP Determine the cumulative distribution function for random variable in Exercise 3.1.16.

*Verify that the following functions are cumulative distribution functions, and determine the probability mass function and the requested probabilities.*

**3.2.7** WP $F(x) = \begin{cases} 0 & x < 1 \\ 0.5 & 1 \le x < 3 \\ 1 & 3 \le x \end{cases}$

**a.** $P(X \le 3)$ **b.** $P(X \le 2)$
**c.** $P(1 \le X \le 2)$ **d.** $P(X > 2)$

**3.2.8** WP $F(x) = \begin{cases} 0 & x < -10 \\ 0.25 & -10 \le x < 30 \\ 0.75 & 30 \le x < 50 \\ 1 & 50 \le x \end{cases}$

**a.** $P(X \le 50)$ **b.** $P(X \le 40)$
**c.** $P(40 \le X \le 60)$ **d.** $P(X < 0)$
**e.** $P(0 \le X < 10)$ **f.** $P(-10 < X < 10)$

## Exercises for Section 3.3

Problem available in WileyPLUS at instructor's discretion.
GO Tutorial Tutoring problem available in WileyPLUS at ructor's discretion
Student solution available in WileyPLUS.
Student video solution available in WileyPLUS.

1 WP If the range of $X$ is the set {0, 1, 2, 3, 4} and $= x) = 0.2$, determine the mean and variance of the random able.
2 Determine the mean and variance of the random variable in rcise 3.1.10.
3 WP VS Determine the mean and variance of the random able in Exercise 3.1.11.
4 WP Determine the mean and variance of the random able in Exercise 3.1.12.
5 WP Determine the mean and variance of the random able in Exercise 3.1.13.
6 In a NiCd battery, a fully charged cell is composed of nick- hydroxide. Nickel is an element that has multiple oxidation s. Assume the following proportions of the states:

| Nickel Charge | Proportions Found |
|---|---|
| 0 | 0.17 |
| +2 | 0.35 |
| +3 | 0.33 |
| +4 | 0.15 |

**a.** Determine the cumulative distribution function of the nickel charge.
**b.** Determine the mean and variance of the nickel charge.

**3.3.7** WP SS The range of the random variable $X$ is [0, 1, 2, 3, $x$] where $x$ is unknown. If each value is equally likely and the mean of $X$ is 6, determine $x$.

**3.3.8** WP Trees are subjected to different levels of carbon dioxide atmosphere with 6% of them in a minimal growth condition at 350 parts per million (ppm), 10% at 450 ppm (slow growth), 47% at 550 ppm (moderate growth), and 37% at 650 ppm (rapid growth). What are the mean and standard deviation of the carbon dioxide atmosphere (in ppm) for these trees in ppm?



## Exercises for Section 3.4

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**3.4.1** WP Assume that the wavelengths of photosynthetically active radiations (PAR) are uniformly distributed at integer nanometers in the red spectrum from 675 to 700 nm.
**a.** What are the mean and variance of the wavelength distribution for this radiation?
**b.** If the wavelengths are uniformly distributed at integer nanometers from 75 to 100 nanometers, how do the mean and variance of the wavelength distribution compare to the previous part? Explain.

**3.4.2** Let the random variable $X$ have a discrete uniform distribution on the integers $0 \le x \le 99$. Determine the mean and variance of $X$.

**3.4.3** WP SS Suppose that $X$ has a discrete uniform distribution on the integers 0 through 9. Determine the mean, variance, and standard deviation of the random variable $Y = 5X$ and compare to the corresponding results for $X$.

**3.4.4** WP Thickness measurements of a coating process are made to the nearest hundredth of a millimeter. The thickness measurements are uniformly distributed with values 0.15, 0.16, 0.17, 0.18, and 0.19. Determine the mean and variance of the coating thickness for this process.

**3.4.5** Suppose that 1000 seven-digit telephone numbers within your area code are dialed randomly. What is the probability that your number is called?

**3.4.6** Each multiple-choice question on an exam has four choices. Suppose that there are 10 questions and the choice is selected randomly and independently for each question. Let $X$ denote the number of questions answered correctly. Does $X$ have a discrete uniform distribution? Why or why not?

## Exercises for Section 3.5

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**3.5.1** For each scenario (a)–(j), state whether or not the binomial distribution is a reasonable model for the random variable and why. State any assumptions you make.
**a.** A production process produces thousands of temperature transducers. Let $X$ denote the number of nonconforming transducers in a sample of size 30 selected at random from the process.
**b.** From a batch of 50 temperature transducers, a sample of size 30 is selected without replacement. Let $X$ denote the number of nonconforming transducers in the sample.
**c.** Four identical electronic components are wired to a controller that can switch from a failed component to one of the remaining spares. Let $X$ denote the number of components that have failed after a specified period of operation.
**d.** Let $X$ denote the number of accidents that occur along the federal highways in Arizona during a one-month period.
**e.** Let $X$ denote the number of correct answers by a student taking a multiple-choice exam in which a student can eliminate some of the choices as being incorrect in some questions and all of the incorrect choices in other questions.
**f.** Defects occur randomly over the surface of a semiconductor chip. However, only 80% of defects can be found by testing. A sample of 40 chips with one defect each is tested. Let $X$ denote the number of chips in which the test finds a defect.
**g.** Errors in a digital communication channel occur in bursts that affect several consecutive bits. Let $X$ denote the number of bits in error in a transmission of 100,000 bits.
**h.** Let $X$ denote the number of surface flaws in a large coil of galvanized steel.

**3.5.2** WP Let $X$ be a binomial random variable with $p = 0.1$ and $n = 10$. Calculate the following probabilities.
**a.** $P(X \le 2)$ **b.** $P(X > 8)$
**c.** $P(X = 4)$ **d.** $P(5 \le X \le 7)$

**3.5.3** WP SS The random variable $X$ has a binomial distribution with $n = 10$ and $p = 0.01$. Determine the following probabilities.
**a.** $P(X = 5)$ **b.** $P(X \le 2)$
**c.** $P(X \ge 9)$ **d.** $P(3 \le X < 5)$

**3.5.4** The random variable $X$ has a binomial distribution with $n = 10$ and $p = 0.5$. Sketch the probability mass function of $X$.
**a.** What value of $X$ is most likely?
**b.** What value(s) of $X$ is(are) least likely?
**c.** Repeat the previous parts with $p = 0.01$.

**5.5** Determine the cumulative distribution function of a binomial random variable with $n = 3$ and $p = 1/4$.

**5.6** WP An electronic product contains 40 integrated circuits. he probability that any integrated circuit is defective is 0.01, and e integrated circuits are independent. The product operates only there are no defective integrated circuits. What is the probability at the product operates?

**5.7** SS VS The phone lines to an airline reservation system e occupied 40% of the time. Assume that the events that the nes are occupied on successive calls are independent. Assume at 10 calls are placed to the airline.

**a.** What is the probability that for exactly three calls, the lines are occupied?

**b.** What is the probability that for at least one call, the lines are not occupied?

**c.** What is the expected number of calls in which the lines are all occupied?

**5.8** WP A multiple-choice test contains 25 questions, each th four answers. Assume that a student just guesses on each estion.

**a.** What is the probability that the student answers more than 20 questions correctly?

**b.** What is the probability that the student answers fewer than 5 questions correctly?

**.9** WP Samples of rejuvenated mitochondria are mutated efective) in 1% of cases. Suppose that 15 samples are studied d can be considered to be independent for mutation. Determine following probabilities.

**a.** No samples are mutated.

**b.** At most one sample is mutated.

**c.** More than half the samples are mutated.

**.10** WP Heart failure is due to either natural occurrences %) or outside factors (13%). Outside factors are related to uced substances or foreign objects. Natural occurrences are sed by arterial blockage, disease, and infection. Suppose t 20 patients will visit an emergency room with heart fail-. Assume that causes of heart failure for the individuals are ependent.

**a.** What is the probability that three individuals have conditions caused by outside factors?

**b.** What is the probability that three or more individuals have conditions caused by outside factors?

**c.** What are the mean and standard deviation of the number of individuals with conditions caused by outside factors?

**.11** A computer system uses passwords that are exactly six racters and each character is one of the 26 letters (a–z) or integers (0–9). Suppose that 10,000 users of the system have que passwords. A hacker randomly selects (with replacement) ,000 passwords from the potential set, and a match to a user's sword is called a *hit*.

**a.** What is the distribution of the number of hits?

**b.** What is the probability of no hits?

**c.** What are the mean and variance of the number of hits?

**12** WP Samples of 20 parts from a metal punching process selected every hour. Typically, 1% of the parts require rework. Let $X$ denote the number of parts in the sample of 20 that require rework. A process problem is suspected if $X$ exceeds its mean by more than 3 standard deviations.

**a.** If the percentage of parts that require rework remains at 1%, what is the probability that $X$ exceeds its mean by more than 3 standard deviations?

**b.** If the rework percentage increases to 4%, what is the probability that $X$ exceeds 1?

**c.** If the rework percentage increases to 4%, what is the probability that $X$ exceeds 1 in at least one of the next five hours of samples?

**3.5.13** WP Because all airline passengers do not show up for their reserved seat, an airline sells 125 tickets for a flight that holds only 120 passengers. The probability that a passenger does not show up is 0.10, and the passengers behave independently.

**a.** What is the probability that every passenger who shows up can take the flight?

**b.** What is the probability that the flight departs with empty seats?

**3.5.14** WP This exercise illustrates that poor quality can affect schedules and costs. A manufacturing process has 100 customer orders to fill. Each order requires one component part that is purchased from a supplier. However, typically, 2% of the components are identified as defective, and the components can be assumed to be independent.

**a.** If the manufacturer stocks 100 components, what is the probability that the 100 orders can be filled without reordering components?

**b.** If the manufacturer stocks 102 components, what is the probability that the 100 orders can be filled without reordering components?

**c.** If the manufacturer stocks 105 components, what is the probability that the 100 orders can be filled without reordering components?

**3.5.15** WP Consider the lengths of stay at a hospital's emergency department in Exercise 3.1.19. Assume that five persons independently arrive for service.

**a.** What is the probability that the length of stay of exactly one person is less than or equal to 4 hours?

**b.** What is the probability that exactly two people wait more than 4 hours?

**c.** What is the probability that at least one person waits more than 4 hours?

**3.5.16** The probability that a visitor to a Web site provides contact data for additional information is 0.01. Assume that 1000 visitors to the site behave independently. Determine the following probabilities:

**a.** No visitor provides contact data.

**b.** Exactly 10 visitors provide contact data.

**c.** More than 3 visitors provide contact data.

**3.5.17** Consider the time to recharge the flash in cell-phone cameras as in Example 3.1. Assume that the probability that a camera passes the test is 0.8 and the cameras perform independently. What is the smallest sample size needed so that the probability of at least one camera failing is at least 95%?



## Exercises for Section 3.6

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**3.6.1** WP SS Suppose that the random variable $X$ has a geometric distribution with $p = 0.5$. Determine the following probabilities:

**a.** $P(X = 1)$ **b.** $P(X = 4)$ **c.** $P(X = 8)$
**d.** $P(X \leq 2)$ **e.** $P(X > 2)$

**3.6.2** WP Suppose that $X$ is a negative binomial random variable with $p = 0.2$ and $r = 4$. Determine the following:

**a.** $E(X)$ **b.** $P(X = 20)$
**c.** $P(X = 19)$ **d.** $P(X = 21)$
**e.** The most likely value for $X$

**3.6.3** WP Consider a sequence of independent Bernoulli trials with $p = 0.2$.

**a.** What is the expected number of trials to obtain the first success?

**b.** After the eighth success occurs, what is the expected number of trials to obtain the ninth success?

**3.6.4** WP In a clinical study, volunteers are tested for a gene that has been found to increase the risk for a disease. The probability that a person carries the gene is 0.1. People are assumed to be independent with respect to the gene.

**a.** What is the probability that exactly four people need to be tested to detect two with the gene?

**b.** What is the probability that four or more people need to be tested to detect two with the gene?

**c.** What is the expected number of people to test to detect two with the gene?

**3.6.5** WP Assume that each of your calls to a popular radio station has a probability of 0.02 of connecting, that is, of not obtaining a busy signal. Assume that your calls are independent.

**a.** What is the probability that your first call that connects is your 10th call?

**b.** What is the probability that it requires more than five calls for you to connect?

**c.** What is the mean number of calls needed to connect?

**3.6.6** WP A player of a video game is confronted with a series of opponents and has an 80% probability of defeating each one. Success with any opponent is independent of previous encounters. Until defeated, the player continues to contest opponents.

**a.** What is the probability mass function of the number of opponents contested in a game?

**b.** What is the probability that a player defeats at least two opponents in a game?

**c.** What is the expected number of opponents contested in a game?

**d.** What is the probability that a player contests four or more opponents in a game?

**e.** What is the expected number of game plays until a player contests four or more opponents?

**3.6.7** WP Heart failure is due to either natural occurrences (87%) or outside factors (13%). Outside factors are related to induced substances or foreign objects. Natural occurrences are caused by arterial blockage, disease, and infection. Assume that causes of heart failure for the individuals are independent.

**a.** What is the probability that the first patient with heart failure who enters the emergency room has the condition due to outside factors?

**b.** What is the probability that the third patient with heart failure who enters the emergency room is the first one due to outside factors?

**c.** What is the mean number of heart failure patients with the condition due to natural causes who enter the emergency room before the first patient with heart failure from outside factors?

**3.6.8** Assume that 20 parts are checked each hour and that $X$ denotes the number of parts in the sample of 20 that require rework. Parts are assumed to be independent with respect to rework.

**a.** If the percentage of parts that require rework remains at 1%, what is the probability that hour 10 is the first sample at which $X$ exceeds 1?

**b.** If the rework percentage increases to 4%, what is the probability that hour 10 is the first sample at which $X$ exceeds 1?

**c.** If the rework percentage increases to 4%, what is the expected number of hours until $X$ exceeds 1?

**3.6.9** SS VS A trading company uses eight computers to trade on the New York Stock Exchange (NYSE). The probability of a computer failing in a day is 0.005, and the computers fail independently. Computers are repaired in the evening, and each day is an independent trial.

**a.** What is the probability that all eight computers fail in a day?

**b.** What is the mean number of days until a specific computer fails?

**c.** What is the mean number of days until all eight computers fail on the same day?

**3.6.10** WP In the process of meiosis, a single parent diploid cell goes through eight different phases. However, only 60% of the processes pass the first six phases and only 40% pass all eight. Assume that the results from each phase are independent.

**a.** If the probability of a successful pass of each one of the first six phases is constant, what is the probability of a successful pass of a single one of these phases?

**b.** If the probability of a successful pass of each one of the last two phases is constant, what is the probability of a successful pass of a single one of these phases?

**6.11** A Web site randomly selects among 10 products to discount each day. The color printer of interest to you is discounted day.

**a.** What is the expected number of days until this product is again discounted?

**b.** What is the probability that this product is first discounted again exactly 10 days from now?

**c.** If the product is not discounted for the next five days, what is the probability that it is first discounted again 15 days from now?

**d.** What is the probability that this product is first discounted again within three or fewer days?

**6.12** Consider the visits that result in leave without being seen (LWBS) at an emergency department in Example 2.6. Assume that people independently arrive for service at hospital l.

**a.** What is the probability that the fifth visit is the first one to LWBS?

**b.** What is the probability that either the fifth or sixth visit is the first one to LWBS?

**c.** What is the probability that the first visit to LWBS is among the first four visits?

**d.** What is the expected number of visits until the third LWBS occurs?

**6.13** Consider the time to recharge the flash in cell-phone meras as in Example 3.1. Assume that the probability that a camera passes the test is 0.8 and the cameras perform independently. Determine the following:

**a.** Probability that the second failure occurs on the 10th camera tested.

**b.** Probability that the second failure occurs in tests of four or fewer cameras.

**c.** Expected number of cameras tested to obtain the third failure.

**3.6.14** Customers visit a Web site, and the probability of an order if a customer views five or fewer pages is 0.01. However, if a customer views more than five pages, the probability of an order is 0.1. The probability a customer views five or more pages is 0.25. The customers behave independently.

**a.** Is the number of customers who visit the site until an order is obtained a geometric random variable? Why or why not?

**b.** What is the probability that the first order is obtained from the 10th customer to visit the site?

**3.6.15** An array of 30 LED bulbs is used in an automotive light. The probability that a bulb is defective is 0.001 and defective bulbs occur independently. Determine the following:

**a.** Probability that an automotive light has two or more defective bulbs.

**b.** Expected number of automotive lights to check to obtain one with two or more defective bulbs.

## Exercises for Section 3.7

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutoring Tutoring problem available in WileyPLUS at structor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**7.1** WP Suppose that $X$ has a hypergeometric distribution th $N = 100$, $n = 4$, and $K = 20$. Determine the following:

**a.** $P(X = 1)$ **b.** $P(X = 6)$

**c.** $P(X = 4)$ **d.** Mean and variance of $X$

**7.2** WP Suppose that $X$ has a hypergeometric distribution th $N = 10$, $n = 3$, and $K = 4$. Sketch the probability mass function of $X$. Determine the cumulative distribution function for $X$.

**7.3** WP SS A research study uses 800 men under the age of Suppose that 30% carry a marker on the male chromosome t indicates an increased risk for high blood pressure.

**a.** If 10 men are selected randomly and tested for the marker, what is the probability that exactly 1 man has the marker?

**b.** If 10 men are selected randomly and tested for the marker, what is the probability that more than 1 has the marker?

**7.4** WP Printed circuit cards are placed in a functional test er being populated with semiconductor chips. A lot contains cards, and 20 are selected without replacement for functional ting.

**a.** If 20 cards are defective, what is the probability that at least 1 defective card is in the sample?

**b.** If 5 cards are defective, what is the probability that at least 1 defective card appears in the sample?

**3.7.5** WP The analysis of results from a leaf transmutation experiment (turning a leaf into a petal) is summarized by the type of transformation completed:

| | | Total Textural Transformation | |
|---|---|---|---|
| | | Yes | No |
| Total Color | Yes | 243 | 26 |
| Transformation | No | 13 | 18 |

A naturalist randomly selects three leaves from this set without replacement. Determine the following probabilities.

**a.** Exactly one has undergone both types of transformations.

**b.** At least one has undergone both transformations.

**c.** Exactly one has undergone one but not both transformations.

**d.** At least one has undergone at least one transformation.

**3.7.6** WP A state runs a lottery in which six numbers are randomly selected from 40 without replacement. A player chooses six numbers before the state's sample is selected.

**a.** What is the probability that the six numbers chosen by a player match all six numbers in the state's sample?

**b.** What is the probability that five of the six numbers chosen by a player appear in the state's sample?

**c.** What is the probability that four of the six numbers chosen by a player appear in the state's sample?

**d.** If a player enters one lottery each week, what is the expected number of weeks until a player matches all six numbers in the state's sample?



**3.7.7** WP A slitter assembly contains 48 blades. Five blades are selected at random and evaluated each day for sharpness. If any dull blade is found, the assembly is replaced with a newly sharpened set of blades.

**a.** If 10 of the blades in an assembly are dull, what is the probability that the assembly is replaced the first day it is evaluated?

**b.** If 10 of the blades in an assembly are dull, what is the probability that the assembly is not replaced until the third day of evaluation? [*Hint*: Assume that the daily decisions are independent, and use the geometric distribution.]

**c.** Suppose that on the first day of evaluation, 2 of the blades are dull; on the second day of evaluation, 6 are dull; and on the third day of evaluation, 10 are dull. What is the probability that the assembly is not replaced until the third day of evaluation? [*Hint*: Assume that the daily decisions are independent. However, the probability of replacement changes every day.]

**3.7.8** **a.** For Exercise 3.7.1, calculate $P(X = 1)$ and $P(X = 4)$, assuming that $X$ has a binomial distribution, and compare these results to results derived from the hypergeometric distribution.

**b.** Use the binomial approximation to the hypergeometric distribution to approximate the probabilities in Exercise 3.7.4. What is the finite population correction in this exercise?

**3.7.9** SS Suppose that a healthcare provider selects 20 patients randomly (without replacement) from among 500 to evaluate adherence to a medication schedule. Suppose that 10% of the 500 patients fail to adhere with the schedule. Determine the following:

**a.** Probability that exactly 10% of the patients in the sample fail to adhere.

**b.** Probability that fewer than 10% of the patients in the sample fail to adhere.

**c.** Probability that more than 10% of the patients in the sample fail to adhere.

**d.** Mean and variance of the number of patients in the sample who fail to adhere.

**3.7.10** A utility company might offer electrical rates based on time-of-day consumption to decrease the peak demand in a day. Enough customers need to accept the plan for it to be successful. Suppose that among 50 major customers, 15 would accept the plan. The utility selects 10 major customers randomly (without replacement) to contact and promote the plan.

**a.** What is the probability that exactly two of the selected major customers accept the plan?

**b.** What is the probability that at least one of the selected major customers accepts the plan?

**c.** Instead of 15 customers, what is the minimum number of major customers that would need to accept the plan to meet the following objective? The probability that at least one selected major customer accepts the plan is greater than or equal to 0.95.

**3.7.11** Suppose that lesions are present at 5 sites among 50 in a patient. A biopsy selects 8 sites randomly (without replacement).

**a.** What is the probability that lesions are present in at least one selected site?

**b.** What is the probability that lesions are present in two or more selected sites?

**c.** Instead of eight sites, what is the minimum number of sites that need to be selected to meet the following objective? The probability that at least one site has lesions present is greater than or equal to 0.9.

## Exercises for Section 3.8

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutoring Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**3.8.1** WP Suppose that $X$ has a Poisson distribution. Determine the following probabilities when the mean of $X$ is 4 and repeat for a mean of 0.4:

**a.** $P(X = 0)$ **b.** $P(X \leq 2)$

**c.** $P(X = 4)$ **d.** $P(X = 8)$

**3.8.2** The number of telephone calls that arrive at a phone exchange is often modeled as a Poisson random variable. Assume that on the average there are 10 calls per hour. Determine the following probabilities:

**a.** exactly 5 calls in one hour

**b.** 3 or fewer calls in one hour

**c.** exactly 15 calls in two hours

**d.** exactly 5 calls in 30 minutes

**3.8.3** WP Suppose that the number of customers who enter a store in an hour is a Poisson random variable, and suppose that $P(X = 0) = 0.05$. Determine the mean and variance of $X$.

**3.8.4** Data from *www.centralhudsonlabs.com* determined the mean number of insect fragments in 225-gram chocolate bars was 14.4, but three brands had insect contamination more than twice the average. See the U.S. Food and Drug Administration–Center for Food Safety and Applied Nutrition for Defect Action Levels for food products. Assume that the number of fragments (contaminants) follows a Poisson distribution.

**a.** If you consume a 225-gram bar from a brand at the mean contamination level, what is the probability of no insect contaminants?

**b.** Suppose that you consume a bar that is one-fifth the size tested (45 grams) from a brand at the mean contamination level. What is the probability of no insect contaminants?

**c.** If you consume seven 28.35-gram (one-ounce) bars this week from a brand at the mean contamination level, what is

**a.** What is the mean number of assemblies that need to be checked to obtain five defective assemblies?
**b.** What is the standard deviation of the number of assemblies that need to be checked to obtain five defective assemblies?
**c.** Determine the minimum number of assemblies that need to be checked so that the probability that at least one defective assembly is obtained exceeds 0.95.

**3.S28** WP A manufacturer of a consumer electronics product expects 2% of units to fail during the warranty period. A sample of 500 independent units is tracked for warranty performance.
**a.** What is the probability that none fails during the warranty period?
**b.** What is the expected number of failures during the warranty period?
**c.** What is the probability that more than two units fail during the warranty period?

**3.S29** Determine the probability mass function for the random variable with the following cumulative distribution function:

$$F(x) = \begin{cases} 0 & x < 2 \\ 0.2 & 2 \le x < 5.7 \\ 0.5 & 5.7 \le x < 6.5 \\ 0.8 & 6.5 \le x < 8.5 \\ 1 & 8.5 \le x \end{cases}$$

**3.S30** WP VS The random variable $X$ has the following probability distribution:

| $x$ | 2 | 3 | 5 | 8 |
|---|---|---|---|---|
| Probability | 0.2 | 0.4 | 0.3 | 0.1 |

Determine the following:
**a.** $P(X \le 3)$ **b.** $P(X > 2.5)$
**c.** $P(2.7 < X < 5.1)$ **d.** $E(X)$ **e.** $V(X)$

**3.S31** SS Assume that the number of errors along a magnetic recording surface is a Poisson random variable with a mean of one error every $10^5$ bits. A sector of data consists of 4096 eight-bit bytes.
**a.** What is the probability of more than one error in a sector?
**b.** What is the mean number of sectors until an error occurs?

**3.S32** WP Each main bearing cap in an engine contains 4 bolts. The bolts are selected at random without replacement from a parts bin that contains 30 bolts from one supplier and 70 bolts from another.
**a.** What is the probability that a main bearing cap contains all bolts from the same supplier?
**b.** What is the probability that exactly 3 bolts are from the same supplier?

**3.S33** From 500 customers, a major appliance manufacturer randomly selects a sample without replacement. The company estimates that 25% of the customers will reply to the survey. If this estimate is correct, what is the probability mass function of the number of customers that will reply?
**a.** Assume that the company samples 5 customers.
**b.** Assume that the company samples 10 customers.

**3.S34** WP SS An installation technician for a specialized communication system is dispatched to a city only when three or more orders have been placed. Suppose that orders follow a Poisson distribution with a mean of 0.25 per week for a city with a population of 100,000, and suppose that your city contains a population of 800,000.
**a.** What is the probability that a technician is required after a one-week period?
**b.** If you are the first one in the city to place an order, what is the probability that you have to wait more than two weeks from the time you place your order until a technician is dispatched?

**3.S35** Saguaro cacti are large cacti indigenous to the southwestern United States and Mexico. Assume that the number of saguaro cacti in a region follows a Poisson distribution with a mean of 280 per square kilometer. Determine the following:
**a.** Mean number of cacti per 10,000 square meters.
**b.** Probability of no cacti in 10,000 square meters.
**c.** Area of a region such that the probability of at least two cacti in the region is 0.9.

**3.S36** WP It is suspected that some of the totes containing chemicals purchased from a supplier exceed the moisture content target. Assume that the totes are independent with respect to moisture content. Determine the proportion of totes from the supplier that must exceed the moisture content target so that the probability is 0.90 that at least 1 tote in a sample of 30 fails the test.

# Chapter 4 Exercises

## Exercises for Section 4.1

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**4.1.1** WP Suppose that $f(x) = e^{-x}$ for $0 < x$. Determine the following:
**a.** $P(1 < X)$ **b.** $P(1 < X < 2.5)$ **c.** $P(X = 3)$
**d.** $P(X < 4)$ **e.** $P(3 \le X)$
**f.** $x$ such that $P(x < X) = 0.10$
**g.** $x$ such that $P(X \le x) = 0.10$



**4.1.2** WP The diameter of a particle of contamination (in micrometers) is modeled with the probability density function $f(x) = 2/x^3$ for $x > 1$. Determine the following:
**a.** $P(X < 2)$ **b.** $P(X > 5)$
**c.** $P(4 < X < 8)$ **d.** $P(X < 4 \text{ or } X > 8)$
**e.** $x$ such that $P(X < x) = 0.95$

**4.1.3** SS Suppose that $f(x) = 0.5 \cos x$ for $-\pi/2 < x < \pi/2$. Determine the following:
**a.** $P(X < 0)$ **b.** $P(X < -\pi/4)$
**c.** $P(-\pi/4 < X < \pi/4)$ **d.** $P(X > -\pi/4)$
**e.** $x$ such that $P(X < x) = 0.95$

**4.1.4** WP Suppose that $f(x) = x/8$ for $3 < x < 5$. Determine the following probabilities:
**a.** $P(X < 4)$ **b.** $P(X > 3.5)$
**c.** $P(4 < X < 5)$ **d.** $P(X < 4.5)$
**e.** $P(X < 3.5 \text{ or } X > 4.5)$

**4.1.5** WP Suppose that $f(x) = 1.5x^2$ for $-1 < x < 1$. Determine the following:
**a.** $P(0 < X)$ **b.** $P(0.5 < X)$
**c.** $P(-0.5 \le X \le 0.5)$ **d.** $P(X < -2)$
**e.** $P(X < 0 \text{ or } X > -0.5)$
**f.** $x$ such that $P(x < X) = 0.05$.

**4.1.6** WP The probability density function of the net weight in pounds of a packaged chemical herbicide is $f(x) = 2.0$ for $49.75 < x < 50.25$ pounds.
**a.** Determine the probability that a package weighs more than 50 pounds.
**b.** How much chemical is contained in 90% of all packages?

**4.1.7** WP SS The probability density function of the length of a metal rod is $f(x) = 2$ for $2.3 < x < 2.8$ meters.
**a.** If the specifications for this process are from 2.25 to 2.75 meters, what proportion of rods fail to meet the specifications?
**b.** Assume that the probability density function is $f(x) = 2$ for an interval of length 0.5 meters. Over what value should the density be centered to achieve the greatest proportion of rods within specifications?

**4.1.8** An article in *Electric Power Systems Research* ["Modeling Real-Time Balancing Power Demands in Wind Power Systems Using Stochastic Differential Equations" (2010, Vol. 80(8), pp. 966–974)] considered a new probabilistic model to balance power demand with large amounts of wind power. In this model, the power loss from shutdowns is assumed to have a triangular distribution with probability density function

$$f(x) = \begin{cases} -5.56 \times 10^{-4} + 5.56 \times 10^{-6} x, & x \in [100,\ 500] \\ 4.44 \times 10^{-3} - 4.44 \times 10^{-6} x, & x \in [500, 1000] \\ 0, & \text{otherwise} \end{cases}$$

Determine the following:
**a.** $P(X < 90)$ **b.** $P(100 < X \le 200)$
**c.** $P(X > 800)$ **d.** Value exceeded with probability 0.1.

**4.1.9** The waiting time for service at a hospital emergency department (in hours) follows a distribution with probability density function $f(x) = 0.5 \exp(-0.5x)$ for $0 < x$. Determine the following:
**a.** $P(X < 0.5)$ **b.** $P(X > 2)$
**c.** Value $x$ (in hours) exceeded with probability 0.05.

**4.1.10** The distribution of $X$ is approximated with a triangular probability density function $f(x) = 0.0025x - 0.075$ for $30 < x < 50$ and $f(x) = -0.0025x + 0.175$ for $50 < x < 70$. Determine the following:
**a.** $P(X \le 40)$ **b.** $P(40 < X \le 60)$
**c.** Value $x$ exceeded with probability 0.99.

## Exercises for Section 4.2

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**4.2.1** WP Suppose that the cumulative distribution function of the random variable $X$ is

$$F(x) = \begin{cases} 0 & x < 0 \\ 0.25x & 0 \le x < 5 \\ 1 & 5 \le x \end{cases}$$

Determine the following:
**a.** $P(X < 2.8)$ **b.** $P(X > 1.5)$
**c.** $P(X < -2)$ **d.** $P(X > 6)$

**4.2.2** WP Determine the cumulative distribution function for the distribution in Exercise 4.1.1.

**4.2.3** Determine the cumulative distribution function for the distribution in Exercise 4.1.3.

**4.2.4** Determine the cumulative distribution function for the distribution in Exercise 4.1.2.

**4.2.5** SS Determine the cumulative distribution function for the distribution in Exercise 4.1.4.

**4.2.6** WP The probability density function of the time you arrive at a terminal (in minutes after 8:00 A.M.) is $f(x) = 0.1 \exp(-0.1x)$ for $0 < x$. Determine the probability that
**a.** You arrive by 9:00 A.M.
**b.** You arrive between 8:15 A.M. and 8:30 A.M.
**c.** You arrive before 8:40 A.M. on two or more days of five days. Assume that your arrival times on different days are independent.
**d.** Determine the cumulative distribution function and use the cumulative distribution function to determine the probability that you arrive between 8:15 A.M. and 8:30 A.M.

**4.2.7** WP Determine the cumulative distribution function for the distribution in Exercise 4.1.7. Use the cumulative distribution function to determine the probability that a length exceeds 2.7 meters.

*etermine the probability density function for each of the follow-g cumulative distribution functions.*

**2.8** WP $F(x) = 1 - e^{-2x} \quad x > 0$

**2.9** WP SS

$$F(x) = \begin{cases} 0 & x < 0 \\ 0.2x & 0 \le x < 4 \\ 0.04x + 0.64 & 4 \le x < 9 \\ 1 & 9 \le x \end{cases}$$

**2.10** Determine the cumulative distribution function for the ndom variable in Exercise 4.1.10. Use the cumulative distribution function to determine the probability that the random variable is less than 55.

**4.2.11** Determine the cumulative distribution function for the random variable in Exercise 4.1.9. Use the cumulative distribution function to determine the probability that the waiting time is less than 1 hour.

**4.2.12** Determine the cumulative distribution function for the random variable in Exercise 4.1.8. Use the cumulative distribution function to determine the probability that $400 < X < 500$.

## Exercises for Section 4.3

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at structor's discretion
S Student solution available in WileyPLUS.
S Student video solution available in WileyPLUS.

**3.1** WP SS VS Suppose that $f(x) = 1.5x^2$ for $-1 < x < 1$. etermine the mean and variance of $X$.

**3.2** WP Suppose that $f(x) = 0.125x$ for $0 < x < 4$. Determine e mean and variance of $X$.

**3.3** Determine the mean and variance of the random variable in xercise 4.1.1.

**3.4** Determine the mean and variance of the random variable in xercise 4.1.8.

**3.5** Determine the mean and variance of the random variable in xercise 4.1.10.

**4.3.6** WP Suppose that the probability density function of the length of computer cables is $f(x) = 0.1$ from 1200 to 1210 millimeters.

**a.** Determine the mean and standard deviation of the cable length.

**b.** If the length specifications are $1195 < x < 1205$ millimeters, what proportion of cables is within specifications?

**4.3.7** WP SS The thickness of a conductive coating in micrometers has a density function of $600x^{-2}$ for 100 μm $< x <$ 120 μm.

**a.** Determine the mean and variance of the coating thickness.

**b.** If the coating costs \$0.50 per micrometer of thickness on each part, what is the average cost of the coating per part?

## Exercises for Section 4.4

P Problem available in WileyPLUS at instructor's discretion.
P GO Tutorial Tutoring problem available in WileyPLUS at structor's discretion
S Student solution available in WileyPLUS.
S Student video solution available in WileyPLUS.

**4.1** WP Suppose $X$ has a continuous uniform distribution over e interval $[-1, 1]$. Determine the following:

**a.** Mean, variance, and standard deviation of $X$

**b.** Value for $x$ such that $P(-x < X < x) = 0.90$

**c.** Cumulative distribution function

**4.2** WP VS The thickness of a flange on an aircraft compo-nt is uniformly distributed between 0.95 and 1.05 millimeters. etermine the following:

**a.** Cumulative distribution function of flange thickness

**b.** Proportion of flanges that exceeds 1.02 millimeters

**c.** Thickness exceeded by 90% of the flanges

**d.** Mean and variance of flange thickness

**4.3** WP The thickness of photoresist applied to wafers in semi-nductor manufacturing at a particular location on the wafer is uniformly distributed between 0.2050 and 0.2150 micrometers. Determine the following:

**a.** Cumulative distribution function of photoresist thickness

**b.** Proportion of wafers that exceeds 0.2125 micrometers in photoresist thickness

**c.** Thickness exceeded by 10% of the wafers

**d.** Mean and variance of photoresist thickness

**4.4.4** WP An adult can lose or gain two pounds of water in the course of a day. Assume that the changes in water weight are uniformly distributed between minus two and plus two pounds in a day. What is the standard deviation of a person's weight over a day?

**4.4.5** WP SS A show is scheduled to start at 9:00 A.M., 9:30 A.M., and 10:00 A.M. Once the show starts, the gate will be closed. A visitor will arrive at the gate at a time uniformly distributed between 8:30 A.M. and 10:00 A.M. Determine the following:

**a.** Cumulative distribution function of the time (in minutes) between arrival and 8:30 A.M.

**b.** Mean and variance of the distribution in the previous part

**c.** Probability that a visitor waits less than 10 minutes for a show

**d.** Probability that a visitor waits more than 20 minutes for a show

**4.4.6** The volume of a shampoo filled into a container is uniformly distributed between 374 and 380 milliliters.

**a.** What are the mean and standard deviation of the volume of shampoo?

**b.** What is the probability that the container is filled with less than the advertised target of 375 milliliters?

**c.** What is the volume of shampoo that is exceeded by 95% of the containers?

**d.** Every milliliter of shampoo costs the producer \$0.002. Any shampoo more than 375 milliliters in the container is an extra cost to the producer. What is the mean extra cost?

**4.4.7** An e-mail message will arrive at a time uniformly distributed between 9:00 A.M. and 11:00 A.M. You check e-mail at 9:15 A.M. and every 30 minutes afterward.



**a.** What is the standard deviation of arrival time (in minutes)?

**b.** What is the probability that the message arrives less than 10 minutes before you view it?

**c.** What is the probability that the message arrives more than 15 minutes before you view it?

**4.4.8** An electron emitter produces electron beams with changing kinetic energy that is uniformly distributed between 3 and 7 joules. Suppose that it is possible to adjust the upper limit of the kinetic energy (currently set to 7 joules).

**a.** What is the mean kinetic energy?

**b.** What is the variance of the kinetic energy?

**c.** What is the probability that an electron beam has a kinetic energy of exactly 3.2 joules?

**d.** What should be the upper limit so that the mean kinetic energy increases to 8 joules?

**e.** What should be the upper limit so that the variance of kinetic energy decreases to 0.75 joules?

## Exercises for Section 4.5

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**4.5.1** WP Use Appendix Table III to determine the following probabilities for the standard normal random variable $Z$:

**a.** $P(Z < 1.32)$ **b.** $P(Z < 3.0)$

**c.** $P(Z > 1.45)$ **d.** $P(Z > -2.15)$

**e.** $P(-2.34 < Z < 1.76)$

**4.5.2** WP Assume that $Z$ has a standard normal distribution. Use Appendix Table III to determine the value for $z$ that solves each of the following:

**a.** $P(-z < Z < z) = 0.95$ **b.** $P(-z < Z < z) = 0.99$

**c.** $P(-z < Z < z) = 0.68$ **d.** $P(-z < Z < z) = 0.9973$

**4.5.3** WP SS Assume that $X$ is normally distributed with a mean of 10 and a standard deviation of 2. Determine the following:

**a.** $P(Z < 13)$ **b.** $P(Z > 9)$

**c.** $P(6 < X < 14)$ **d.** $P(2 < X < 4)$

**e.** $P(-2 < X < 8)$

**4.5.4** WP Assume that $X$ is normally distributed with a mean of 10 and a standard deviation of 2. Determine the value for $x$ that solves each of the following:

**a.** $P(X > x) = 0.5$ **b.** $P(X > x) = 0.95$

**c.** $P(x < X < 10) = 0$ **d.** $P(-x < X - 10 < x) = 0.95$

**e.** $P(-x < X - 10 < x) = 0.99$

**4.5.5** An article in *Knee Surgery Sports Traumatology Arthroscopy* ["Effect of Provider Volume on Resource Utilization for Surgical Procedures" (2005, Vol. 13, pp. 273–279)] showed a mean time of 129 minutes and a standard deviation of 14 minutes for anterior cruciate ligament (ACL) reconstruction surgery at high-volume hospitals (with more than 300 such surgeries per year).

**a.** What is the probability that your ACL surgery at a high-volume hospital requires a time more than 2 standard deviations above the mean?

**b.** What is the probability that your ACL surgery at a high-volume hospital is completed in less than 100 minutes?

**c.** The probability of a completed ACL surgery at a high-volume hospital is equal to 95% at what time?

**d.** If your surgery requires 199 minutes, what do you conclude about the volume of such surgeries at your hospital? Explain.

**4.5.6** WP The time until recharge for a battery in a laptop computer under common conditions is normally distributed with a mean of 260 minutes and a standard deviation of 50 minutes.

**a.** What is the probability that a battery lasts more than four hours?

**b.** What are the quartiles (the 25% and 75% values) of battery life?

**c.** What value of life in minutes is exceeded with 95% probability?

**4.5.7** WP SS VS A driver's reaction time to visual stimulus is normally distributed with a mean of 0.4 seconds and a standard deviation of 0.05 seconds.

**a.** What is the probability that a reaction requires more than 0.5 seconds?

**b.** What is the probability that a reaction requires between 0.4 and 0.5 seconds?

**c.** What reaction time is exceeded 90% of the time?

**4.5.8** Cholesterol is a fatty substance that is an important part of the outer lining (membrane) of cells in the body of animals. Its normal range for an adult is 120–240 mg/dl. The Food and Nutrition Institute of the Philippines found that the total cholesterol level for Filipino adults has a mean of 159.2 mg/dl and

84.1% of adults have a cholesterol level less than 200 mg/dl (*www.fnri.dost.gov.ph/*). Suppose that the total cholesterol level is normally distributed.

**a.** Determine the standard deviation of this distribution.

**b.** What are the quartiles (the 25% and 75% percentiles) of this distribution?

**c.** What is the value of the cholesterol level that exceeds 90% of the population?

**d.** An adult is at moderate risk if cholesterol level is more than 1 but less than 2 standard deviations above the mean. What percentage of the population is at moderate risk according to this criterion?

**e.** An adult whose cholesterol level is more than 2 standard deviations above the mean is thought to be at high risk. What percentage of the population is at high risk?

**f.** An adult whose cholesterol level is less than 1 standard deviation below the mean is thought to be at low risk. What percentage of the population is at low risk?

**4.5.9** In an accelerator center, an experiment needs a 1.41-cm-thick aluminum cylinder (*http://puhep1.princeton.edu/mumu/target/Solenoid_Coil.pdf*). Suppose that the thickness of a cylinder has a normal distribution with a mean of 1.41 cm and a standard deviation of 0.01 cm.

**a.** What is the probability that a thickness is greater than 1.42 cm?

**b.** What thickness is exceeded by 95% of the samples?

**c.** If the specifications require that the thickness is between 1.39 cm and 1.43 cm, what proportion of the samples meets specifications?

**4.5.10** In 2002, the average height of a woman aged 20–74 years was 64 inches, with an increase of approximately 1 inch from 1960 (*http://usgovinfo.about.com/od/healthcare*). Suppose the height of a woman is normally distributed with a standard deviation of 2 inches.

**a.** What is the probability that a randomly selected woman in this population is between 58 inches and 70 inches?

**b.** What are the quartiles of this distribution?

**c.** Determine the height that is symmetric about the mean that includes 90% of this population.

**d.** What is the probability that five women selected at random from this population all exceed 68 inches?

**4.5.11** The life of a semiconductor laser at a constant power is normally distributed with a mean of 7000 hours and a standard deviation of 600 hours.

**a.** What is the probability that a laser fails before 5000 hours?

**b.** What is the life in hours that 95% of the lasers exceed?

**c.** If three lasers are used in a product and they are assumed to fail independently, what is the probability that all three are still operating after 7000 hours?

**4.5.12** The demand for water use in Phoenix in 2003 hit a high of about 442 million gallons per day on June 27 (*http://phoenix.gov/WATER/wtrfacts.html*). Water use in the summer is normally distributed with a mean of 310 million gallons per day and a standard deviation of 45 million gallons per day. City reservoirs have a combined storage capacity of nearly 350 million gallons.

**a.** What is the probability that a day requires more water than is stored in city reservoirs?

**b.** What reservoir capacity is needed so that the probability that it is exceeded is 1%?

**c.** What amount of water use is exceeded with 95% probability?

**d.** Water is provided to approximately 1.4 million people. What is the mean daily consumption per person at which the probability that the demand exceeds the current reservoir capacity is 1%? Assume that the standard deviation of demand remains the same.

**4.5.13** SS The weight of a running shoe is normally distributed with a mean of 12 ounces and a standard deviation of 0.5 ounce.

**a.** What is the probability that a shoe weighs more than 13 ounces?

**b.** What must the standard deviation of weight be in order for the company to state that 99.9% of its shoes weighs less than 13 ounces?

**c.** If the standard deviation remains at 0.5 ounce, what must the mean weight be for the company to state that 99.9% of its shoes weighs less than 13 ounces?

**4.5.14** The diameter of the dot produced by a printer is normally distributed with a mean diameter of 0.002 inch and a standard deviation of 0.0004 inch.

**a.** What is the probability that the diameter of a dot exceeds 0.0026?

**b.** What is the probability that a diameter is between 0.0014 and 0.0026?

**c.** What standard deviation of diameters is needed so that the probability in part (b) is 0.995?

**4.5.15** Assume that a random variable is normally distributed with a mean of 24 and a standard deviation of 2. Consider an interval of length one unit that starts at the value $a$ so that the interval is $[a, a + 1]$. For what value of $a$ is the probability of the interval greatest? Does the standard deviation affect that choice of interval?

**4.5.16** The length of stay at a specific emergency department in Phoenix, Arizona, in 2009 had a mean of 4.6 hours with a standard deviation of 2.9. Assume that the length of stay is normally distributed.

**a.** What is the probability of a length of stay greater than 10 hours?

**b.** What length of stay is exceeded by 25% of the visits?

**c.** From the normally distributed model, what is the probability of a length of stay less than 0 hours? Comment on the normally distributed assumption in this example.

**4.5.17** An article in *Atmospheric Chemistry and Physics* ["Relationship Between Particulate Matter and Childhood Asthma—Basis of a Future Warning System for Central Phoenix" (2012, Vol. 12, pp. 2479–2490)] reported the use of PM10 (particulate matter <10 μm diameter) air quality data measured hourly from sensors in Phoenix, Arizona. The 24-hour (daily) mean PM10 for a centrally located sensor was 50.9 $\mu g/m^3$ with a standard deviation of 25.0. Assume that the daily mean of PM10 is normally distributed.

**a.** What is the probability of a daily mean of PM10 greater than 100 $\mu g/m^3$?

**b.** What is the probability of a daily mean of PM10 less than 25 $\mu g/m^3$?



**c.** What daily mean of PM10 value is exceeded with probability 5%?

**4.5.18** An article in *Microelectronics Reliability* ["Advanced Electronic Prognostics through System Telemetry and Pattern Recognition Methods" (2007, Vol. 47(12), pp. 1865–1873)] presented an example of electronic prognosis. The objective was to detect faults to decrease the system downtime and the number of unplanned repairs in high-reliability systems. Previous measurements of the power supply indicated that the signal is normally distributed with a mean of 1.5 V and a standard deviation of 0.02 V.

**a.** Suppose that lower and upper limits of the predetermined specifications are 1.45 V and 1.55 V, respectively. What is the probability that a signal is within these specifications?

**b.** What is the signal value that is exceeded with 95% probability?

**c.** What is the probability that a signal value exceeds the mean by two or more standard deviations?

**4.5.19** A signal in a communication channel is detected when the voltage is higher than 1.5 volts in absolute value. Assume that the voltage is normally distributed with a mean of 0. What is the standard deviation of voltage such that the probability of a false signal is 0.005?

**4.5.20** An article in the *Journal of Cardiovascular Magnetic Resonance* ["Right Ventricular Ejection Fraction Is Better Reflected by Transverse Rather Than Longitudinal Wall Motion in Pulmonary Hypertension" (2010, Vol. 12(35)] discussed a study of the regional right-ventricle transverse wall motion in patients with pulmonary hypertension (PH). The right-ventricle ejection fraction (EF) was approximately normally distributed with a mean and a standard deviation of 36 and 12, respectively, for PH subjects, and with mean and standard deviation of 56 and 8, respectively, for control subjects.

**a.** What is the EF for PH subjects exceeded with 5% probability?

**b.** What is the probability that the EF of a control subject is less than the value in part (a)?

**c.** Comment on how well the control and PH subjects can be distinguished by EF measurements.

## Exercises for Section 4.6

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**4.6.1** WP SS Suppose that $X$ is a binomial random variable with $n = 200$ and $p = 0.4$. Approximate the following probabilities:

**a.** $P(X \leq 70)$ **b.** $P(70 < X < 90)$ **c.** $P(X = 80)$

**4.6.2** WP Suppose that $X$ is a Poisson random variable with $\lambda = 6$.

**a.** Compute the exact probability that $X$ is less than four.

**b.** Approximate the probability that $X$ is less than four and compare to the result in part (a).

**c.** Approximate the probability that $8 < X < 12$.

**4.6.3** WP There were 49.7 million people with some type of long-lasting condition or disability living in the United States in 2000. This represented 19.3 percent of the majority of civilians aged five and over (*http://factfinder.census.gov*). A sample of 1000 persons is selected at random.

**a.** Approximate the probability that more than 200 persons in the sample have a disability.

**b.** Approximate the probability that between 180 and 300 people in the sample have a disability.

**4.6.4** WP The manufacturing of semiconductor chips produces 2% defective chips. Assume that the chips are independent and that a lot contains 1000 chips. Approximate the following probabilities:

**a.** More than 25 chips are defective.

**b.** Between 20 and 30 chips are defective.

**4.6.5** VS Hits to a high-volume Web site are assumed to follow a Poisson distribution with a mean of 10,000 per day. Approximate each of the following:

**a.** Probability of more than 20,000 hits in a day.

**b.** Probability of less than 9900 hits in a day.

**c.** Value such that the probability that the number of hits in a day exceeds the value is 0.01.

**d.** Expected number of days in a year (365 days) that exceed 10,200 hits.

**e.** Probability that over a year (365 days), each of the more than 15 days has more than 10,200 hits.

**4.6.6** Phoenix water is provided to approximately 1.4 million people who are served through more than 362,000 accounts (*http://phoenix.gov/WATER/wtrfacts.html*). All accounts are metered and billed monthly. The probability that an account has an error in a month is 0.001, and accounts can be assumed to be independent.

**a.** What are the mean and standard deviation of the number of account errors each month?

**b.** Approximate the probability of fewer than 350 errors in a month.

**c.** Approximate a value so that the probability that the number of errors exceeds this value is 0.05.

**d.** Approximate the probability of more than 400 errors per month in the next two months. Assume that results between months are independent.

**4.6.7** SS An article in *Atmospheric Chemistry and Physics* ["Relationship Between Particulate Matter and Childhood Asthma—Basis of a Future Warning System for Central Phoenix" (2012, Vol. 12, pp. 2479–2490)] linked air quality to childhood asthma incidents. The study region in central Phoenix, Arizona,

recorded 10,500 asthma incidents in children in a 21-month period. Assume that the number of asthma incidents follows a Poisson distribution.

**a.** Approximate the probability of more than 550 asthma incidents in a month.

**b.** Approximate the probability of 450 to 550 asthma incidents in a month.

**c.** Approximate the number of asthma incidents exceeded with probability 5%.

**d.** If the number of asthma incidents was greater during the winter than the summer, what would this imply about the Poisson distribution assumption?

**4.6.8** An acticle in *Biometrics* ["Integrative Analysis of Transcriptomic and Proteomic Data of *Desulfovibrio Vulgaris:* A Nonlinear Model to Predict Abundance of Undetected Proteins" (2009, Vol. 25(15), pp. 1905–1914)] reported that protein abundance from an operon (a set of biologically related genes) was less dispersed than from randomly selected genes. In the research, 1000 sets of genes were randomly constructed, and of these sets, 75% were more disperse than a specific opteron. If the probability that a random set is more disperse than this opteron is truly 0.5, approximate the probability that 750 or more random sets exceed the opteron. From this result, what do you conclude about the dispersion in the opteron versus random genes?

**4.6.9** A set of 200 independent patients take antiacid medication at the start of symptoms, and 80 experience moderate to substantial relief within 90 minutes. Historically, 30% of patients experience relief within 90 minutes with no medication. If the medication has no effect, approximate the probability that 80 or more patients experience relief of symptoms. What can you conclude about the effectiveness of this medication?

**4.6.10** Cabs pass your workplace according to a Poisson process with a mean of five cabs per hour.

**a.** Determine the mean and standard deviation of the number of cabs per 10-hour day.

**b.** Approximate the probability that more than 65 cabs pass within a 10-hour day.

**c.** Approximate the probability that between 50 and 65 cabs pass in a 10-hour day.

**d.** Determine the mean hourly rate so that the probability is approximately 0.95 that 100 or more cabs pass in a 10-hour day.

**4.6.11** The number of (large) inclusions in cast iron follows a Poisson distribution with a mean of 2.5 per cubic millimeter. Approximate the following probabilities:

**a.** Determine the mean and standard deviation of the number of inclusions in a cubic centimeter (cc).

**b.** Approximate the probability that fewer than 2600 inclusions occur in a cc.

**c.** Approximate the probability that more than 2400 inclusions occur in a cc.

**d.** Determine the mean number of inclusions per cubic millimeter such that the probability is approximately 0.9 that 500 or fewer inclusions occur in a cc.

## Exercises for Section 4.7

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**4.7.1** WP SS Suppose that $X$ has an exponential distribution with mean equal to 10. Determine the following:

**a.** $P(X > 10)$ **b.** $P(X > 20)$ **c.** $P(X < 30)$

**d.** Find the value of $x$ such that $P(X < x) = 0.95$.

**4.7.2** WP Suppose that $X$ has an exponential distribution with a mean of 10. Determine the following:

**a.** $P(X < 5)$ **b.** $P(X < 15 \mid X > 10)$

**c.** Compare the results in parts (a) and (b) and comment on the role of the lack of memory property.

**4.7.3** WP Suppose that the log-ons to a computer network follow a Poisson process with an average of three counts per minute.

**a.** What is the mean time between counts?

**b.** What is the standard deviation of the time between counts?

**c.** Determine $x$ such that the probability that at least one count occurs before time $x$ minutes is 0.95.

**4.7.4** WP The life of automobile voltage regulators has an exponential distribution with a mean life of 6 years. You purchase a 6-year-old automobile with a working voltage regulator and plan to own it for 6 years.

**a.** What is the probability that the voltage regulator fails during your ownership?

**b.** If your regulator fails after you own the automobile 3 years and it is replaced, what is the mean time until the next failure?

**4.7.5** WP SS VS The time between calls to a plumbing supply business is exponentially distributed with a mean time between calls of 15 minutes.

**a.** What is the probability that there are no calls within a 30-minute interval?

**b.** What is the probability that at least one call arrives within a 10-minute interval?

**c.** What is the probability that the first call arrives within 5 to 10 minutes after opening?

**d.** Determine the length of an interval of time such that the probability of at least one call in the interval is 0.90.

**4.7.6** The number of stork sightings on a route in South Carolina follows a Poisson process with a mean of 2.3 per year.

**a.** What is the mean time between sightings?

**b.** What is the probability that there are no sightings within three months (0.25 years)?

**c.** What is the probability that the time until the first sighting exceeds six months?

**d.** What is the probability of no sighting within 3 years?

**4.7.7** WP VS The time between arrivals of taxis at a busy intersection is exponentially distributed with a mean of 10 minutes.



**a.** What is the probability that you wait longer than 1 hour for a taxi?

**b.** Suppose that you have already been waiting for 1 hour for a taxi. What is the probability that one arrives within the next 10 minutes?

**c.** Determine $x$ such that the probability that you wait more than $x$ minutes is 0.10.

**d.** Determine $x$ such that the probability that you wait less than $x$ minutes is 0.90.

**e.** Determine $x$ such that the probability that you wait less than $x$ minutes is 0.50.

**4.7.8** WP The distance between major cracks in a highway follows an exponential distribution with a mean of 5 miles.

**a.** What is the probability that there are no major cracks in a 10-mile stretch of the highway?

**b.** What is the probability that there are two major cracks in a 10-mile stretch of the highway?

**c.** What is the standard deviation of the distance between major cracks?

**d.** What is the probability that the first major crack occurs between 12 and 15 miles of the start of inspection?

**e.** What is the probability that there are no major cracks in two separate 5-mile stretches of the highway?

**f.** Given that there are no cracks in the first 5 miles inspected, what is the probability that there are no major cracks in the next 10 miles inspected?

**4.7.9** According to results from the analysis of chocolate bars in Chapter 3, the mean number of insect fragments was 14.4 in 225 grams. Assume that the number of fragments follows a Poisson distribution.

**a.** What is the mean number of grams of chocolate until a fragment is detected?

**b.** What is the probability that there are no fragments in a 28.35-gram (one-ounce) chocolate bar?

**c.** Suppose you consume seven one-ounce (28.35-gram) bars this week. What is the probability of no insect fragments?

**4.7.10** WP The time between arrivals of small aircraft at a county airport is exponentially distributed with a mean of one hour.

**a.** What is the probability that more than three aircraft arrive within an hour?

**b.** If 30 separate one-hour intervals are chosen, what is the probability that no interval contains more than three arrivals?

**c.** Determine the length of an interval of time (in hours) such that the probability that no arrivals occur during the interval is 0.10.

**d.** What is the probability that the first occurrence in which the time between two arrivals exceeds 1.5 hours is at the third arrival?

**4.7.11** The time between calls to a corporate office is exponentially distributed with a mean of 10 minutes.

**a.** What is the probability that there are more than three calls in one-half hour?

**b.** What is the probability that there are no calls within one-half hour?

**c.** Determine $x$ such that the probability that there are no calls within $x$ hours is 0.01.

**d.** What is the probability that there are no calls within a two-hour interval?

**e.** If four nonoverlapping one-half-hour intervals are selected, what is the probability that none of these intervals contains any call?

**f.** What is the probability that the first occurrence in which the time between two calls exceeds 10 minutes is at the fifth call?

**4.7.12** Derive the formula for the mean and variance of an exponential random variable.

**4.7.13** SS If the random variable $X$ has an exponential distribution with mean $\theta$, determine the following:

**a.** $P(X > \theta)$ **b.** $P(X > 2\theta)$ **c.** $P(X > 3\theta)$

**d.** How do the results depend on $\theta$?

**4.7.14** The length of stay at a specific emergency department in a hospital in Phoenix, Arizona, had a mean of 4.6 hours. Assume that the length of stay is exponentially distributed.

**a.** What is the standard deviation of the length of stay?

**b.** What is the probability of a length of stay of more than 10 hours?

**c.** What length of stay is exceeded by 25% of the visits?

**4.7.15** An article in the *Journal of the National Cancer Institute* ["Breast Cancer Screening Policies in Developing Countries: A Cost-Effectiveness Analysis for India" (2008, Vol. 100(18), pp. 1290–1300)] presented a screening analysis model of breast cancer based on data from India. In this analysis, the time that a breast cancer case stays in a preclinical state is modeled to be exponentially distributed with a mean depending on the state. For example, the time that a cancer case stays in the state of T1C (tumor size of 11–20 mm) is exponentially distributed with a mean of 1.48 years.

**a.** What is the probability that a breast cancer case in India stays in the state of T1C for more than 2.0 years?

**b.** What is the proportion of breast cancer cases in India that spend at least 1.0 year in the state of T1C?

**c.** Assume that a person in India is diagnosed to be in the state of T1C. What is the probability that the patient is in the same state six months later?

**4.7.16** An article in *Ad Hoc Networks* ["Underwater Acoustic Sensor Networks: Target Size Detection and Performance Analysis" (2009, Vol. 7(4), pp. 803–808)] discussed an underwater acoustic sensor network to monitor a given area in an ocean. The network does not use cables and does not interfere with shipping activities. The arrival of clusters of signals generated by the same pulse is taken as a Poisson arrival process with a mean of $\lambda$ per unit time. Suppose that for a specific underwater acoustic sensor network, this Poisson process has a rate of 2.5 arrivals per unit time.

**a.** What is the mean time between 2.0 consecutive arrivals?

**b.** What is the probability that there are no arrivals within 0.3 time units?

**c.** What is the probability that the time until the first arrival exceeds 1.0 unit of time?

**d.** Determine the mean arrival rate such that the probability is 0.9 that there are no arrivals in 0.3 time units.

**4.7.17** An article in *Vaccine* ["Modeling the Effects of Influenza Vaccination of Health Care Workers in Hospital Departments" (2009, Vol. 27(44), pp. 6261–6267)] considered the immunization of healthcare workers to reduce the hazard rate of influenza

Determine the following:

**a.** $P(X < 2.5, Y < 3)$ **b.** $P(X < 2.5)$
**c.** $P(Y < 3)$ **d.** $P(X > 1.8, Y > 4.7)$
**e.** $E(X), E(Y), V(X), V(Y)$
**f.** Marginal probability distribution of $X$

**5.1.2** WP Show that the following function satisfies the properties of a joint probability mass function.

| $x$ | $y$ | $f_{XY}(x, y)$ |
|---|---|---|
| −1.0 | −2 | 1/8 |
| −0.5 | −1 | 1/4 |
| 0.5 | 1 | 1/2 |
| 1.0 | 2 | 1/8 |

Determine the following:

**a.** $P(X < 0.5, Y < 1.5)$ **b.** $P(X < 0.5)$
**c.** $P(Y < 1.5)$ **d.** $P(X > 0.25, Y < 4.5)$
**e.** $E(X), E(Y), V(X), V(Y)$
**f.** Marginal probability distribution of $X$

**5.1.3** WP SS VS In the transmission of digital information, the probability that a bit has high, moderate, and low distortion is 0.01, 0.04, and 0.95, respectively. Suppose that three bits are transmitted and that the amount of distortion of each bit is assumed to be independent. Let $X$ and $Y$ denote the number of bits with high and moderate distortion out of the three, respectively. Determine:

**a.** $f_{XY}(x, y)$ **b.** $f_X(x)$ **c.** $E(X)$

**5.1.4** WP A small-business Web site contains 100 pages and 60%, 30%, and 10% of the pages contain low, moderate, and high graphic content, respectively. A sample of four pages is selected randomly without replacement, and $X$ and $Y$ denote the number of pages in the sample with moderate and high graphics output. Determine:

**a.** $f_{XY}(x, y)$ **b.** $f_X(x)$ **c.** $E(X)$

**5.1.5** WP A manufacturing company employs two devices to inspect output for quality control purposes. The first device is able to accurately detect 99.3% of the defective items it receives, whereas the second is able to do so in 99.7% of the cases. Assume that four defective items are produced and sent out for inspection. Let $X$ and $Y$ denote the number of items that will be identified as defective by inspecting devices 1 and 2, respectively. Assume that the devices are independent. Determine:

**a.** $f_{XY}(x, y)$ **b.** $f_X(x)$ **c.** $E(X)$

**5.1.6** WP An article in the *Journal of Database Management* ["Experimental Study of a Self-Tuning Algorithm for DBMS Buffer Pools" (2005, Vol. 16, pp. 1–20)] provided the workload used in the TPC-C OLTP (Transaction Processing Performance Council's Version C On-Line Transaction Processing) benchmark, which simulates a typical order entry application. See the table at the bottom of the page. The frequency of each type of transaction (in the second column) can be used as the percentage of each type of transaction. Let $X$ and $Y$ denote the average number of *selects* and *updates* operations, respectively, required for each type transaction. Determine the following:

**a.** $P(X < 5)$ **b.** $E(X)$

**5.1.7** An engineering statistics class has 40 students; 60% are electrical engineering majors, 10% are industrial engineering majors, and 30% are mechanical engineering majors. A sample of four students is selected randomly without replacement for a project team. Let $X$ and $Y$ denote the number of industrial and mechanical engineering majors in the sample, respectively. Determine the following:

**a.** $f_{XY}(x, y)$ **b.** $f_X(x)$ **c.** $E(X)$

**5.1.8** WP Determine the value of $c$ that makes the function $f(x, y) = c(x + y)$ a joint probability density function over the range $0 < x < 3$ and $x < y < x + 2$. Determine the following:

**a.** $P(X < 1, Y < 2)$ **b.** $P(1 < X < 2)$ **c.** $P(Y > 1)$
**d.** $P(X < 2, Y < 2)$ **e.** $E(X)$ **f.** $V(X)$
**g.** Marginal probability distribution of $X$

**5.1.9** WP SS VS Determine the value of $c$ such that the function $f(x, y) = cxy$ for $0 < x < 3$ and $0 < y < 3$ satisfies the properties of a joint probability density function. Determine the following:

**a.** $P(X < 2, Y < 3)$ **b.** $P(X < 2.5)$
**c.** $P(1 < Y < 2.5)$ **d.** $P(X > 1.8, 1 < Y < 2.5)$
**e.** $E(X)$ **f.** $P(X < 0, Y < 4)$
**g.** Marginal probability distribution of $X$

**5.1.10** VS Determine the value of $c$ that makes the function $f(x, y) = ce^{-2x-3y}$ a joint probability density function over the range $0 < x$ and $x < y$. Determine the following:

**a.** $P(X < 1, Y < 2)$ **b.** $P(1 < X < 2)$
**c.** $P(Y > 3)$ **d.** $P(X < 2, Y < 2)$
**e.** $E(X)$ **f.** $E(Y)$
**g.** Marginal probability distribution of $X$

**Average Frequencies and Operations in TPC-C**

| Transaction | Frequency | Selects | Updates | Inserts | Deletes | Non-unique Selects | Joins |
|---|---|---|---|---|---|---|---|
| New order | 43 | 23.0 | 11 | 12 | 0 | 0 | 0 |
| Payment | 44 | 4.2 | 3 | 1 | 0 | 0.6 | 0 |
| Order status | 4 | 11.4 | 0 | 0 | 0 | 0.6 | 0 |
| Delivery | 5 | 130.0 | 120 | 0 | 10 | 0 | 0 |
| Stock level | 4 | 0 | 0 | 0 | 0 | 0 | 1 |



## Exercises for Section 5.2

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**5.2.1** The conditional probability distribution of $Y$ given $X = x$ is $f_{Y|x}(y) = xe^{-xy}$ for $y > 0$, and the marginal probability distribution of $X$ is a continuous uniform distribution over 0 to 10.

**a.** Graph $f_{Y|x}(y) = xe^{-xy}$ for $y > 0$ for several values of $x$. Determine:
**b.** $P(Y < 2 \mid X = 2)$ **c.** $E(Y \mid X = 2)$
**d.** $E(Y \mid X = x)$ **e.** $f_{XY}(x, y)$
**f.** $f_Y(y)$

**5.2.2** WP Consider the joint distribution in Exercise 5.1.1. Determine the following:

**a.** Conditional probability distribution of $Y$ given that $X = 1.5$
**b.** Conditional probability distribution of $X$ that $Y = 2$
**c.** $E(Y \mid X = 1.5)$
**d.** Are $X$ and $Y$ independent?

**5.2.3** WP Consider the joint distribution in Exercise 5.1.4. Determine the following:

**a.** $f_{Y|3}(y)$ **b.** $E(Y \mid X = 3)$ **c.** $V(Y \mid X = 3)$

**5.2.4** WP A manufacturer of electroluminescent lamps knows that the amount of luminescent ink deposited on one of its products is normally distributed with a mean of 1.2 grams and a standard deviation of 0.03 gram. Any lamp with less than 1.14 grams of luminescent ink fails to meet customers' specifications. A random sample of 25 lamps is collected and the mass of luminescent ink on each is measured.

**a.** What is the probability that at least one lamp fails to meet specifications?
**b.** What is the probability that five or fewer lamps fail to meet specifications?
**c.** What is the probability that all lamps conform to specifications?
**d.** Why is the joint probability distribution of the 25 lamps not needed to answer the previous questions?

**5.2.5** SS The lengths of the minor and major axes are used to summarize dust particles that are approximately elliptical in shape. Let $X$ and $Y$ denote the lengths of the minor and major axes (in micrometers), respectively. Suppose that $f_X(x) = \exp(-x), 0 < x$ and the conditional distribution $f_{Y|x}(y) = \exp[-(y - x)], x < y$. Answer or determine the following:

**a.** Show that $f_{Y|x}(y)$ is a probability density function for any value of $x$
**b.** $P(X < Y)$
**c.** Joint probability density function $f_{XY}(x, y)$
**d.** Conditional probability density function of $X$ given $Y = y$
**e.** $P(Y < 2 \mid X = 1)$ **f.** $E(Y \mid X = 1)$
**g.** $P(X < 1, Y < 1)$ **h.** $P(Y < 2)$
**i.** $c$ such that $P(Y < c) = 0.9$
**j.** Are $X$ and $Y$ independent?

**5.2.6** An article in *Health Economics* ["Estimation of the Transition Matrix of a Discrete-Time Markov Chain" (2002, Vol. 11, pp. 33–42)] considered the changes in CD4 white blood cell counts from one month to the next. The CD4 count is an important clinical measure to determine the severity of HIV infections. The CD4 count was grouped into three distinct categories: 0–49, 50–74, and ≥ 75. Let $X$ and $Y$ denote the (category minimum) CD4 count at a month and the following month, respectively. The conditional probabilities for $Y$ given values for $X$ were provided by a transition probability matrix shown in the following table.

| $X$ | $Y$ | | |
|---|---|---|---|
| | 0 | 50 | 75 |
| 0 | 0.9819 | 0.0122 | 0.0059 |
| 50 | 0.1766 | 0.7517 | 0.0717 |
| 75 | 0.0237 | 0.0933 | 0.8830 |

This table is interpreted as follows. For example, $P(Y = 50 \mid X = 75) = 0.0933$. Suppose also that the probability distribution for $X$ is $P(X = 75) = 0.9, P(X = 50) = 0.08, P(X = 0) = 0.02$. Determine the following:

**a.** $P(Y \leq 50 \mid X = 50)$
**b.** $P(X = 0, Y = 75)$
**c.** $E(Y \mid X = 50)$ **d.** $f_Y(y)$
**e.** $f_{XY}(x, y)$ **f.** Are $X$ and $Y$ independent?

**5.2.7** An article in *Clinical Infectious Diseases* ["Strengthening the Supply of Routinely Administered Vaccines in the United States: Problems and Proposed Solutions" (2006, Vol. 42(3), pp. S97–S103)] reported that recommended vaccines for infants and children were periodically unavailable or in short supply in the United States. Although the number of doses demanded each month is a discrete random variable, the large demands can be approximated with a continuous probability distribution. Suppose that the monthly demands for two of those vaccines, namely measles–mumps–rubella (MMR) and varicella (for chickenpox), are independently, normally distributed with means of 1.1 and 0.55 million doses and standard deviations of 0.3 and 0.1 million doses, respectively. Also suppose that the inventory levels at the beginning of a given month for MMR and varicella vaccines are 1.2 and 0.6 million doses, respectively.

**a.** What is the probability that there is no shortage of either vaccine in a month without any vaccine production?
**b.** To what should inventory levels be set so that the probability is 90% that there is no shortage of either vaccine in a month without production? Can there be more than one answer? Explain.

**5.2.8** The systolic and diastolic blood pressure values (mm Hg) are the pressures when the heart muscle contracts and relaxes (denoted as $Y$ and $X$, respectively). Over a collection of individuals, the distribution of diastolic pressure is normal with mean 73 and standard deviation 8. The systolic pressure is conditionally

normally distributed with mean $1.6x$ when $X = x$ and standard deviation of 10. Determine the following:

**a.** Conditional probability density function of $Y$ given $X = 73$

**b.** $P(Y < 115 \mid X = 73)$

**c.** $E(Y \mid X = 73)$

**d.** Recognize the distribution $f_{XY}(x, y)$ and identify the mean and variance of $Y$

**5.2.9** WP SS VS Consider the joint distribution in Exercise 5.1.10. Determine the following:

**a.** $f_{Y|2}(y)$ **b.** $E(Y \mid X = 2)$

**c.** $V(Y \mid X = 2)$ **d.** Are $X$ and $Y$ independent?

## Exercises for Section 5.3

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**5.3.1** WP SS For the Transaction Processing Performance Council's benchmark in Exercise 5.1.6, let $X$, $Y$, and $Z$ denote the average number of *selects*, *updates*, and *inserts* operations required for each type of transaction, respectively. Calculate the following:

**a.** $f_{XYZ}(x, y, z)$

**b.** Conditional probability mass function for $X$ and $Y$ given $Z = 0$

**c.** $P(X < 6, Y < 6 \mid Z = 0)$ **d.** $E(X \mid Y = 0, Z = 0)$

**5.3.2** WP Suppose that the random variables $X$, $Y$, and $Z$ have the following joint probability distribution.

| $x$ | $y$ | $z$ | $f(x, y, z)$ |
|---|---|---|---|
| 1 | 1 | 1 | 0.05 |
| 1 | 1 | 2 | 0.10 |
| 1 | 2 | 1 | 0.15 |
| 1 | 2 | 2 | 0.20 |
| 2 | 1 | 1 | 0.20 |
| 2 | 1 | 2 | 0.15 |
| 2 | 2 | 1 | 0.10 |
| 2 | 2 | 2 | 0.05 |

Determine the following:

**a.** $P(X = 2)$ **b.** $P(X = 1, Y = 2)$

**c.** $P(Z < 1.5)$ **d.** $P(X = 1 \text{ or } Z = 2)$

**e.** $E(X)$ **f.** $P(X = 1 \mid Y = 1)$

**g.** $P(X = 1, Y = 1 \mid Z = 2)$ **h.** $P(X = 1 \mid Y = 1, Z = 2)$

**i.** Conditional probability distribution of $X$ given that $Y = 1$ and $Z = 2$

**5.3.3** WP Suppose that the random variables $X$, $Y$, and $Z$ have the joint probability density function $f(x, y, z) = 8xyz$ for $0 < x < 1$, $0 < y < 1$, and $0 < z < 1$. Determine the following:

**a.** $P(X < 0.5)$ **b.** $P(X < 0.5, Y < 0.5)$

**c.** $P(Z < 2)$ **d.** $P(X < 0.5 \text{ or } Z < 2)$

**e.** $E(X)$ **f.** $P(X < 0.5 \mid Y = 0.5)$

**g.** $P(X < 0.5, Y < 0.5 \mid Z = 0.8)$

**h.** Conditional probability distribution of $X$ given that $Y = 0.5$ and $Z = 0.8$

**i.** $P(X < 0.5 \mid Y = 0.5, Z = 0.8)$

**5.3.4** Suppose that the random variables $X$, $Y$, and $Z$ have the joint probability density function $f_{XYZ}(x, y, z) = c$ over the cylinder $x^2 + y^2 < 4$ and $0 < z < 4$. Determine the constant $c$ so that $f_{XYZ}(x, y, z)$ is a probability density function.

Determine the following:

**a.** $P(X^2 + Y^2 < 2)$ **b.** $P(Z < 2)$

**c.** $E(X)$ **d.** $P(X < 1 \mid Y = 1)$

**e.** $P(X^2 + Y^2 < 1 \mid Z = 1)$

**f.** Conditional probability distribution of $Z$ given that $X = 1$ and $Y = 1$.

## Exercises for Section 5.4

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**5.4.1** WP SS VS Determine the covariance and correlation for the following joint probability distribution:

| $x$ | 1 | 1 | 2 | 4 |
|---|---|---|---|---|
| $y$ | 3 | 4 | 5 | 6 |
| $f_{XY}(x, y)$ | 1/8 | 1/4 | 1/2 | 1/8 |

**5.4.2** WP Determine the value for $c$ and the covariance and correlation for the joint probability mass function $f_{XY}(x, y) = c(x + y)$ for $x = 1, 2, 3$ and $y = 1, 2, 3$.

**5.4.3** Patients are given a drug treatment and then evaluated. Symptoms either improve, degrade, or remain the same with probabilities 0.4, 0.1, 0.5, respectively. Assume that four independent patients are treated and let $X$ and $Y$ denote the number of patients who improve and degrade, respectively. Are $X$ and $Y$ independent? Calculate the covariance and correlation between $X$ and $Y$.



**5.4.4** For the Transaction Processing Performance Council's benchmark in Exercise 5.1.6, let $X$, $Y$, and $Z$ denote the average number of *selects*, *updates*, and *inserts* operations required for each type of transaction, respectively. Calculate the following:

**a.** Correlation between $X$ and $Y$

**b.** Correlation between $X$ and $Z$

**5.4.5** WP Determine the value for $c$ and the covariance and correlation for the joint probability density function $f_{XY}(x, y) = cxy$ over the range $0 < x < 3$ and $0 < y < x$.

**5.4.6** WP Determine the covariance and correlation for the joint probability density function $f_{XY}(x, y) = e^{-x-y}$ over the range $0 < x$ and $0 < y$.

**5.4.7** SS The joint probability distribution is

| $x$ | −1 | 0 | 0 | 1 |
|---|---|---|---|---|
| $y$ | 0 | −1 | 1 | 0 |
| $f_{XY}(x, y)$ | 1/4 | 1/4 | 1/4 | 1/4 |

Show that the correlation between $X$ and $Y$ is zero but $X$ and $Y$ are not independent.

**5.4.8** Determine the covariance and correlation for the CD4 counts in a month and the following month in Exercise 5.2.6.

**5.4.9** Determine the covariance and correlation for the lengths of the minor and major axes in Exercise 5.2.5.

**5.4.10** WP Suppose that the correlation between $X$ and $Y$ is $\rho$. For constants $a$, $b$, $c$, and $d$, what is the correlation between the random variables $U = aX + b$ and $V = cY + d$?

## Exercises for Section 5.5

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
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**5.5.1** WP Based on the number of voids, a ferrite slab is classified as either high, medium, or low. Historically, 5% of the slabs are classified as high, 85% as medium, and 10% as low. A group of 20 slabs that are independent regarding voids is selected for testing. Let $X$, $Y$, and $Z$ denote the number of slabs that are classified as high, medium, and low, respectively.

**a.** What are the name and the values of the parameters of the joint probability distribution of $X$, $Y$, and $Z$?

**b.** What is the range of the joint probability distribution of $X$, $Y$, and $Z$?

**c.** What are the name and the values of the parameters of the marginal probability distribution of $X$?

**d.** Determine $E(X)$ and $V(X)$.

Determine the following:

**e.** $P(X = 1, Y = 17, Z = 3)$

**f.** $P(X \le 1, Y = 17, Z = 3)$

**g.** $P(X \le 1)$ **h.** $E(Y)$

**i.** $P(X = 2, Z = 3 \mid Y = 17)$ **j.** $P(X = 2 \mid Y = 17)$

**k.** $E(X \mid Y = 17)$

**5.5.2** A Web site uses ads to route visitors to one of four landing pages. The probabilities for each landing page are equal. Consider 20 independent visitors and let the random variables $W$, $X$, $Y$, and $Z$ denote the number of visitors routed to each page.

Calculate the following:

**a.** $P(W = 5, X = 5, Y = 5, Z = 5)$

**b.** $P(W = 5, X = 5, Y = 5, Z = 5)$

**c.** $P(W = 7, X = 7, Y = 6 \mid Z = 3)$

**d.** $P(W = 7, X = 7, Y = 3 \mid Z = 3)$

**e.** $P(W \le 2)$ **f.** $E(W)$

**g.** $P(W = 5, X = 5)$ **h.** $P(W = 5 \mid X = 5)$

**5.5.3** WP Suppose that $X$ and $Y$ have a bivariate normal distribution with $\sigma_X = 0.04$, $\sigma_Y = 0.08$, $\mu_X = 3.00$, $\mu_Y = 7.70$, and $\rho = 0$.

Determine the following:

**a.** $P(2.95 < X < 3.05)$ **b.** $P(7.60 < Y < 7.80)$

**c.** $P(2.95 < X < 3.05, 7.60 < Y < 7.80)$

**5.5.4** Let $X$ and $Y$ represent the concentration and viscosity of a chemical product. Suppose that $X$ and $Y$ have a bivariate normal distribution with $\sigma_X = 4$, $\sigma_Y = 1$, $\mu_X = 2$, and $\mu_Y = 1$. Draw a rough contour plot of the joint probability density function for each of the following values of $\rho$:

**a.** $\rho = 0$ **b.** $\rho = 0.8$ **c.** $\rho = -0.8$

**5.5.5** WP SS VS In the manufacture of electroluminescent lamps, several different layers of ink are deposited onto a plastic substrate. The thickness of these layers is critical if specifications regarding the final color and intensity of light are to be met. Let $X$ and $Y$ denote the thickness of two different layers of ink. It is known that $X$ is normally distributed with a mean of 0.1 millimeter and a standard deviation of 0.00031 millimeter, and $Y$ is normally distributed with a mean of 0.23 millimeter and a standard deviation of 0.00017 millimeter. The value of $\rho$ for these variables is equal to 0. Specifications call for a lamp to have a thickness of the ink corresponding to $X$ in the range of 0.099535 to 0.100465 millimeter and $Y$ in the range of 0.22966 to 0.23034 millimeter. What is the probability that a randomly selected lamp will conform to specifications?

**5.5.6** WP In an acid-base titration, a base or acid is gradually added to the other until they have completely neutralized each other. Let $X$ and $Y$ denote the milliliters of acid and base needed for equivalence, respectively. Assume that $X$ and $Y$ have

a bivariate normal distribution with $\sigma_X = 5$ mL, $\sigma_Y = 2$ mL, $\mu_X = 120$ mL, $\mu_Y = 100$ mL, and $\rho = 0.6$.
Determine the following:

**a.** Covariance between $X$ and $Y$
**b.** Marginal probability distribution of $X$
**c.** $P(X < 116)$
**d.** Conditional probability distribution of $X$ given that $Y = 102$
**e.** $P(X < 116 \mid Y = 102)$

**5.5.7** If $X$ and $Y$ have a bivariate normal distribution with $\rho = 0$, show that $X$ and $Y$ are independent.

**5.5.8** Patients given drug therapy either improve, remain the same, or degrade with probabilities 0.5, 0.4, and 0.1, respectively. Suppose that 20 patients (assumed to be independent) are given the therapy. Let $X_1, X_2$, and $X_3$ denote the number of patients who improved, stayed the same, or became degraded, respectively. Determine the following:

**a.** Are $X_1, X_2, X_3$ independent?
**b.** $P(X_1 = 10)$
**c.** $P(X_1 = 10, X_2 = 8, X_3 = 2)$
**d.** $P(X_1 = 5 \mid X_2 = 12)$
**e.** $E(X_1)$

## Exercises for Section 5.6

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutoring Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
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**5.6.1** $X$ and $Y$ are independent, normal random variables with $E(X) = 2, V(X) = 5, E(Y) = 6$, and $V(Y) = 8$.
Determine the following:

**a.** $E(3X + 2Y)$
**b.** $V(3X + 2Y)$
**c.** $P(3X + 2Y < 18)$
**d.** $P(3X + 2Y < 28)$

**5.6.2** WP Suppose that the random variable $X$ represents the length of a punched part in centimeters. Let $Y$ be the length of the part in millimeters. If $E(X) = 5$ and $V(X) = 0.25$, what are the mean and variance of $Y$?

**5.6.3** WP SS In the manufacture of electroluminescent lamps, several different layers of ink are deposited onto a plastic substrate. The thickness of these layers is critical if specifications regarding the final color and intensity of light are to be met. Let $X$ and $Y$ denote the thickness of two different layers of ink. It is known that $X$ is normally distributed with a mean of 0.1 mm and a standard deviation of 0.00031 mm, and $Y$ is also normally distributed with a mean of 0.23 mm and a standard deviation of 0.00017 mm. Assume that these variables are independent.

**a.** If a particular lamp is made up of these two inks only, what is the probability that the total ink thickness is less than 0.2337 mm?
**b.** A lamp with a total ink thickness exceeding 0.2405 mm lacks the uniformity of color that the customer demands. Find the probability that a randomly selected lamp fails to meet customer specifications.

**5.6.4** WP Making handcrafted pottery generally takes two major steps: wheel throwing and firing. The time of wheel throwing and the time of firing are normally distributed random variables with means of 40 minutes and 60 minutes and standard deviations of 2 minutes and 3 minutes, respectively.

**a.** What is the probability that a piece of pottery will be finished within 95 minutes?
**b.** What is the probability that it will take longer than 110 minutes?

**5.6.5** WP An article in *Knee Surgery Sports Traumatology, Arthroscopy* ["Effect of Provider Volume on Resource Utilization for Surgical Procedures" (2005, Vol. 13, pp. 273–279)] showed a mean time of 129 minutes and a standard deviation of 14 minutes for ACL reconstruction surgery for high-volume hospitals (with more than 300 such surgeries per year). If a high-volume hospital needs to schedule 10 surgeries, what are the mean and variance of the total time to complete these surgeries? Assume that the times of the surgeries are independent and normally distributed.

**5.6.6** WP An automated filling machine fills soft-drink cans, and the standard deviation is 0.5 fluid ounce. Assume that the fill volumes of the cans are independent, normal random variables.

**a.** What is the standard deviation of the average fill volume of 100 cans?
**b.** If the mean fill volume is 12.1 oz, what is the probability that the average fill volume of 100 cans is less than 12 oz?
**c.** What should the mean fill volume equal so that the probability that the average of 100 cans is less than 12 oz is 0.005?
**d.** If the mean fill volume is 12.1 oz, what should the standard deviation of fill volume equal so that the probability that the average of 100 cans is less than 12 oz is 0.005?
**e.** Determine the number of cans that need to be measured such that the probability that the average fill volume is less than 12 oz is 0.01.

**5.6.7** WP SS VS Assume that the weights of individuals are independent and normally distributed with a mean of 160 pounds and a standard deviation of 30 pounds. Suppose that 25 people squeeze into an elevator that is designed to hold 4300 pounds.

**a.** What is the probability that the load (total weight) exceeds the design limit?
**b.** What design limit is exceeded by 25 occupants with probability 0.0001?

**5.6.8** A U-shaped component is to be formed from the three parts $A$, $B$, and $C$. See Figure 5.14. The length of $A$ is normally distributed with a mean of 10 mm and a standard deviation of 0.1 mm. The thicknesses of parts $B$ and $C$ are each normally distributed with a mean of 2 mm and a standard deviation of 0.05 mm. Assume that all dimensions are independent.



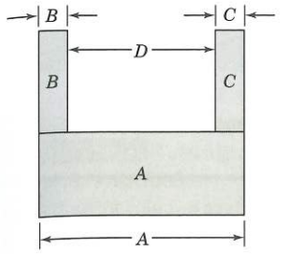


**FIGURE 5.14**
**Illustration for the U-shaped component.**

**a.** Determine the mean and standard deviation of gap D.
**b.** What is the probability that gap D is less than 5.9 mm?

**5.6.9** Consider the perimeter of a part in Example 5.27. Let $X_1$ and $X_2$ denote the length and width of a part with standard deviations 0.1 and 0.2 centimeters, respectively. Suppose that the covariance between $X_1$ and $X_2$ is 0.02. Determine the variance of the perimeter $Y = 2X_1 + 2X_2$ of a part. Compare and comment on the result here and in the example.

**5.6.10** The rate of return of an asset is the change in price divided by the initial price (denoted as $r$). Suppose that \$10,000 is used to purchase shares in three stocks with rates of returns $X_1, X_2, X_3$. Initially, \$2500, \$3000, and \$4500 are allocated to each one, respectively. After one year, the distribution of the rate of return for each is normally distributed with the following parameters: $\mu_1 = 0.12$, $\sigma_1 = 0.14$, $\mu_2 = 0.04$, $\sigma_2 = 0.02$, $\mu_3 = 0.07$, $\sigma_3 = 0.08$.

**a.** Assume that these rates of return are independent. Determine the mean and variance of the rate of return after one year for the entire investment of \$10,000.
**b.** Assume that $X_1$ is independent of $X_2$ and $X_3$ but that the covariance between $X_2$ and $X_3$ is −0.005. Repeat part (a).
**c.** Compare the means and variances obtained in parts (a) and (b) and comment on any benefits from negative covariances between the assets.

**5.6.11** In Exercise 5.2.7, the monthly demand for MMR vaccine was assumed to be approximately normally distributed with a mean and standard deviation of 1.1 and 0.3 million doses, respectively. Suppose that the demands for different months are independent, and let $Z$ denote the demand for a year (in millions of doses). Determine the following:

**a.** Mean, variance, and distribution of $Z$
**b.** $P(Z < 13.2)$
**c.** $P(11 < Z < 15)$
**d.** Value for $c$ such that $P(Z < c) = 0.99$

## Exercises for Section 5.7

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutoring Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
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**5.7.1** WP Suppose that $X$ is a random variable with probability distribution

$$f_X(x) = 1/4, \qquad x = 1, 2, 3, 4$$

Determine the probability distribution of $Y = 2X + 1$.

**5.7.2** WP Suppose that $X$ is a continuous random variable with probability distribution

$$f_X(x) = \frac{x}{18}, \qquad 0 \le x \le 6$$

**a.** Determine the probability distribution of the random variable $Y = 2X + 10$.
**b.** Determine the expected value of $Y$.

**5.7.3** WP A random variable $X$ has the probability distribution

$$f_X(x) = e^{-x}, \qquad x \ge 0$$

Determine the probability distribution for the following:

**a.** $Y = X^2$
**b.** $Y = X^{1/2}$
**c.** $Y = \ln X$

**5.7.4** WP The velocity of a particle in a gas is a random variable $V$ with probability distribution

$$f_V(v) = av^2e^{-bv}, \qquad v > 0$$

where $b$ is a constant that depends on the temperature of the gas and the mass of the particle.

**a.** Determine the value of the constant $a$.
**b.** The kinetic energy of the particle is $W = mV^2/2$. Determine the probability distribution of $W$.

**5.7.5** WP Suppose that $X$ has the probability distribution

$$f_X(x) = 1, \qquad 1 \le x \le 2$$

Determine the probability distribution of the random variable $Y = e^X$.

**5.7.6** Derive the probability density function for a lognormal random variable $Y$ from the relationship that $Y = \exp(W)$ for a normal random variable $W$ with mean $\theta$ and variance $\omega^2$.

**5.7.7** SS An aircraft is flying at a constant altitude with velocity magnitude $r_1$ (relative to the air) and angle $\theta_1$ (in a two-dimensional coordinate system). The magnitude and direction of the wind are $r_2$ and $\theta_2$, respectively. Suppose that the wind angle is uniformly distributed between 10 and 20 degrees and all other parameters are constant. Determine the probability density function of the magnitude of the resultant vector $r = [r_1^2 + r_2^2 + r_1r_2 (\cos \theta_1 - \cos \theta_2)]^{0.5}$.

**5.7.8** Power meters enable cyclists to obtain power measurements nearly continuously. The meters also calculate the average power generated over a time interval. Professional riders can generate 6.6 watts per kilogram of body weight for extended periods of time. Some meters calculate a normalized power measurement to

components are independent and normally distributed with the following means and standard deviations (in ounces):

| Component | Mean | Standard Deviation |
|---|---|---|
| Left case | 4.0 | 0.4 |
| Right case | 5.5 | 0.5 |
| Bearing assembly | 10.0 | 0.2 |
| Bolt assembly | 8.0 | 0.5 |

**a.** What is the probability that the weight of an assembly exceeds 29.5 ounces?

**b.** What is the probability that the mean weight of eight independent assemblies exceeds 29 ounces?

**5.S21** WP The permeability of a membrane used as a moisture barrier in a biological application depends on the thickness of two integrated layers. The layers are normally distributed with means of 0.5 and 1 millimeters, respectively. The standard deviations of layer thickness are 0.1 and 0.2 millimeters, respectively. The correlation between layers is 0.7.

**a.** Determine the mean and variance of the total thickness of the two layers.

**b.** What is the probability that the total thickness is less than 1 millimeter?

**c.** Let $X_1$ and $X_2$ denote the thickness of layers 1 and 2, respectively. A measure of performance of the membrane is a function of $2X_1 + 3X_2$ of the thickness. Determine the mean and variance of this performance measure.

**5.S22** A small company is to decide what investments to use for cash generated from operations. Each investment has a mean and standard deviation associated with a percentage return. The first security has a mean return of 5% with a standard deviation of 2%, and the second security provides the same mean of 5% with a standard deviation of 4%. The securities have a correlation of –0.5. If the company invests \$2 million with half in each security, what are the mean and standard deviation of the percentage return? Compare the standard deviation of this strategy to one that invests the \$2 million into the first security only.

**5.S23** WP A marketing company performed a risk analysis for a manufacturer of synthetic fibers and concluded that new competitors present no risk 13% of the time (due mostly to the diversity of fibers manufactured), moderate risk 72% of the time (some overlapping of products), and very high risk (competitor manufactures the exact same products) 15% of the time. It is known that 12 international companies are planning to open new facilities for the manufacture of synthetic fibers within the next 3 years. Assume that the companies are independent. Let $X$, $Y$, and $Z$ denote the number of new competitors that will pose no, moderate, and very high risk for the interested company, respectively.

Determine the following:

**a.** Range of the joint probability distribution of $X$, $Y$, and $Z$

**b.** $P(X = 1, Y = 3, Z = 1)$ **c.** $P(Z \leq 2)$

**d.** $P(Z = 2 \mid Y = 1, X = 10)$ **e.** $P(Z \leq 1 \mid X = 10)$

**f.** $P(Y \leq 1, Z \leq 1 \mid X = 10)$ **g.** $E(Z \mid X = 10)$

**5.S24** Suppose $X$ has a lognormal distribution with parameters $\theta$ and $\omega$. Determine the probability density function and the parameters values for $Y = X^{\gamma}$ for a constant $\gamma > 0$. What is the name of this distribution?

**5.S25** The power in a DC circuit is $P = I^2/R$ where $I$ and $R$ denote the current and resistance, respectively. Suppose that $I$ is approximately normally distributed with mean of 200 mA and standard deviation 0.2 mA and $R$ is a constant. Determine the probability density function of power.

**5.S26** The intensity (mW/mm$^2$) of a laser beam on a surface theoretically follows a bivariate normal distribution with maximum intensity at the center, equal variance $\sigma$ in the $x$ and $y$ directions, and zero covariance. There are several definitions for the width of the beam. One definition is the diameter at which the intensity is 50% of its peak. Suppose that the beam width is 1.6 mm under this definition. Determine $\sigma$. Also determine the beam width when it is defined as the diameter where the intensity equals $1/e^2$ of the peak.

# Chapter 6 Exercises

## Exercises for Section 6.1

WP Problem available in WileyPLUS at instructor's discretion.
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**6.1.1** Will the sample mean always correspond to one of the observations in the sample?

**6.1.2** Will exactly half of the observations in a sample fall below the mean?

**6.1.3** Will the sample mean always be the most frequently occurring data value in the sample?

**6.1.4** For any set of data values, is it possible for the sample standard deviation to be larger than the sample mean? If so, give an example.

**6.1.5** Can the sample standard deviation be equal to zero? If so, give an example.

**6.1.6** WP GO Tutorial In *Applied Life Data Analysis* (Wiley, 1982), Wayne Nelson presents the breakdown time of an insulating fluid between electrodes at 34 kV. The times, in minutes, are as follows: 0.19, 0.78, 0.96, 1.31, 2.78, 3.16, 4.15, 4.67, 4.85, 6.50, 7.35, 8.01, 8.27, 12.06, 31.75, 32.52, 33.91, 36.71, and 72.89. Calculate the sample mean and sample standard deviation.



**6.1.7** WP SS VS An article in *Human Factors* ["Visual Performance on CRT Screens and Hard-Copy Displays" (1989, Vol. 31(3), pp. 247–257)] presented data on visual accommodation (a function of eye movement) when recognizing a speckle pattern on a high-resolution CRT screen. The data are as follows: 36.45, 67.90, 38.77, 42.18, 26.72, 50.77, 39.30, and 49.71. Calculate the sample mean and sample standard deviation. Construct a dot diagram of the data.

**6.1.8** Preventing fatigue crack propagation in aircraft structures is an important element of aircraft safety. An engineering study to investigate fatigue crack in $n = 9$ cyclically loaded wing boxes reported the following crack lengths (in mm): 2.13, 2.96, 3.02, 1.82, 1.15, 1.37, 2.04, 2.47, 2.60. Calculate the sample mean and sample standard deviation. Prepare a dot diagram of the data.

**6.1.9** An article in the *Journal of Physiology* ["Response of Rat Muscle to Acute Resistance Exercise Defined by Transcriptional and Translational Profiling" (2002, Vol. 545, pp. 27–41)] studied gene expression as a function of resistance exercise. Expression data (measures of gene activity) from one gene are shown in the following table. One group of rats was exercised for 6 hours while the other received no exercise. Compute the sample mean and standard deviation of the exercise and no-exercise groups separately. Construct a dot diagram for the exercise and no-exercise groups separately. Comment on any differences for the groups.

| 6 Hours of Exercise | 6 Hours of Exercise | No Exercise | No Exercise |
|---|---|---|---|
| 425.313 | 208.475 | 485.396 | 406.921 |
| 223.306 | 286.484 | 159.471 | 335.209 |
| 388.793 | 244.242 | 478.314 | |
| 139.262 | 408.099 | 245.782 | |
| 212.565 | 157.743 | 236.212 | |
| 324.024 | 436.37 | 252.773 | |

**6.1.10** WP Exercise 6.1.7 describes data from an article in *Human Factors* on visual accommodation from an experiment involving a high-resolution CRT screen.

Data from a second experiment using a low-resolution screen were also reported in the article. They are 8.85, 35.80, 26.53, 64.63, 9.00, 15.38, 8.14, and 8.24. Prepare a dot diagram for this second sample and compare it to the one for the first sample. What can you conclude about CRT resolution in this situation?

**6.1.11** WP SS The following data are the joint temperatures of the O-rings (°F) for each test firing or actual launch of the Space Shuttle rocket motor (from *Presidential Commission on the Space Shuttle Challenger Accident*, Vol. 1, pp. 129–131): 84, 49, 61, 40, 83, 67, 45, 66, 70, 69, 80, 58, 68, 60, 67, 72, 73, 70, 57, 63, 70, 78, 52, 67, 53, 67, 75, 61, 70, 81, 76, 79, 75, 76, 58, 31.

**a.** Compute the sample mean and sample standard deviation and construct a dot diagram of the temperature data.

**b.** Set aside the smallest observation (31°F) and recompute the quantities in part (a). Comment on your findings. How "different" are the other temperatures from this last value?

**6.1.12** Cloud seeding, a process in which chemicals such as silver iodide and frozen carbon dioxide are introduced by aircraft into clouds to promote rainfall, was widely used in the 20th century. Recent research has questioned its effectiveness ["Reassessment of Rain Enhancement Experiments and Operations in Israel Including Synoptic Considerations," *Journal of Atmospheric Research* (2010, Vol. 97(4), pp. 513–525)]. An experiment was performed by randomly assigning 52 clouds to be seeded or not. The amount of rain generated was then measured in acre-feet. Here are the data for the unseeded and seeded clouds:

Unseeded:

81.2 26.1 95.0 41.1 28.6 21.7 11.5 68.5 345.5 321.2
1202.6 1.0 4.9 163.0 372.4 244.3 47.3 87.0 26.3 24.4
830.1 4.9 36.6 147.8 17.3 29.0

Seeded:

274.7 302.8 242.5 255.0 17.5 115.3 31.4 703.4 334.1
1697.8 118.3 198.6 129.6 274.7 119.0 1656.0 7.7 430.0
40.6 92.4 200.7 32.7 4.1 978.0 489.1 2745.6

Find the sample mean, sample standard deviation, and range of rainfall for

**a.** All 52 clouds

**b.** The unseeded clouds

**c.** The seeded clouds

**6.1.13** Construct dot diagrams of the seeded and unseeded clouds and compare their distributions in a couple of sentences.

**6.1.14** In the 2000 Sydney Olympics, a special program initiated by the IOC president allowed developing countries to send athletes to the Olympics without the usual qualifying procedure. Here are the 71 times for the first round of the 100-meter men's swim (in seconds).

60.39 49.93 53.40 51.82 50.46 51.34 50.28 50.19 52.14
50.56 52.72 50.95 49.74 49.16 52.57 52.53 52.09 52.40
49.75 54.06 53.50 50.63 51.93 51.62 52.58 53.55 51.07
49.76 49.73 50.90 59.26 49.29 52.78 112.72 49.79 49.83
52.43 51.28 52.22 49.76 49.70 52.90 50.19 54.33 62.45
51.93 52.24 52.82 50.96 48.64 51.11 50.87 52.18 54.12
50.49 49.84 52.91 52.52 50.32 51.52 52.0 52.85 52.24
49.45 51.28 49.09 58.79 49.74 49.32 50.62 49.45

**a.** Find the sample mean and sample standard deviation of these 100-meter swim times.

**b.** Construct a dot diagram of the data.

**c.** Comment on anything unusual that you see.

## Exercises for Section 6.2

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**6.2.1** A back-to-back stem-and-leaf display on two data sets is conducted by hanging the data on both sides of the same stems. Here is a back-to-back stem-and-leaf display for the cloud seeding data in Exercise 6.1.12 showing the unseeded clouds on the left and the seeded clouds on the right.

```
65098754433332221000 | 0 | 01233492223
                     | 2 | 00467703
                     | 4 | 39
                     | 6 | 0
                   3 | 8 | 8
                     | 10 |
                   0 | 12 |
                     | 14 |
                     | 16 | 60
                     | 18 |
                     | 20 |
                     | 22 |
                     | 24 |
                     | 26 | 5
```

How does the back-to-back stem-and-leaf display show the differences in the data set in a way that the dotplot cannot?

**6.2.2** When will the median of a sample be equal to the sample mean?

**6.2.3** When will the median of a sample be equal to the mode?

**6.2.4** WP An article in *Technometrics* ["Validation of Regression Models: Methods and Examples" (1977, Vol. 19(4), p. 425)] presented the following data on the motor fuel octane ratings of several blends of gasoline:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 88.5 | 98.8 | 89.6 | 92.2 | 92.7 | 88.4 | 87.5 | 90.9 |
| 94.7 | 88.3 | 90.4 | 83.4 | 87.9 | 92.6 | 87.8 | 89.9 |
| 84.3 | 90.4 | 91.6 | 91.0 | 93.0 | 93.7 | 88.3 | 91.8 |
| 90.1 | 91.2 | 90.7 | 88.2 | 94.4 | 96.5 | 89.2 | 89.7 |
| 89.0 | 90.6 | 88.6 | 88.5 | 90.4 | 84.3 | 92.3 | 92.2 |
| 89.8 | 92.2 | 88.3 | 93.3 | 91.2 | 93.2 | 88.9 | |
| 91.6 | 87.7 | 94.2 | 87.4 | 86.7 | 88.6 | 89.8 | |
| 90.3 | 91.1 | 85.3 | 91.1 | 94.2 | 88.7 | 92.7 | |
| 90.0 | 86.7 | 90.1 | 90.5 | 90.8 | 92.7 | 93.3 | |
| 91.5 | 93.4 | 89.3 | 100.3 | 90.1 | 89.3 | 86.7 | |
| 89.9 | 96.1 | 91.1 | 87.6 | 91.8 | 91.0 | 91.0 | |

Construct a stem-and-leaf display for these data. Calculate the median and quartiles of these data.

**6.2.5** WP GO Tutorial SS The following data are the numbers of cycles to failure of aluminum test coupons subjected to repeated alternating stress at 21,000 psi, 18 cycles per second.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1115 | 865 | 1015 | 885 | 1594 | 1000 | 1416 | 1501 |
| 1310 | 2130 | 845 | 1223 | 2023 | 1820 | 1560 | 1238 |
| 1540 | 1421 | 1674 | 375 | 1315 | 1940 | 1055 | 990 |
| 1502 | 1109 | 1016 | 2265 | 1269 | 1120 | 1764 | 1468 |
| 1258 | 1481 | 1102 | 1910 | 1260 | 910 | 1330 | 1512 |
| 1315 | 1567 | 1605 | 1018 | 1888 | 1730 | 1608 | 1750 |
| 1085 | 1883 | 706 | 1452 | 1782 | 1102 | 1535 | 1642 |
| 798 | 1203 | 2215 | 1890 | 1522 | 1578 | 1781 | |
| 1020 | 1270 | 785 | 2100 | 1792 | 758 | 1750 | |

Construct a stem-and-leaf display for these data. Calculate the median and quartiles of these data. Does it appear likely that a coupon will "survive" beyond 2000 cycles? Justify your answer.

**6.2.6** WP Calculate the sample median, mode, and mean of the data in Exercise 6.2.4. Explain how these three measures of location describe different features of the data.

**6.2.7** Calculate the sample median, mode, and mean of the data in Exercise 6.2.5. Explain how these three measures of location describe different features in the data.

**6.2.8** WP The female students in an undergraduate engineering core course at ASU self-reported their heights to the nearest inch. The data follow. Construct a stem-and-leaf diagram for the height data and comment on any important features that you notice. Calculate the sample mean, the sample standard deviation, and the sample median of height.

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 62 | 64 | 61 | 67 | 65 | 68 | 61 | 65 | 60 | 65 | 64 | 63 | 59 |
| 68 | 64 | 66 | 68 | 69 | 65 | 67 | 62 | 66 | 68 | 67 | 66 | 65 |
| 69 | 65 | 69 | 65 | 67 | 67 | 65 | 63 | 64 | 67 | 65 | | |

**6.2.9** The shear strengths of 100 spot welds in a titanium alloy follow. Construct a stem-and-leaf diagram for the weld strength data and comment on any important features that you notice. What is the 95th percentile of strength?

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 5408 | 5431 | 5475 | 5442 | 5376 | 5388 | 5459 | 5422 | 5416 | 5435 |
| 5420 | 5429 | 5401 | 5446 | 5487 | 5416 | 5382 | 5357 | 5388 | 5457 |
| 5407 | 5469 | 5416 | 5377 | 5454 | 5375 | 5409 | 5459 | 5445 | 5429 |
| 5463 | 5408 | 5481 | 5453 | 5422 | 5354 | 5421 | 5406 | 5444 | 5466 |
| 5399 | 5391 | 5477 | 5447 | 5329 | 5473 | 5423 | 5441 | 5412 | 5384 |
| 5445 | 5436 | 5454 | 5453 | 5428 | 5418 | 5465 | 5427 | 5421 | 5396 |
| 5381 | 5425 | 5388 | 5388 | 5378 | 5481 | 5387 | 5440 | 5482 | 5406 |
| 5401 | 5411 | 5399 | 5431 | 5440 | 5413 | 5406 | 5342 | 5452 | 5420 |
| 5458 | 5485 | 5431 | 5416 | 5431 | 5390 | 5399 | 5435 | 5387 | 5462 |
| 5383 | 5401 | 5407 | 5385 | 5440 | 5422 | 5448 | 5366 | 5430 | 5418 |

**6.2.10** An important quality characteristic of water is the concentration of suspended solid material. Following are 60 measurements on suspended solids from a certain lake. Construct a stem-and-leaf diagram for these data and comment on any important features that you notice. Compute the sample mean, the sample standard deviation, and the sample median. What is the 90th percentile of concentration?

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 42.4 | 65.7 | 29.8 | 58.7 | 52.1 | 55.8 | 57.0 | 68.7 | 67.3 | 67.3 |
| 54.3 | 54.0 | 73.1 | 81.3 | 59.9 | 56.9 | 62.2 | 69.9 | 66.9 | 59.0 |
| 56.3 | 43.3 | 57.4 | 45.3 | 80.1 | 49.7 | 42.8 | 42.4 | 59.6 | 65.8 |
| 61.4 | 64.0 | 64.2 | 72.6 | 72.5 | 46.1 | 53.1 | 56.1 | 67.2 | 70.7 |
| 42.6 | 77.4 | 54.7 | 57.1 | 77.3 | 39.3 | 76.4 | 59.3 | 51.1 | 73.8 |
| 61.4 | 73.1 | 77.3 | 48.5 | 89.8 | 50.7 | 52.0 | 59.6 | 66.1 | 31.6 |

**6.2.11** WP A group of wine enthusiasts taste-tested a pinot noir wine from Oregon. The evaluation was to grade the wine on a 0-to-100-point scale. The results follow. Construct a stem-and-leaf diagram for these data and comment on any important features that you notice. Compute the sample mean, the sample standard deviation, and the sample median. A wine rated above 90 is considered truly exceptional. What proportion of the taste-tasters considered this particular pinot noir truly exceptional?

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 94 | 90 | 92 | 91 | 91 | 86 | 89 | 91 | 91 | 90 |
| 90 | 93 | 87 | 90 | 91 | 92 | 89 | 86 | 89 | 90 |
| 88 | 95 | 91 | 88 | 89 | 92 | 87 | 89 | 95 | 92 |
| 85 | 91 | 85 | 89 | 88 | 84 | 85 | 90 | 90 | 83 |



**6.2.12** WP In their book *Introduction to Linear Regression Analysis* (5th edition, Wiley, 2012), Montgomery, Peck, and Vining presented measurements on $NbOCl_3$ concentration from a tube-flow reactor experiment. The data, in gram-mole per liter $\times 10^{-3}$, are as follows. Construct a stem-and-leaf diagram for these data and comment on any important features that you notice. Compute the sample mean, the sample standard deviation, and the sample median.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 450 | 450 | 473 | 507 | 457 | 452 | 453 | 1215 | 1256 |
| 1145 | 1085 | 1066 | 1111 | 1364 | 1254 | 1396 | 1575 | 1617 |
| 1733 | 2753 | 3186 | 3227 | 3469 | 1911 | 2588 | 2635 | 2725 |

**6.2.13** In Exercise 6.2.8, we presented height data that were self-reported by female undergraduate engineering students in a core course at ASU. In the same class, the male students self-reported their heights as follows. Construct a comparative stem-and-leaf diagram by listing the stems in the center of the display and then placing the female leaves on the left and the male leaves on the right. Comment on any important features that you notice in this display.

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 69 | 67 | 69 | 70 | 65 | 68 | 69 | 70 | 71 | 69 | 66 | 67 | 69 | 75 | 68 | 67 | 68 |
| 69 | 70 | 71 | 72 | 68 | 69 | 69 | 70 | 71 | 68 | 72 | 69 | 69 | 68 | 69 | 73 | 70 |
| 73 | 68 | 69 | 71 | 67 | 68 | 65 | 68 | 68 | 69 | 70 | 74 | 71 | 69 | 70 | 69 | |

## Exercises for Section 6.3

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**6.3.1** VS Construct a frequency distribution and histogram using the failure data from Exercise 6.2.5.

**6.3.2** WP Construct a frequency distribution and histogram for the motor fuel octane data from Exercise 6.2.4. Use eight bins.

**6.3.3** SS Construct histograms with 8 and 16 bins for the data in Exercise 6.2.5. Compare the histograms. Do both histograms display similar information?

**6.3.4** WP VS Construct frequency distributions and histograms with 8 bins and 16 bins for the motor fuel octane data in Exercise 6.2.4. Compare the histograms. Do both histograms display similar information?

**6.3.5** Construct a histogram for the spot weld shear strength data in Exercise 6.2.9. Comment on the shape of the histogram. Does it convey the same information as the stem-and-leaf display?

**6.3.6** WP Construct a histogram for the female student height data in Exercise 6.2.8.

**6.3.7** Construct a histogram for the pinot noir wine rating data in Exercise 6.2.11. Comment on the shape of the histogram. Does it convey the same information as the stem-and-leaf display?

**6.3.8** WP Construct a histogram for the water quality data in Exercise 6.2.10. Comment on the shape of the histogram. Does it convey the same information as the stem-and-leaf display?

**6.3.9** WP GO Tutorial SS VS **The Pareto Chart**. An important variation of a histogram for categorical data is the Pareto chart. This chart is widely used in quality improvement efforts, and the categories usually represent different types of defects, failure modes, or product/process problems. The categories are ordered so that the category with the largest frequency is on the left, followed by the category with the second largest frequency, and so forth. These charts are named after Italian economist Vilfredo Pareto, and they usually exhibit "Pareto's law"; that is, most of the defects can be accounted for by only a few categories. Suppose that the following information on structural defects in automobile doors is obtained: dents, 4; pits, 4; parts assembled out of sequence, 6; parts undertrimmed, 21; missing holes/slots, 8; parts not lubricated, 5; parts out of contour, 30; and parts not deburred, 3. Construct and interpret a Pareto chart.

## Exercises for Section 6.4

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**6.4.1** Using the data from Exercise 6.1.12 on cloud seeding,
**a.** Find the median and quartiles for the unseeded cloud data.
**b.** Find the median and quartiles for the seeded cloud data.
**c.** Make two side-by-side box plots, one for each group on the same plot.
**d.** Compare the distributions from what you can see in the side-by-side box plots.

**6.4.2** Using the data from Exercise 6.1.14 on swim times,
**a.** Find the median and quartiles for the data.
**b.** Make a box plot of the data.
**c.** Repeat (a) and (b) for the data without the extreme outlier and comment.
**d.** Compare the distribution of the data with and without the extreme outlier.

**6.4.3** WP GO Tutorial SS VS The "cold start ignition time" of an automobile engine is being investigated by a gasoline manufacturer. The following times (in seconds) were obtained for a test vehicle: 1.75, 1.92, 2.62, 2.35, 3.09, 3.15, 2.53, 1.91.
**a.** Calculate the sample mean, sample variance, and sample standard deviation.
**b.** Construct a box plot of the data.

**6.4.4** An article in *Transactions of the Institution of Chemical Engineers* (1956, Vol. 34, pp. 280–293) reported data from an experiment investigating the effect of several process variables on the vapor phase oxidation of naphthalene. A sample of the percentage mole conversion of naphthalene to maleic anhydride follows: 4.2, 4.7, 4.7, 5.0, 3.8, 3.6, 3.0, 5.1, 3.1, 3.8, 4.8, 4.0, 5.2, 4.3, 2.8, 2.0, 2.8, 3.3, 4.8, 5.0.
**a.** Calculate the sample mean, sample variance, and sample standard deviation.
**b.** Construct a box plot of the data.

**6.4.5** Exercise 6.1.11 presented the joint temperatures of the O-rings (°F) for each test firing or actual launch of the Space Shuttle rocket motor. In that exercise, you were asked to find the sample mean and sample standard deviation of temperature.
**a.** Find the median and the upper and lower quartiles of temperature.
**b.** Set aside the lowest observation (31°F) and recompute the quantities in part (a). Comment on your findings. How "different" are the other temperatures from this lowest value?
**c.** Construct a box plot of the data and comment on the possible presence of outliers.

**6.4.6** WP Reconsider the water quality data in Exercise 6.2.10. Construct a box plot of the concentrations and write an interpretation of the plot. How does the box plot compare in interpretive value to the original stem-and-leaf diagram?

**6.4.7** Reconsider the weld strength data in Exercise 6.2.9. Construct a box plot of the data and write an interpretation of the plot. How does the box plot compare in interpretive value to the original stem-and-leaf diagram?

**6.4.8** WP In Exercise 6.4.3, data were presented on the cold start ignition time of a particular gasoline used in a test vehicle. A second formulation of the gasoline was tested in the same vehicle with the following times (in seconds): 1.83, 1.99, 3.13, 3.29, 2.65, 2.87, 3.40, 2.46, 1.89, and 3.35. Use these new data along with the cold start times reported in Exercise 6.4.3 to construct comparative box plots. Write an interpretation of the information that you see in these plots.

**6.4.9** An article in *Nature Genetics* ["Treatment-specific Changes in Gene Expression Discriminate In Vivo Drug Response in Human Leukemia Cells" (2003, Vol. 34(1), pp. 85–90)] studied gene expression as a function of treatments for leukemia. One group received a high dose of the drug, while the control group received no treatment. Expression data (measures of gene activity) from one gene are shown in the table on the next page. Construct a box plot for each group of patients. Write an interpretation to compare the information in these plots.

| High Dose | Control | Control | Control |
|---|---|---|---|
| 16.1 | 297.1 | 25.1 | 131.1 |
| 134.9 | 491.8 | 820.1 | 166.5 |
| 52.7 | 1332.9 | 82.5 | 2258.4 |
| 14.4 | 1172 | 713.9 | 497.5 |
| 124.3 | 1482.7 | 785.6 | 263.4 |
| 99 | 335.4 | 114 | 252.3 |
| 24.3 | 528.9 | 31.9 | 351.4 |
| 16.3 | 24.1 | 86.3 | 678.9 |
| 15.2 | 545.2 | 646.6 | 3010.2 |
| 47.7 | 92.9 | 169.9 | 67.1 |
| 12.9 | 337.1 | 20.2 | 318.2 |
| 72.7 | 102.3 | 280.2 | 2476.4 |
| 126.7 | 255.1 | 194.2 | 181.4 |
| 46.4 | 100.5 | 408.4 | 2081.5 |
| 60.3 | 159.9 | 155.5 | 424.3 |
| 23.5 | 168 | 864.6 | 188.1 |
| 43.6 | 95.2 | 355.4 | 563 |
| 79.4 | 132.5 | 634 | 149.1 |
| 38 | 442.6 | 2029.9 | 2122.9 |
| 58.2 | 15.8 | 362.1 | 1295.9 |
| 26.5 | 175.6 | | |



## Exercises for Section 6.5

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**6.5.1** WP In their book *Time Series Analysis, Forecasting, and Control* (Prentice Hall, 1994), G. E. P. Box, G. M. Jenkins, and G. C. Reinsel present chemical concentration readings made every 2 hours. Some of these data follow (read down the column, then left to right).

| | | | | |
|---|---|---|---|---|
| 17.0 | 16.7 | 17.1 | 17.5 | 17.6 |
| 16.6 | 17.4 | 17.4 | 18.1 | 17.5 |
| 16.3 | 17.2 | 17.4 | 17.5 | 16.5 |
| 16.1 | 17.4 | 17.5 | 17.4 | 17.8 |
| 17.1 | 17.4 | 17.4 | 17.4 | 17.3 |
| 16.9 | 17.0 | 17.6 | 17.1 | 17.3 |
| 16.8 | 17.3 | 17.4 | 17.6 | 17.1 |
| 17.4 | 17.2 | 17.3 | 17.7 | 17.4 |
| 17.1 | 17.4 | 17.0 | 17.4 | 16.9 |
| 17.0 | 16.8 | 17.8 | 17.8 | 17.3 |

Construct and interpret either a digidot plot or a separate stem-and-leaf and time series plot of these data.

**6.5.2** The following table shows U.S. petroleum imports as a percentage of the totals, and Persian Gulf imports as a percentage of all imports by year since 1973 (*Source*: U.S. Department of Energy Web site, *www.eia.doe.gov*). Construct and interpret either a digidot plot or a separate stem-and-leaf and time series plot for each column of data.

| Year | Petroleum Imports (thousand barrels per day) | Total Petroleum Imports as Percent of Petroleum Products Supplied | Petroleum Imports from Persian Gulf as Percent of Total Petroleum Imports |
|---|---|---|---|
| 1973 | 6256 | 36.1 | 13.5 |
| 1974 | 6112 | 36.7 | 17.0 |
| 1975 | 6055 | 37.1 | 19.2 |
| 1976 | 7313 | 41.8 | 25.1 |
| 1977 | 8807 | 47.7 | 27.8 |
| 1978 | 8363 | 44.3 | 26.5 |
| 1979 | 8456 | 45.6 | 24.4 |
| 1980 | 6909 | 40.5 | 21.9 |
| 1981 | 5996 | 37.3 | 20.3 |
| 1982 | 5113 | 33.4 | 13.6 |
| 1983 | 5051 | 33.1 | 8.7 |
| 1984 | 5437 | 34.5 | 9.3 |
| 1985 | 5067 | 32.2 | 6.1 |
| 1986 | 6224 | 38.2 | 14.6 |
| 1987 | 6678 | 40.0 | 16.1 |
| 1988 | 7402 | 42.8 | 20.8 |
| 1989 | 8061 | 46.5 | 23.0 |
| 1990 | 8018 | 47.1 | 24.5 |
| 1991 | 7627 | 45.6 | 24.1 |
| 1992 | 7888 | 46.3 | 22.5 |
| 1993 | 8620 | 50.0 | 20.6 |
| 1994 | 8996 | 50.7 | 19.2 |

| Year | Petroleum Imports (thousand barrels per day) | Total Petroleum Imports as Percent of Petroleum Products Supplied | Petroleum Imports from Persian Gulf as Percent of Total Petroleum Imports |
|---|---|---|---|
| 1995 | 8835 | 49.8 | 17.8 |
| 1996 | 9478 | 51.7 | 16.9 |
| 1997 | 10,162 | 54.5 | 17.2 |
| 1998 | 10,708 | 56.6 | 19.9 |
| 1999 | 10,852 | 55.5 | 22.7 |
| 2000 | 11,459 | 58.1 | 21.7 |
| 2001 | 11,871 | 60.4 | 23.2 |
| 2002 | 11,530 | 58.3 | 19.6 |
| 2003 | 12,264 | 61.2 | 20.3 |
| 2004 | 13,145 | 63.4 | 18.9 |
| 2005 | 13,714 | 65.9 | 17.0 |
| 2006 | 13,707 | 66.3 | 16.1 |
| 2007 | 13,468 | 65.1 | 16.1 |
| 2008 | 12,915 | 66.2 | 18.4 |
| 2009 | 11,691 | 62.3 | 14.4 |
| 2010 | 11,793 | 61.5 | 14.5 |
| 2011 | 11,436 | 60.6 | 16.3 |
| 2012 | 10,627 | 57.3 | 20.3 |
| 2013 | 9859 | 52.0 | 20.4 |
| 2014 | 9241 | 48.4 | 20.3 |
| 2015 | 9449 | 48.4 | 15.9 |

**6.5.3** WP SS The 100 annual Wolfer sunspot numbers from 1770 to 1869 follow. (For an interesting analysis and interpretation of these numbers, see the book by Box, Jenkins, and Reinsel referenced in Exercise 6.5.1. Their analysis requires some advanced knowledge of statistics and statistical model building.) Read down, then left to right. The 1869 result is 74. Construct and interpret either a digidot plot or a stem-and-leaf and time series plot of these data.

| | | | | | |
|---|---|---|---|---|---|
| 101 | 31 | 154 | 38 | 83 | 90 |
| 82 | 7 | 125 | 23 | 132 | 67 |
| 66 | 20 | 85 | 10 | 131 | 60 |
| 35 | 92 | 68 | 24 | 118 | 47 |
| 41 | 10 | 16 | 8 | 62 | 94 |
| 21 | 8 | 7 | 13 | 98 | 96 |
| 16 | 2 | 4 | 57 | 124 | 77 |
| 6 | 0 | 2 | 122 | 96 | 59 |
| 4 | 1 | 8 | 138 | 66 | 44 |
| 7 | 5 | 17 | 103 | 64 | 47 |
| 14 | 12 | 36 | 86 | 54 | 30 |
| 34 | 14 | 50 | 63 | 39 | 16 |
| 45 | 35 | 62 | 37 | 21 | 7 |
| 43 | 46 | 67 | 24 | 7 | 37 |
| 48 | 41 | 71 | 11 | 4 | 74 |
| 42 | 30 | 48 | 15 | 23 | |
| 28 | 24 | 28 | 40 | 55 | |

**6.5.4** The table on the next page contains the global mean surface air temperature anomaly and the global $CO_2$ concentration for the years 1880–2004 (*Source:* http://data.giss.nasa.gov/gistemp/). The temperature is measured at a number of locations around the world and averaged annually, and then subtracted from a base period average (1951–1980) and the result reported as an anomaly.

**a.** Construct a time series plot of the global mean surface air temperature anomaly data and comment on any features that you observe.

**b.** Construct a time series plot of the global $CO_2$ concentration data and comment on any features that you observe.

**c.** Overlay the two plots on the same set of axes and comment on the plot.



| Year | Anomaly, °C | $CO_2$, ppmv | Year | Anomaly, °C | $CO_2$, ppmv | Year | Anomaly, °C | $CO_2$, ppmv |
|---|---|---|---|---|---|---|---|---|
| 1880 | −0.11 | 290.7 | 1922 | −0.09 | 303.8 | 1964 | −0.25 | 319.2 |
| 1881 | −0.13 | 291.2 | 1923 | −0.16 | 304.1 | 1965 | −0.15 | 320.0 |
| 1882 | −0.01 | 291.7 | 1924 | −0.11 | 304.5 | 1966 | −0.07 | 321.1 |
| 1883 | −0.04 | 292.1 | 1925 | −0.15 | 305.0 | 1967 | −0.02 | 322.0 |
| 1884 | −0.42 | 292.6 | 1926 | 0.04 | 305.4 | 1968 | −0.09 | 322.9 |
| 1885 | −0.23 | 293.0 | 1927 | −0.05 | 305.8 | 1969 | 0.00 | 324.2 |
| 1886 | −0.25 | 293.3 | 1928 | 0.01 | 306.3 | 1970 | 0.04 | 325.2 |
| 1887 | −0.45 | 293.6 | 1929 | −0.22 | 306.8 | 1971 | −0.10 | 326.1 |
| 1888 | −0.23 | 293.8 | 1930 | −0.03 | 307.2 | 1972 | −0.05 | 327.2 |
| 1889 | 0.04 | 294.0 | 1931 | 0.03 | 307.7 | 1973 | 0.18 | 328.8 |
| 1890 | −0.22 | 294.2 | 1932 | 0.04 | 308.2 | 1974 | −0.06 | 329.7 |
| 1891 | −0.55 | 294.3 | 1933 | −0.11 | 308.6 | 1975 | −0.02 | 330.7 |
| 1892 | −0.40 | 294.5 | 1934 | 0.05 | 309.0 | 1976 | −0.21 | 331.8 |
| 1893 | −0.39 | 294.6 | 1935 | −0.08 | 309.4 | 1977 | 0.16 | 333.3 |
| 1894 | −0.32 | 294.7 | 1936 | 0.01 | 309.8 | 1978 | 0.07 | 334.6 |
| 1895 | −0.32 | 294.8 | 1937 | 0.12 | 310.0 | 1979 | 0.13 | 336.9 |
| 1896 | −0.27 | 294.9 | 1938 | 0.15 | 310.2 | 1980 | 0.27 | 338.7 |
| 1897 | −0.15 | 295.0 | 1939 | −0.02 | 310.3 | 1981 | 0.40 | 339.9 |
| 1898 | −0.21 | 295.2 | 1940 | 0.14 | 310.4 | 1982 | 0.10 | 341.1 |
| 1899 | −0.25 | 295.5 | 1941 | 0.11 | 310.4 | 1983 | 0.34 | 342.8 |
| 1900 | −0.05 | 295.8 | 1942 | 0.10 | 310.3 | 1984 | 0.16 | 344.4 |
| 1901 | −0.05 | 296.1 | 1943 | 0.06 | 310.2 | 1985 | 0.13 | 345.9 |
| 1902 | −0.30 | 296.5 | 1944 | 0.10 | 310.1 | 1986 | 0.19 | 347.2 |
| 1903 | −0.35 | 296.8 | 1945 | −0.01 | 310.1 | 1987 | 0.35 | 348.9 |
| 1904 | −0.42 | 297.2 | 1946 | 0.01 | 310.1 | 1988 | 0.42 | 351.5 |
| 1905 | −0.25 | 297.6 | 1947 | 0.12 | 310.2 | 1989 | 0.28 | 352.9 |
| 1906 | −0.15 | 298.1 | 1948 | −0.03 | 310.3 | 1990 | 0.49 | 354.2 |
| 1907 | −0.41 | 298.5 | 1949 | −0.09 | 310.5 | 1991 | 0.44 | 355.6 |
| 1908 | −0.30 | 298.9 | 1950 | −0.17 | 310.7 | 1992 | 0.16 | 356.4 |
| 1909 | −0.31 | 299.3 | 1951 | −0.02 | 311.1 | 1993 | 0.18 | 357.0 |
| 1910 | −0.21 | 299.7 | 1952 | 0.03 | 311.5 | 1994 | 0.31 | 358.9 |
| 1911 | −0.25 | 300.1 | 1953 | 0.12 | 311.9 | 1995 | 0.47 | 360.9 |
| 1912 | −0.33 | 300.4 | 1954 | −0.09 | 312.4 | 1996 | 0.36 | 362.6 |
| 1913 | −0.28 | 300.8 | 1955 | −0.09 | 313.0 | 1997 | 0.40 | 363.8 |
| 1914 | −0.02 | 301.1 | 1956 | −0.18 | 313.6 | 1998 | 0.71 | 366.6 |
| 1915 | 0.06 | 301.4 | 1957 | 0.08 | 314.2 | 1999 | 0.43 | 368.3 |
| 1916 | −0.20 | 301.7 | 1958 | 0.10 | 314.9 | 2000 | 0.41 | 369.5 |
| 1917 | −0.46 | 302.1 | 1959 | 0.05 | 315.8 | 2001 | 0.56 | 371.0 |
| 1918 | −0.33 | 302.4 | 1960 | −0.02 | 316.6 | 2002 | 0.70 | 373.1 |
| 1919 | −0.09 | 302.7 | 1961 | 0.10 | 317.3 | 2003 | 0.66 | 375.6 |
| 1920 | −0.15 | 303.0 | 1962 | 0.05 | 318.1 | 2004 | 0.60 | 377.4 |
| 1921 | −0.04 | 303.4 | 1963 | 0.03 | 318.7 | | | |

## Exercises for Section 6.6

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**6.6.1** SS VS An article in the *Journal of Pharmaceuticals Sciences* ["Statistical Analysis of the Extended Hansen Method Using the Bootstrap Technique" (1991, Vol. 80, pp. 971–977)] presented data on the observed mole fraction solubility of a solute at a constant temperature and the dispersion, dipolar, and hydrogen-bonding Hansen partial solubility parameters. The data are as shown in the table below, where $y$ is the negative logarithm of the mole fraction solubility, $x_1$ is the dispersion partial solubility, $x_2$ is the dipolar partial solubility, and $x_3$ is the hydrogen bonding partial solubility.

**a.** Construct a matrix of scatter plots for these variables.

**b.** Comment on the apparent relationships among $y$ and the other three variables.

**6.6.2** Consider the petroleum imports in Exercise 6.5.2.

**a.** Construct a matrix of scatter plots for the three variables and compute the correlation coefficients.

**b.** Comment on the apparent relationships among the variables.

| Observation Number | $y$ | $x_1$ | $x_2$ | $x_3$ |
|---|---|---|---|---|
| 1 | 0.22200 | 7.3 | 0.0 | 0.0 |
| 2 | 0.39500 | 8.7 | 0.0 | 0.3 |
| 3 | 0.42200 | 8.8 | 0.7 | 1.0 |
| 4 | 0.43700 | 8.1 | 4.0 | 0.2 |
| 5 | 0.42800 | 9.0 | 0.5 | 1.0 |
| 6 | 0.46700 | 8.7 | 1.5 | 2.8 |
| 7 | 0.44400 | 9.3 | 2.1 | 1.0 |
| 8 | 0.37800 | 7.6 | 5.1 | 3.4 |
| 9 | 0.49400 | 10.0 | 0.0 | 0.3 |
| 10 | 0.45600 | 8.4 | 3.7 | 4.1 |
| 11 | 0.45200 | 9.3 | 3.6 | 2.0 |
| 12 | 0.11200 | 7.7 | 2.8 | 7.1 |
| 13 | 0.43200 | 9.8 | 4.2 | 2.0 |
| 14 | 0.10100 | 7.3 | 2.5 | 6.8 |
| 15 | 0.23200 | 8.5 | 2.0 | 6.6 |
| 16 | 0.30600 | 9.5 | 2.5 | 5.0 |
| 17 | 0.09230 | 7.4 | 2.8 | 7.8 |
| 18 | 0.11600 | 7.8 | 2.8 | 7.7 |
| 19 | 0.07640 | 7.7 | 3.0 | 8.0 |
| 20 | 0.43900 | 10.3 | 1.7 | 4.2 |
| 21 | 0.09440 | 7.8 | 3.3 | 8.5 |
| 22 | 0.11700 | 7.1 | 3.9 | 6.6 |
| 23 | 0.07260 | 7.7 | 4.3 | 9.5 |
| 24 | 0.04120 | 7.4 | 6.0 | 10.9 |
| 25 | 0.25100 | 7.3 | 2.0 | 5.2 |
| 26 | 0.00002 | 7.6 | 7.8 | 20.7 |

## Exercises for Section 6.7

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**6.7.1** WP Construct a normal probability plot of the visual accommodation data in Exercise 6.1.7. Does it seem reasonable to assume that visual accommodation is normally distributed?

**6.7.2** WP Construct a normal probability plot of the octane rating data in Exercise 6.2.4. Does it seem reasonable to assume that octane rating is normally distributed?

**6.7.3** SS VS Construct a normal probability plot of the cycles to failure data in Exercise 6.2.5. Does it seem reasonable to assume that cycles to failure are normally distributed?

**6.7.4** WP Construct a normal probability plot of the suspended solids concentration data in Exercise 6.2.10. Does it seem reasonable to assume that the concentration of suspended solids in water from this particular lake is normally distributed?

**6.7.5** Construct two normal probability plots for the height data in Exercises 6.2.8 and 6.2.13. Plot the data for female and male students on the same axes. Does height seem to be normally distributed for either group of students? If both populations have the same variance, the two normal probability plots should have identical slopes. What conclusions would you draw about the heights of the two groups of students from visual examination of the normal probability plots?

**6.7.6** It is possible to obtain a "quick-and-dirty" estimate of the mean of a normal distribution from the 50th percentile value on a normal probability plot. Provide an argument why this is so. It is also possible to obtain an estimate of the standard deviation of a normal distribution by subtracting the 84th percentile value from the 50th percentile value. Provide an argument explaining why this is so.



## Supplemental Exercises for Chapter 6

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**6.S7** WP SS The table below shows unemployment data for the United States that are seasonally adjusted. Construct a time series plot of these data and comment on any features (*Source:* U.S. Bureau of Labor Web site, http://data.bls.gov).

**6.S8** A sample of six resistors yielded the following resistances (ohms): $x_1 = 45$, $x_2 = 38$, $x_3 = 47$, $x_4 = 41$, $x_5 = 35$, and $x_6 = 43$.

**a.** Compute the sample variance and sample standard deviation.

**b.** Subtract 35 from each of the original resistance measurements and compute $s^2$ and $s$. Compare your results with those obtained in part (a) and explain your findings.

**c.** If the resistances were 450, 380, 470, 410, 350, and 430 ohms, could you use the results of previous parts of this problem to find $s^2$ and $s$?

**6.S9** WP Consider the following two samples:

Sample 1: 10, 9, 8, 7, 8, 6, 10, 6
Sample 2: 10, 6, 10, 6, 8, 10, 8, 6

**a.** Calculate the sample range for both samples. Would you conclude that both samples exhibit the same variability? Explain.

**b.** Calculate the sample standard deviations for both samples. Do these quantities indicate that both samples have the same variability? Explain.

**c.** Write a short statement contrasting the sample range versus the sample standard deviation as a measure of variability.

**6.S10** WP The following data are the temperatures of effluent at discharge from a sewage treatment facility on consecutive days:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 43 | 47 | 51 | 48 | 52 | 50 | 46 | 49 |
| 45 | 52 | 46 | 51 | 44 | 49 | 46 | 51 |
| 49 | 45 | 44 | 50 | 48 | 50 | 49 | 50 |

**a.** Calculate the sample mean, sample median, sample variance, and sample standard deviation.

**b.** Construct a box plot of the data and comment on the information in this display.

**6.S11** WP SS VS The total net electricity consumption of the United States by year from 1980 to 2007 (in billion kilowatt-hours) is in the table below (*Source:* U.S. Department of Energy Web site, www.eia.doe.gov/emeu/international/contents.html#InternationalElectricity). Net consumption excludes the energy consumed by the generating units.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1980 | 2094.4 | 1981 | 2147.1 | 1982 | 2086.4 | 1983 | 2151.0 |
| 1984 | 2285.8 | 1985 | 2324.0 | 1986 | 2368.8 | 1987 | 2457.3 |
| 1988 | 2578.1 | 1989 | 2755.6 | 1990 | 2837.1 | 1991 | 2886.1 |
| 1992 | 2897.2 | 1993 | 3000.7 | 1994 | 3080.9 | 1995 | 3164.0 |
| 1996 | 3253.8 | 1997 | 3301.8 | 1998 | 3425.1 | 1999 | 3483.7 |
| 2000 | 3592.4 | 2001 | 3557.1 | 2002 | 3631.7 | 2003 | 3662.0 |
| 2004 | 3715.9 | 2005 | 3811.0 | 2006 | 3816.8 | 2007 | 3891.7 |

Construct a time series plot of these data. Construct and interpret a stem-and-leaf display of these data.

| Year | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1999 | 4.3 | 4.4 | 4.2 | 4.3 | 4.2 | 4.3 | 4.3 | 4.2 | 4.2 | 4.1 | 4.1 | 4.0 |
| 2000 | 4.0 | 4.1 | 4.0 | 3.8 | 4.0 | 4.0 | 4.0 | 4.1 | 3.9 | 3.9 | 3.9 | 3.9 |
| 2001 | 4.2 | 4.2 | 4.3 | 4.4 | 4.3 | 4.5 | 4.6 | 4.9 | 5.0 | 5.3 | 5.5 | 5.7 |
| 2002 | 5.7 | 5.7 | 5.7 | 5.9 | 5.8 | 5.8 | 5.8 | 5.7 | 5.7 | 5.7 | 5.9 | 6.0 |
| 2003 | 5.8 | 5.9 | 5.9 | 6.0 | 6.1 | 6.3 | 6.2 | 6.1 | 6.1 | 6.0 | 5.8 | 5.7 |
| 2004 | 5.7 | 5.6 | 5.8 | 5.6 | 5.6 | 5.6 | 5.5 | 5.4 | 5.4 | 5.5 | 5.4 | 5.4 |
| 2005 | 5.2 | 5.4 | 5.2 | 5.2 | 5.1 | 5.1 | 5.0 | 4.9 | 5.0 | 5.0 | 5.0 | 4.8 |
| 2006 | 4.7 | 4.8 | 4.7 | 4.7 | 4.7 | 4.6 | 4.7 | 4.7 | 4.5 | 4.4 | 4.5 | 4.4 |
| 2007 | 4.6 | 4.5 | 4.4 | 4.5 | 4.5 | 4.6 | 4.7 | 4.7 | 4.7 | 4.8 | 4.7 | 4.9 |
| 2008 | 4.9 | 4.8 | 5.1 | 5.0 | 5.5 | 5.6 | 5.8 | 6.2 | 6.2 | 6.6 | 6.8 | 7.2 |
| 2009 | 7.6 | 8.1 | 8.5 | 8.9 | 9.4 | 9.5 | 9.4 | 9.7 | 9.8 | | | |

**6.S12** Transformations. In some data sets, a transformation by some mathematical function applied to the original data, such as $\sqrt{y}$ or $\log y$, can result in data that are simpler to work with statistically than the original data. To illustrate the effect of a transformation, consider the following data, which represent cycles to failure for a yarn product: 675, 3650, 175, 1150, 290, 2000, 100, 375.

**a.** Construct a normal probability plot and comment on the shape of the data distribution.

**b.** Transform the data using logarithms; that is, let $y^*$(new value) = log $y$(old value). Construct a normal probability plot of the transformed data and comment on the effect of the transformation.

**6.S13** In 1879, A. A. Michelson made 100 determinations of the velocity of light in air using a modification of a method proposed by French physicist Léon Foucault. Michelson made the measurements in five trials of 20 measurements each. The observations (in kilometers per second) are in the table below. Each value has 299,000 subtracted from it.

The currently accepted true velocity of light in a vacuum is 299,792.5 kilometers per second. Stigler ["Do Robust Estimators Work with Real Data?" *Annals of Statistics* (1977, Vol. 5(6), pp. 1055–1098)] reported that the "true" value for comparison to these measurements is 734.5 because of air effects. Construct comparative box plots of these measurements. Does it seem that all five trials are consistent with respect to the variability of the measurements? Are all five trials centered on the same value? How does each group of trials compare to the true value? Could there have been "startup" effects in the experiment that Michelson performed? Could there have been bias in the measuring instrument?

| Trial 1 | | | | |
|---|---|---|---|---|
| 850 | 900 | 930 | 950 | 980 |
| 1000 | 930 | 760 | 1000 | 960 |
| 740 | 1070 | 850 | 980 | 880 |
| 980 | 650 | 810 | 1000 | 960 |
| **Trial 2** | | | | |
| 960 | 960 | 880 | 850 | 900 |
| 830 | 810 | 880 | 800 | 760 |
| 940 | 940 | 800 | 880 | 840 |
| 790 | 880 | 830 | 790 | 800 |
| **Trial 3** | | | | |
| 880 | 880 | 720 | 620 | 970 |
| 880 | 850 | 840 | 850 | 840 |
| 880 | 860 | 720 | 860 | 950 |
| 910 | 870 | 840 | 840 | 840 |
| **Trial 4** | | | | |
| 890 | 810 | 800 | 760 | 750 |
| 910 | 890 | 880 | 840 | 850 |
| 810 | 820 | 770 | 740 | 760 |
| 920 | 860 | 720 | 850 | 780 |
| **Trial 5** | | | | |
| 890 | 780 | 760 | 790 | 820 |
| 870 | 810 | 810 | 950 | 810 |
| 840 | 810 | 810 | 810 | 850 |
| 870 | 740 | 940 | 800 | 870 |

**6.S14** WP In 1789, Henry Cavendish estimated the density of the Earth by using a torsion balance. His 29 measurements follow, expressed as a multiple of the density of water.

| | | | | | |
|---|---|---|---|---|---|
| 5.50 | 5.30 | 5.47 | 5.10 | 5.29 | 5.65 |
| 5.55 | 5.61 | 5.75 | 5.63 | 5.27 | 5.44 |
| 5.57 | 5.36 | 4.88 | 5.86 | 5.34 | 5.39 |
| 5.34 | 5.53 | 5.29 | 4.07 | 5.85 | 5.46 |
| 5.42 | 5.79 | 5.62 | 5.58 | 5.26 | |

**a.** Calculate the sample mean, sample standard deviation, and median of the Cavendish density data.

**b.** Construct a normal probability plot of the data. Comment on the plot. Does there seem to be a "low" outlier in the data?

**c.** Would the sample median be a better estimate of the density of the earth than the sample mean? Why?

**6.S15** In their book *Introduction to Time Series Analysis and Forecasting* (Wiley, 2008), Montgomery, Jennings, and Kulahci presented the data on the drowning rate for children between one and four years old per 100,000 of population in Arizona from 1970 to 2004. The data are: 19.9, 16.1, 19.5, 19.8, 21.3, 15.0, 15.5, 16.4, 18.2, 15.3, 15.6, 19.5, 14.0, 13.1, 10.5, 11.5, 12.9, 8.4, 9.2, 11.9, 5.8, 8.5, 7.1, 7.9, 8.0, 9.9, 8.5, 9.1, 9.7, 6.2, 7.2, 8.7, 5.8, 5.7, and 5.2.

**a.** Perform an appropriate graphical analysis of the data.

**b.** Calculate and interpret the appropriate numerical summaries.

**c.** Notice that the rate appears to decrease dramatically starting about 1990. Discuss some potential reasons explaining why this could have happened.

**d.** If there has been a real change in the drowning rate beginning about 1990, what impact does this have on the summary statistics that you calculated in part (b)?

**6.S16** The energy consumption for 90 gas-heated homes during a winter heating season is given in the table below. The variable reported is BTU/number of heating degree days.

**a.** Calculate the sample mean and standard deviation of energy usage.

**b.** Construct a histogram of the energy usage data and comment on the shape of the data distribution.

**c.** Construct a stem-and-leaf diagram of energy usage.

**d.** What proportion of the energy usage data is above the average usage plus 2 standard deviations?



| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 7.87 | 9.43 | 7.16 | 8.67 | 12.31 | 9.84 | 16.90 | 10.04 | 12.62 | 7.62 |
| 11.12 | 13.43 | 9.07 | 6.94 | 10.28 | 9.37 | 7.93 | 13.96 | 6.80 | 4.00 |
| 8.58 | 8.00 | 5.98 | 15.24 | 8.54 | 11.09 | 11.70 | 12.71 | 6.78 | 9.82 |
| 12.91 | 10.35 | 9.60 | 9.58 | 9.83 | 9.52 | 18.26 | 10.64 | 6.62 | 5.20 |
| 12.28 | 7.23 | 2.97 | 8.81 | 9.27 | 11.29 | 8.29 | 9.96 | 10.30 | 16.06 |
| 14.24 | 11.43 | 10.28 | 13.60 | 5.94 | 10.36 | 6.85 | 6.72 | 10.21 | 8.61 |
| 11.62 | 11.21 | 10.95 | 7.62 | 10.40 | 12.92 | 15.12 | 13.47 | 8.47 | 11.70 |
| 7.73 | 8.37 | 7.29 | 10.49 | 8.69 | 8.26 | 7.69 | 12.19 | 5.56 | 9.76 |
| 7.15 | 12.69 | 13.38 | 13.11 | 10.50 | 14.35 | 13.42 | 6.35 | 9.83 | 12.16 |

**6.S17** One of the authors (DCM) has a Mercedes-Benz 500 SL Roadster. It is a 2003 model and has fairly low mileage (currently 60,112 miles on the odometer). He is interested in learning how his car's mileage compares with the mileage on similar SLs. The table below contains the mileage on 100 Mercedes-Benz SLs from the model years 2003–2009 taken from the Cars.com website.

**a.** Calculate the sample mean and standard deviation of the odometer readings.

**b.** Construct a histogram of the odometer readings and comment on the shape of the data distribution.

**c.** Construct a stem-and-leaf diagram of the odometer readings.

**d.** What is the percentile of DCM's mileage?

**6.S18** Consider the global mean surface air temperature anomaly and the global $CO_2$ concentration data originally shown in Exercise 6.5.4.

**a.** Construct a scatter plot of the global mean surface air temperature anomaly versus the global $CO_2$ concentration. Comment on the plot.

**b.** What is the simple correlation coefficient between these two variables?

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 2020 | 8905 | 1698 | 17,971 | 6207 | 22,643 | 4977 | 17,656 | 8940 | 11,508 |
| 7893 | 10,327 | 37,687 | 15,000 | 4166 | 9056 | 19,842 | 15,598 | 33,745 | 22,168 |
| 19,000 | 31,668 | 33,512 | 28,522 | 5824 | 18,327 | 31,845 | 30,015 | 2171 | 36,161 |
| 15,984 | 16,903 | 37,789 | 28,958 | 40,944 | 18,498 | 40,057 | 15,272 | 28,968 | 30,487 |
| 32,271 | 36,889 | 21,564 | 31,000 | 42,915 | 19,377 | 19,634 | 26,313 | 43,049 | 30,396 |
| 38,277 | 72,272 | 3800 | 21,218 | 29,250 | 48,648 | 29,216 | 44,944 | 49,125 | 33,065 |
| 32,524 | 38,139 | 62,940 | 51,326 | 54,126 | 4100 | 45,540 | 26,235 | 46,505 | 34,420 |
| 15,972 | 41,218 | 43,382 | 15,879 | 13,500 | 77,809 | 25,708 | 29,000 | 58,006 | 51,071 |
| 63,249 | 58,526 | 66,325 | 49,489 | 32,800 | 67,000 | 60,499 | 63,260 | 60,449 | 27,422 |
| 60,583 | 83,500 | 56,314 | 67,072 | 62,500 | 47,603 | 51,936 | 65,195 | 64,473 | 85,475 |

# Chapter 7 Exercises

## Exercises for Section 7.2

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**7.2.1** WP Consider the compressive strength data in Table 6.2. What proportion of the specimens exhibit compressive strength of at least 200 psi? What is the population parameter estimated here?

**7.2.2** WP Consider the synthetic fiber in the previous exercise. How is the standard deviation of the sample mean changed when the sample size is increased from $n = 6$ to $n = 49$?

**7.2.3** WP VS A synthetic fiber used in manufacturing carpet has tensile strength that is normally distributed with mean 75.5 psi and standard deviation 3.5 psi. Find the probability that a random sample of $n = 6$ fiber specimens will have sample mean tensile strength that exceeds 75.75 psi.

**7.2.4** WP Suppose that the random variable $X$ has the continuous uniform distribution

$$f(x) = \begin{cases} 1, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

Suppose that a random sample of $n = 12$ observations is selected from this distribution. What is the approximate probability distribution of $\overline{X} - 6$? Find the mean and variance of this quantity.

**7.2.5** WP A normal population has mean 100 and variance 25. How large must the random sample be if you want the standard error of the sample average to be 1.5?

**7.2.6** WP VS The amount of time that a customer spends waiting at an airport check-in counter is a random variable with mean 8.2 minutes and standard deviation 1.5 minutes. Suppose that a random sample of $n = 49$ customers is observed. Find the probability that the average time waiting in line for these customers is

**a.** Less than 10 minutes **b.** Between 5 and 10 minutes

**c.** Less than 6 minutes

**7.2.7** WP SS VS A random sample of size $n_1 = 16$ is selected from a normal population with a mean of 75 and a standard deviation of 8. A second random sample of size $n_2 = 9$ is taken from another normal population with mean 70 and standard deviation 12. Let $\overline{X}_1$ and $\overline{X}_2$ be the two sample means. Find:

**a.** The probability that $\overline{X}_1 - \overline{X}_2$ exceeds 4

**b.** The probability that $3.5 \le \overline{X}_1 - \overline{X}_2 \le 5.5$

**7.2.8** Scientists at the Hopkins Memorial Forest in western Massachusetts have been collecting meteorological and environmental data in the forest data for more than 100 years. In the past few years, sulfate content in water samples from Birch Brook has averaged 7.48 mg/L with a standard deviation of 1.60 mg/L.

**a.** What is the standard error of the sulfate in a collection of 10 water samples?

**b.** If 10 students measure the sulfate in their samples, what is the probability that their average sulfate will be between 6.49 and 8.47 mg/L?

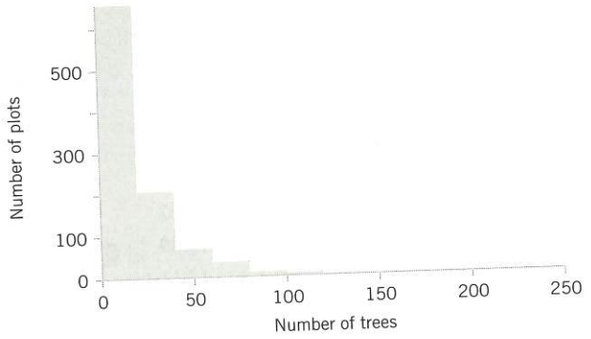
**c.** What do you need to assume for the probability calculated in (b) to be accurate?

**7.2.9** Data on the pH of rain in Ingham County, Michigan, are as follows:

5.47 5.37 5.38 4.63 5.37 3.74 3.71 4.96 4.64 5.11 5.65 5.39 4.16 5.62 4.57 4.64 5.48 4.57 4.57 4.51 4.86 4.56 4.61 4.32 3.98 5.70 4.15 3.98 5.65 3.10 5.04 4.62 4.51 4.34 4.16 4.64 5.12 3.71 4.64

What proportion of the samples has pH below 5.0?

**7.2.10** Researchers in the Hopkins Forest (see Exercise 7.2.8) also count the number of maple trees (genus *acer*) in plots throughout the forest. The following is a histogram of the number of live maples in 1002 plots sampled over the past 20 years. The average number of maples per plot was 19.86 trees with a standard deviation of 23.65 trees.




**a.** If we took the mean of a sample of eight plots, what would be the standard error of the mean?

**b.** Using the central limit theorem, what is the probability that the mean of the eight would be within 1 standard error of the mean?

**c.** Why might you think that the probability that you calculated in (b) might not be very accurate?

**7.2.11** Like hurricanes and earthquakes, geomagnetic storms are natural hazards with possible severe impact on the earth. Severe storms can cause communication and utility breakdowns, leading to possible blackouts. The National Oceanic and Atmospheric Administration beams electron and proton flux data in various energy ranges to various stations on the earth to help forecast possible disturbances. The following are 25 readings of proton flux in the 47-68 kEV range (units are in p/(cm2-sec-ster-MeV)) on the evening of December 28, 2011:

2310 2320 2010 10800 2190 3360 5640 2540 3360 11800 2010 3430 10600 7370 2160 3200 2020 2850 3500 10200 8550 9500 2260 7730 2250

**a.** Find a point estimate of the mean proton flux in this time period.

**b.** Find a point estimate of the standard deviation of the proton flux in this time period.

**c.** Find an estimate of the standard error of the estimate in part (a).

**d.** Find a point estimate for the median proton flux in this time period.

**e.** Find a point estimate for the proportion of readings that are less than 5000 p/(cm2-sec-ster-MeV).

**7.2.12** Wayne Collier designed an experiment to measure the fuel efficiency of his family car under different tire pressures. For each run, he set the tire pressure and then measured the miles he drove on a highway (I-95 between Mills River and Pisgah Forest, NC) until he ran out of fuel using 2 liters of fuel each time. To do this, he made some alterations to the normal flow of gasoline to the engine. In Wayne's words, "I inserted a T-junction into the fuel line just before the fuel filter, and a line into the passenger compartment of my car, where it joined with a graduated 2-liter Rubbermaid© bottle that I mounted in a box where the passenger seat is normally fastened. Then I sealed off the fuel-return line, which under normal operation sends excess fuel from the fuel pump back to the fuel tank."



Suppose that you call the mean miles that he can drive with normal pressure in the tires $\mu$. An unbiased estimate for $\mu$ is the mean of the sample runs, $\bar{x}$. But Wayne has a different idea. He decides to use the following estimator: He flips a fair coin. If the coin comes up heads, he will add 5 miles to each observation. If tails come up, he will subtract 5 miles from each observation.

**a.** Show that Wayne's estimate is, in fact, unbiased.

**b.** Compare the standard deviation of Wayne's estimate with the standard deviation of the sample mean.

**c.** Given your answer to (b), why does Wayne's estimate not make good sense scientifically?

## Exercises for Section 7.3

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**7.3.1** A computer software package calculated some numerical summaries of a sample of data. The results are displayed here:

| Variable | $N$ | Mean | SE Mean | StDev | Variance | Sum | Sum of Squares |
|---|---|---|---|---|---|---|---|
| $x$ | ? | ? | 2.05 | 10.25 | ? | 3761.70 | ? |

**a.** Fill in the missing quantities.

**b.** What is the estimate of the mean of the population from which this sample was drawn?

**7.3.2** Suppose we have a random sample of size $2n$ from a population denoted by $X$, and $E(X) = \mu$ and $V(X) = \sigma^2$. Let

$$\overline{X}_1 = \frac{1}{2n}\sum_{i=1}^{2n} X_i \quad \text{and} \quad \overline{X}_2 = \frac{1}{n}\sum_{i=1}^{n} X_i$$

be two estimators of $\mu$. Which is the better estimator of $\mu$? Explain your choice.

**7.3.3** SS Suppose that we have a random sample $X_1, X_2, \ldots, X_n$ from a population that is $N(\mu, \sigma^2)$. We plan to use $\hat{\Theta} = \sum_{i=1}^{n}(X_i - \overline{X})^2/c$ to estimate $\sigma^2$. Compute the bias in $\hat{\Theta}$ as an estimator of $\sigma^2$ as a function of the constant $c$.

**7.3.4** WP VS Suppose that $\hat{\Theta}_1$ and $\hat{\Theta}_2$ are unbiased estimators of the parameter $\theta$. We know that $V(\hat{\Theta}_1) = 10$ and $V(\hat{\Theta}_2) = 4$. Which estimator is better and in what sense is it better? Calculate the relative efficiency of the two estimators.

**7.3.5** Suppose that $\hat{\Theta}_1, \hat{\Theta}_2$, and $\hat{\Theta}_3$ are estimators of $\theta$. We know that $E(\hat{\Theta}_1) = E(\hat{\Theta}_2) = \theta$, $E(\hat{\Theta}_3) \ne \theta$, $V(\hat{\Theta}_1) = 12$, $V(\hat{\Theta}_2) = 10$, and $E(\hat{\Theta}_3 - \theta)^2 = 6$. Compare these three estimators. Which do you prefer? Why?

**7.3.6** WP Data on pull-off force (pounds) for connectors used in an automobile engine application are as follows: 79.3, 75.1, 78.2, 74.1, 73.9, 75.0, 77.6, 77.3, 73.8, 74.6, 75.5, 74.0, 74.7, 75.9, 72.9, 73.8, 74.2, 78.1, 75.4, 76.3, 75.3, 76.2, 74.9, 78.0, 75.1, 76.8.

**a.** Calculate a point estimate of the mean pull-off force of all connectors in the population. State which estimator you used and why.

**b.** Calculate a point estimate of the pull-off force value that separates the weakest 50% of the connectors in the population from the strongest 50%.

**c.** Calculate point estimates of the population variance and the population standard deviation.

**d.** Calculate the standard error of the point estimate found in part (a). Interpret the standard error.

**e.** Calculate a point estimate of the proportion of all connectors in the population whose pull-off force is less than 73 pounds.

**7.3.7** WP SS VS Data on the oxide thickness of semiconductor wafers are as follows: 425, 431, 416, 419, 421, 436, 418, 410, 431, 433, 423, 426, 410, 435, 436, 428, 411, 426, 409, 437, 422, 428, 413, 416.

**a.** Calculate a point estimate of the mean oxide thickness for all wafers in the population.

**b.** Calculate a point estimate of the standard deviation of oxide thickness for all wafers in the population.

**c.** Calculate the standard error of the point estimate from part (a).

**d.** Calculate a point estimate of the median oxide thickness for all wafers in the population.

**e.** Calculate a point estimate of the proportion of wafers in the population that have oxide thickness of more than 430 angstroms.

**7.3.8** Suppose that $X$ is the number of observed "successes" in a sample of $n$ observations where $p$ is the probability of success on each observation.

**a.** Show that $\hat{P} = X/n$ is an unbiased estimator of $p$.

**b.** Show that the standard error of $\hat{P}$ is $\sqrt{p(1-p)/n}$. How would you estimate the standard error?

**7.3.9** VS $\overline{X}_1$ and $S_1^2$ are the sample mean and sample variance from a population with mean $\mu_1$ and variance $\sigma_1^2$. Similarly, $\overline{X}_2$ and $S_2^2$ are the sample mean and sample variance from a second independent population with mean $\mu_2$ and variance $\sigma_2^2$. The sample sizes are $n_1$ and $n_2$, respectively.

**a.** Show that $\overline{X}_1 - \overline{X}_2$ is an unbiased estimator of $\mu_1 - \mu_2$.

**b.** Find the standard error of $\overline{X}_1 - \overline{X}_2$. How could you estimate the standard error?

**c.** Suppose that both populations have the same variance; that is, $\sigma_1^2 = \sigma_2^2 = \sigma^2$. Show that

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

is an unbiased estimator of $\sigma^2$.

**7.3.10** Consider a normal random variable with mean 10 and standard deviation 4. Suppose that a random sample of size 16 is drawn from this distribution and the sample mean is computed.

We know that the standard error of the sample mean in this case is $\sigma/\sqrt{n} = \sigma/\sqrt{16} = 1$. Use the bootstrap method with $n_B = 200$ bootstrap samples to find the standard error of the sample mean. Compare the bootstrap standard error to the actual standard error.

**7.3.11** Of $n_1$ randomly selected engineering students at Arizona State University, $X_1$ owned an Apple computer, and of $n_2$ randomly selected engineering students at Virginia Tech, $X_2$ owned an Apple computer. Let $p_1$ and $p_2$ be the probability that randomly selected ASU and Virginia Tech engineering students, respectively, own Apple computers.

**a.** Show that an unbiased estimate for $p_1 - p_2$ is $(X_1/n_1) - (X_2/n_2)$.

**b.** What is the standard error of the point estimate in part (a)?

**c.** How would you compute an estimate of the standard error found in part (b)?

**d.** Suppose that $n_1 = 200, X_1 = 150, n_2 = 250$, and $X_2 = 185$. Use the results of part (a) to compute an estimate of $p_1 - p_2$.

**e.** Use the results in parts (b) through (d) to compute an estimate of the standard error of the estimate.

**7.3.12** Suppose that two independent random samples (of size $n_1$ and $n_2$) from two normal distributions are available. Explain how you would estimate the standard error of the difference in sample means $\overline{X}_1 - \overline{X}_2$ with the bootstrap method.

## Exercises for Section 7.4

WP Problem available in WileyPLUS at instructor's discretion.
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SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**7.4.1** WP SS VS Let $X$ be a geometric random variable with parameter $p$. Find the maximum likelihood estimator of $p$ based on a random sample of size $n$.

**7.4.2** WP VS Consider the probability density function

$$f(x) = \frac{1}{\theta^2} x e^{-x/\theta}, \qquad 0 \le x < \infty, \quad 0 < \theta < \infty$$

Find the maximum likelihood estimator for $\theta$.

**7.4.3** Let $X_1, X_2, \ldots, X_n$ be uniformly distributed on the interval 0 to $a$. Show that the moment estimator of $a$ is $\hat{a} = 2\overline{X}$. Is this an unbiased estimator? Discuss the reasonableness of this estimator.

**7.4.4** Consider the probability density function

$$f(x) = c(1 + \theta x), \quad -1 \le x \le 1$$

**a.** Find the value of the constant $c$.
**b.** What is the moment estimator for $\theta$?
**c.** Show that $\hat{\theta} = 3\overline{X}$ is an unbiased estimator for $\theta$.
**d.** Find the maximum likelihood estimator for $\theta$.

**7.4.5** Consider the Weibull distribution

$$f(x) = \begin{cases} \frac{\beta}{\delta}\left(\frac{x}{\delta}\right)^{\beta-1} e^{-\left(\frac{x}{\delta}\right)^{\beta}}, & 0 < x \\ 0, & \text{otherwise} \end{cases}$$

**a.** Find the likelihood function based on a random sample of size $n$. Find the log likelihood.

**b.** Show that the log likelihood is maximized by solving the following equations

$$\beta = \left[\frac{\sum_{i=1}^{n} x_i^{\beta} \ln(x_i)}{\sum_{i=1}^{n} x_i^{\beta}} - \frac{\sum_{i=1}^{n} \ln(x_i)}{n}\right]^{-1} \qquad \delta = \left[\frac{\sum_{i=1}^{n} x_i^{\beta}}{n}\right]^{1/\beta}$$

**c.** What complications are involved in solving the two equations in part (b)?

**7.4.6** Let $X_1, X_2, \ldots, X_n$ be uniformly distributed on the interval 0 to $a$. Recall that the maximum likelihood estimator of $a$ is $\hat{a} = \max(X_i)$.

**a.** Argue intuitively why $\hat{a}$ cannot be an unbiased estimator for $a$.

**b.** Suppose that $E(\hat{a}) = na/(n+1)$. Is it reasonable that $\hat{a}$ consistently underestimates $a$? Show that the bias in the estimator approaches zero as $n$ gets large.

**c.** Propose an unbiased estimator for $a$.

**d.** Let $Y = \max(X_i)$. Use the fact that $Y \le y$ if and only if each $X_i \le y$ to derive the cumulative distribution function of $Y$. Then show that the probability density function of $Y$ is

$$f(y) = \begin{cases} \frac{n y^{n-1}}{a^n}, & 0 \le y \le a \\ 0, & \text{otherwise} \end{cases}$$

Use this result to show that the maximum likelihood estimator for $a$ is biased.

**e.** We have two unbiased estimators for $a$: the moment estimator $\hat{a}_1 = 2\overline{X}$ and $\hat{a}_2 = [(n+1)/n]\max(X_i)$, where $\max(X_i)$ is the largest observation in a random sample of size $n$. It can be shown that $V(\hat{a}_1) = a^2/(3n)$ and that $V(\hat{a}_2) = a^2/[n(n+2)]$. Show that if $n > 1$, $\hat{a}_2$ is a better estimator than $\hat{a}$. In what sense is it a better estimator of $a$?



**7.4.7** SS Reconsider the oxide thickness data in Exercise 7.3.7 and suppose that it is reasonable to assume that oxide thickness is normally distributed.

**a.** Compute the maximum likelihood estimates of $\mu$ and $\sigma^2$.

**b.** Graph the likelihood function in the vicinity of $\hat{\mu}$ and $\hat{\sigma}^2$, the maximum likelihood estimates, and comment on its shape.

**c.** Suppose that the sample size was larger ($n = 40$) but the maximum likelihood estimates were numerically equal to the values obtained in part (a). Graph the likelihood function for $n = 40$, compare it to the one from part (b), and comment on the effect of the larger sample size.

## Supplemental Exercises for Chapter 7

WP Problem available in WileyPLUS at instructor's discretion.
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**7.S8** Suppose that a random variable is normally distributed with mean $\mu$ and variance $\sigma^2$, and we draw a random sample of five observations from this distribution. What is the joint probability density function of the sample?

**7.S9** WP Transistors have a life that is exponentially distributed with parameter $\lambda$. A random sample of $n$ transistors is taken. What is the joint probability density function of the sample?

**7.S10** WP VS A procurement specialist has purchased 25 resistors from vendor 1 and 30 resistors from vendor 2. Let $X_{1,1}, X_{1,2}, \ldots, X_{1,25}$ represent the vendor 1 observed resistances, which are assumed to be normally and independently distributed with mean 100 ohms and standard deviation 1.5 ohms. Similarly, let $X_{2,1}, X_{2,2}, \ldots, X_{2,30}$ represent the vendor 2 observed resistances, which are assumed to be normally and independently distributed with mean 105 ohms and standard deviation of 2.0 ohms. What is the sampling distribution of $\overline{X}_1 - \overline{X}_2$? What is the standard error of $\overline{X}_1 - \overline{X}_2$?

**7.S11** WP SS VS A random sample of 36 observations has been drawn from a normal distribution with mean 50 and standard deviation 12. Find the probability that the sample mean is in the interval $47 \le X \le 53$. Is the assumption of normality important? Why?

**7.S12** A manufacturer of semiconductor devices takes a random sample of 100 chips and tests them, classifying each chip as defective or nondefective. Let $X_i = 0$ if the chip is nondefective and $X_i = 1$ if the chip is defective. The sample fraction defective is

$$\hat{P} = \frac{X_1 + X_2 + \cdots + X_{100}}{100}$$

What is the sampling distribution of the random variable $\hat{P}$?

**7.S13** WP A random variable $x$ has probability density function

$$f(x) = \frac{1}{2\theta^3} x^2 e^{-x/\theta}, \qquad 0 < x < \infty, \quad 0 < \theta < \infty$$

Find the maximum likelihood estimator for $\theta$.

**7.S14** Let $f(x) = (1/\theta)x^{(1-\theta)/\theta}$, $0 < x < 1$, and $0 < \theta < \infty$. Show that $\hat{\Theta} = -(1/n)\sum_{i=1}^{n} \ln(X_i)$ is the maximum likelihood estimator for $\theta$ and that $\hat{\Theta}$ is an unbiased estimator for $q$.

**7.S15** An electric utility has placed special meters on 10 houses in a subdivision that measures the energy consumed (demand) at each hour of the day. The company is interested in the energy demand at one specific hour—the hour at which the system experiences peak consumption. The data from these 10 meters are as follows (in KW): 23.1, 15.6, 17.4, 20.1, 19.8, 26.4, 25.1, 20.5, 21.9, and 28.7. If $\mu$ is the true mean peak demand for the 10 houses in this group of houses having the special meters, estimate $\mu$. Now suppose that the utility wants to estimate the demand at the peak hour for all 5000 houses in this subdivision. Let $\theta$ be this quantity. Estimate $\theta$ using the data given. Estimate the proportion of houses in the subdivision that demand at least 20 KW at the hour of system peak.

# Chapter 8 Exercises

## Exercises for Section 8.1

WP Problem available in WileyPLUS at instructor's discretion.
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**8.1.1** WP For a normal population with known variance $\sigma^2$, answer the following questions:

**a.** What is the confidence level for the interval $\overline{x} - 2.14\sigma/\sqrt{n} \le \mu \le \overline{x} + 2.14\sigma/\sqrt{n}$?

**b.** What is the confidence level for the interval $\bar{x} - 2.49\sigma/\sqrt{n} \bar{x} - 2.49\sigma/\sqrt{n} \le \mu \le \bar{x} + 2.49\sigma/\sqrt{n}$?

**c.** What is the confidence level for the interval $\bar{x} - 1.85\sigma/\sqrt{n} \le \mu \le \bar{x} + 1.85\sigma/\sqrt{n}$?

**d.** What is the confidence level for the interval $\mu \le \bar{x} + 2.00\sigma/\sqrt{n}$?

**e.** What is the confidence level for the interval $\bar{x} - 1.96\sigma/\sqrt{n} \le \mu$?

**8.1.2** WP Consider the one-sided confidence interval expressions for a mean of a normal population.

**a.** What value of $z_\alpha$ would result in a 90% CI?

**b.** What value of $z_\alpha$ would result in a 95% CI?

**c.** What value of $z_\alpha$ would result in a 99% CI?

**8.1.3** A random sample has been taken from a normal distribution and the following confidence intervals constructed using the same data: (38.02, 61.98) and (39.95, 60.05)

**a.** What is the value of the sample mean?

**b.** One of these intervals is a 95% CI and the other is a 90% CI. Which one is the 95% CI and why?

**8.1.4** WP A confidence interval estimate is desired for the gain in a circuit on a semiconductor device. Assume that gain is normally distributed with standard deviation $\sigma = 20$.

**a.** How large must $n$ be if the length of the 95% CI is to be 40?

**b.** How large must $n$ be if the length of the 99% CI is to be 40?

**8.1.5** Suppose that $n = 100$ random samples of water from a freshwater lake were taken and the calcium concentration (milligrams per liter) measured. A 95% CI on the mean calcium concentration is $0.49 \le \mu \le 0.82$.

**a.** Would a 99% CI calculated from the same sample data be longer or shorter?

**b.** Consider the following statement: There is a 95% chance that $\mu$ is between 0.49 and 0.82. Is this statement correct? Explain your answer.

**c.** Consider the following statement: If $n = 100$ random samples of water from the lake were taken and the 95% CI on $\mu$ computed, and this process were repeated 1000 times, 950 of the CIs would contain the true value of $\mu$. Is this statement correct? Explain your answer.

**8.1.6** WP SS The yield of a chemical process is being studied. From previous experience, yield is known to be normally distributed and $\sigma = 3$. The past 5 days of plant operation have resulted in the following percent yields: 91.6, 88.75, 90.8, 89.95, and 91.3. Find a 95% two-sided confidence interval on the true mean yield.

**8.1.7** WP A manufacturer produces piston rings for an automobile engine. It is known that ring diameter is normally distributed with $\sigma = 0.001$ millimeters. A random sample of 15 rings has a mean diameter of $\bar{x} = 74.036$ millimeters.

**a.** Construct a 99% two-sided confidence interval on the mean piston ring diameter.

**b.** Construct a 99% lower-confidence bound on the mean piston ring diameter. Compare the lower bound of this confidence interval with the one in part (a).

**8.1.8** WP VS A civil engineer is analyzing the compressive strength of concrete. Compressive strength is normally distributed with $\sigma^2 = 1000(\text{psi})^2$. A random sample of 12 specimens has a mean compressive strength of $\bar{x} = 3250$ psi.

**a.** Construct a 95% two-sided confidence interval on mean compressive strength.

**b.** Construct a 99% two-sided confidence interval on mean compressive strength. Compare the width of this confidence interval with the width of the one found in part (a).

**8.1.9** Suppose that in Exercise 8.1.8 it is desired to estimate the compressive strength with an error that is less than 15 psi at 99% confidence. What sample size is required?

**8.1.10** If the sample size $n$ is doubled, by how much is the length of the CI on $\mu$ in Equation 8.5 reduced? What happens to the length of the interval if the sample size is increased by a factor of four?

**8.1.11** WP SS By how much must the sample size $n$ be increased if the length of the CI on $\mu$ in Equation 8.5 is to be halved?

**8.1.12** Ishikawa et al. ["Evaluation of Adhesiveness of *Acinetobacter* sp. Tol 5 to Abiotic Surfaces," *Journal of Bioscience and Bioengineering* (Vol. 113(6), pp. 719–725)] studied the adhesion of various biofilms to solid surfaces for possible use in environmental technologies. Adhesion assay is conducted by measuring absorbance at $A_{590}$. Suppose that for the bacterial strain *Acinetobacter*, five measurements gave readings of 2.69, 5.76, 2.67, 1.62, and 4.12 dyne-cm$^2$. Assume that the standard deviation is known to be 0.66 dyne-cm$^2$.

**a.** Find a 95% confidence interval for the mean adhesion.

**b.** If the scientists want the confidence interval to be no wider than 0.55 dyne-cm$^2$, how many observations should they take?

**8.1.13** WP GO Tutorial An article in the *Journal of Agricultural Science* ["The Use of Residual Maximum Likelihood to Model Grain Quality Characteristics of Wheat with Variety, Climatic and Nitrogen Fertilizer Effects" (1997, Vol. 128, pp. 135–142)] investigated means of wheat grain crude protein content (CP) and Hagberg falling number (HFN) surveyed in the United Kingdom. The analysis used a variety of nitrogen fertilizer applications (kg N/ha), temperature (°C), and total monthly rainfall (mm). The following data below describe temperatures for wheat grown at Harper Adams Agricultural College between 1982 and 1993. The temperatures measured in June were obtained as follows:

| | | | | | |
|---|---|---|---|---|---|
| 15.2 | 14.2 | 14.0 | 12.2 | 14.4 | 12.5 |
| 14.3 | 14.2 | 13.5 | 11.8 | 15.2 | |

Assume that the standard deviation is known to be $\sigma = 0.5$.

**a.** Construct a 99% two-sided confidence interval on the mean temperature.

**b.** Construct a 95% lower-confidence bound on the mean temperature.

**c.** Suppose that you wanted to be 95% confident that the error in estimating the mean temperature is less than 2 degrees Celsius. What sample size should be used?

**d.** Suppose that you wanted the total width of the two-sided confidence interval on mean temperature to be 1.5 degrees Celsius at 95% confidence. What sample size should be used?



## Exercises for Section 8.2

WP Problem available in WileyPLUS at instructor's discretion.
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SS Student solution available in WileyPLUS.
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**8.2.1** A random sample has been taken from a normal distribution. Output from a software package follows:

| Variable | N | Mean | SE Mean | StDev | Variance | Sum |
|---|---|---|---|---|---|---|
| x | 10 | ? | 0.507 | 1.605 | ? | 251.848 |

**a.** Fill in the missing quantities.

**b.** Find a 95% CI on the population mean.

**8.2.2** WP SS Find the values of the following percentiles: $t_{0.025,15}$, $t_{0.05,10}$, $t_{0.10,20}$, $t_{0.005,25}$, and $t_{0.001,30}$.

**8.2.3** A research engineer for a tire manufacturer is investigating tire life for a new rubber compound and has built 16 tires and tested them to end-of-life in a road test. The sample mean and standard deviation are 60,139.7 and 3645.94 kilometers. Find a 95% confidence interval on mean tire life.

**8.2.4** WP Determine the $t$-percentile that is required to construct each of the following one-sided confidence intervals:

**a.** Confidence level = 95%, degrees of freedom = 14

**b.** Confidence level = 99%, degrees of freedom = 19

**c.** Confidence level = 99.9%, degrees of freedom = 24

**8.2.5** WP An article in *Obesity Research* ["Impaired Pressure Natriuresis in Obese Youths" (2003, Vol. 11, pp. 745–751)] described a study in which all meals were provided for 14 lean boys for three days followed by one stress test (with a video-game task). The average systolic blood pressure (SBP) during the test was 118.3 mm HG with a standard deviation of 9.9 mm HG. Construct a 99% one-sided upper confidence interval for mean SBP.

**8.2.6** WP An article in *Medicine and Science in Sports and Exercise* ["Maximal Leg-Strength Training Improves Cycling Economy in Previously Untrained Men" (2005, Vol. 37, pp. 131–136)] studied cycling performance before and after 8 weeks of leg-strength training. Seven previously untrained males performed leg-strength training 3 days per week for 8 weeks (with four sets of five replications at 85% of one repetition maximum). Peak power during incremental cycling increased to a mean of 315 watts with a standard deviation of 16 watts. Construct a 95% confidence interval for the mean peak power after training.

**8.2.7** The brightness of a television picture tube can be evaluated by measuring the amount of current required to achieve a particular brightness level. A sample of 10 tubes results in $\bar{x} = 317.2$ and $s = 15.7$. Find (in microamps) a 99% confidence interval on mean current required. State any necessary assumptions about the underlying distribution of the data.

**8.2.8** An article in the *Journal of Composite Materials* (December 1989, Vol. 23(12), pp. 1200–1215) describes the effect of delamination on the natural frequency of beams made from composite laminates. Five such delaminated beams were subjected to loads, and the resulting frequencies (in hertz) were as follows:

230.66, 233.05, 232.58, 229.48, 232.58

Check the assumption of normality in the population. Calculate a 90% two-sided confidence interval on mean natural frequency.

**8.2.9** The compressive strength of concrete is being tested by a civil engineer who tests 12 specimens and obtains the following data:

| | | | |
|---|---|---|---|
| 2216 | 2237 | 2249 | 2204 |
| 2225 | 2301 | 2281 | 2263 |
| 2318 | 2255 | 2275 | 2295 |

**a.** Check the assumption that compressive strength is normally distributed. Include a graphical display in your answer.

**b.** Construct a 95% two-sided confidence interval on the mean strength.

**c.** Construct a 95% lower confidence bound on the mean strength. Compare this bound with the lower bound of the two-sided confidence interval and discuss why they are different.

**8.2.10** WP SS An article in *Computers & Electrical Engineering* ["Parallel Simulation of Cellular Neural Networks" (1996, Vol. 22, pp. 61–84)] considered the speedup of cellular neural networks (CNNs) for a parallel general-purpose computing architecture based on six transputers in different areas. The data follow:

| | | | | |
|---|---|---|---|---|
| 3.775302 | 3.350679 | 4.217981 | 4.030324 | 4.639692 |
| 4.139665 | 4.395575 | 4.824257 | 4.268119 | 4.584193 |
| 4.930027 | 4.315973 | 4.600101 | | |

**a.** Is there evidence to support the assumption that speedup of CNN is normally distributed? Include a graphical display in your answer.

**b.** Construct a 95% two-sided confidence interval on the mean speedup.

**c.** Construct a 95% lower confidence bound on the mean speedup.

**8.2.11** An article in *Nuclear Engineering International* (February 1988, p. 33) describes several characteristics of fuel rods used in a reactor owned by an electric utility in Norway. Measurements on the percentage of enrichment of 12 rods were reported as follows:

| | | | | | |
|---|---|---|---|---|---|
| 2.94 | 3.00 | 2.90 | 2.75 | 3.00 | 2.95 |
| 2.90 | 2.75 | 2.95 | 2.82 | 2.81 | 3.05 |

**a.** Use a normal probability plot to check the normality assumption.

**b.** Find a 99% two-sided confidence interval on the mean percentage of enrichment. Are you comfortable with the statement that the mean percentage of enrichment is 2.95%? Why?

**8.2.12** A healthcare provider monitors the number of CAT scans performed each month in each of its clinics. The most recent year of data for a particular clinic follows (the reported variable is the number of CAT scans each month expressed as the number of CAT scans per thousand members of the health plan):

2.31, 2.09, 2.36, 1.95, 1.98, 2.25, 2.16, 2.07, 1.88, 1.94, 1.97, 2.02

**a.** Find a 95% two-sided CI on the mean number of CAT scans performed each month at this clinic.
**b.** Historically, the mean number of scans performed by all clinics in the system has been 1.95. Is there any evidence that this particular clinic performs more CAT scans on average than the overall system average?

## Exercises for Section 8.3

WP Problem available in WileyPLUS at instructor's discretion.
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**8.3.1** WP The percentage of titanium in an alloy used in aerospace castings is measured in 51 randomly selected parts. The sample standard deviation is $s = 0.37$. Construct a 95% two-sided confidence interval for $\sigma$.

**8.3.2** WP VS Determine the values of the following percentiles:

$\chi^2_{0.05,10}$, $\chi^2_{0.025,15}$, $\chi^2_{0.01,12}$, $\chi^2_{0.95,20}$, $\chi^2_{0.99,18}$, $\chi^2_{0.995,16}$, and $\chi^2_{0.005,25}$

**8.3.3** WP GO Tutorial VS SS An article in *Urban Ecosystems* ["Urbanization and Warming of Phoenix (Arizona, USA): Impacts, Feedbacks and Mitigation" (2002, Vol. 6, pp. 183–203)] mentions that Phoenix is ideal to study the effects of an urban heat island because it has grown from a population of 300,000 to approximately 3 million over the last 50 years, which is a period with a continuous, detailed climate record. The 50-year averages of the mean annual temperatures at eight sites in Phoenix follow. Check the assumption of normality in the population with a probability plot. Construct a 95% confidence interval for the standard deviation over the sites of the mean annual temperatures.

| Site | Average Mean Temperature (°C) |
|---|---|
| Sky Harbor Airport | 23.3 |
| Phoenix Greenway | 21.7 |
| Phoenix Encanto | 21.6 |
| Waddell | 21.7 |
| Litchfield | 21.3 |
| Laveen | 20.7 |
| Maricopa | 20.9 |
| Harlquahala | 20.1 |

**8.3.4** WP An article in *Medicine and Science in Sports and Exercise* ["Electrostimulation Training Effects on the Physical Performance of Ice Hockey Players" (2005, Vol. 37, pp. 455–460)] considered the use of electromyostimulation (EMS) as a method to train healthy skeletal muscle. EMS sessions consisted of 30 contractions (4-second duration, 85 Hz) and were carried out three times per week for 3 weeks on 17 ice hockey players. The 10-meter skating performance test showed a standard deviation of 0.09 seconds. Construct a 95% confidence interval of the standard deviation of the skating performance test.

**8.3.5** WP An article in *Technometrics* ["Two-Way Random Effects Analyses and Gauge R&R Studies" (1999, Vol. 41(3), pp. 202–211)] studied the capability of a gauge by measuring the weight of paper. The data for repeated measurements of one sheet of paper are in the following table. Construct a 95% one-sided upper confidence interval for the standard deviation of these measurements. Check the assumption of normality of the data and comment on the assumptions for the confidence interval.

| Observations | | | | |
|---|---|---|---|---|
| 3.481 | 3.448 | 3.485 | 3.475 | 3.472 |
| 3.477 | 3.472 | 3.464 | 3.472 | 3.470 |
| 3.470 | 3.470 | 3.477 | 3.473 | 3.474 |

**8.3.6** WP An article in *Cancer Research* ["Analyses of Litter-Matched Time-to-Response Data, with Modifications for Recovery of Interlitter Information" (1977, Vol. 37, pp. 3863–3868)] tested the tumorigenesis of a drug. Rats were randomly selected from litters and given the drug. The times of tumor appearance were recorded as follows:

101, 104, 104, 77, 89, 88, 104, 96, 82, 70, 89, 91, 39, 103, 93, 85, 104, 104, 81, 67, 104, 104, 104, 87, 104, 89, 78, 104, 86, 76, 103, 102, 80, 45, 94, 104, 104, 76, 80, 72, 73

Calculate a 95% confidence interval on the standard deviation of time until a tumor appearance. Check the assumption of normality of the population and comment on the assumptions for the confidence interval.

**8.3.7** From the data on the pH of rain in Ingham County, Michigan:

5.47 5.37 5.38 4.63 5.37 3.74 3.71 4.96 4.64 5.11 5.65 5.39 4.16 5.62 4.57 4.64 5.48 4.57 4.57 4.51 4.86 4.56 4.61 4.32 3.98 5.70 4.15 3.98 5.65 3.10 5.04 4.62 4.51 4.34 4.16 4.64 5.12 3.71 4.64

Find a two-sided 95% confidence interval for the standard deviation of pH.



## Exercises for Section 8.4

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**8.4.1** WP The 2004 presidential election exit polls from the critical state of Ohio provided the following results. The exit polls had 2020 respondents, 768 of whom were college graduates. Of the college graduates, 412 voted for George Bush.
**a.** Calculate a 95% confidence interval for the proportion of college graduates in Ohio who voted for George Bush.
**b.** Calculate a 95% lower confidence bound for the proportion of college graduates in Ohio who voted for George Bush.

**8.4.2** WP VS An article in *Knee Surgery, Sports Traumatology, Arthroscopy* ["Arthroscopic Meniscal Repair with an Absorbable Screw: Results and Surgical Technique" (2005, Vol. 13, pp. 273–279)] showed that only 25 out of 37 tears (67.6%) located between 3 and 6 mm from the meniscus rim were healed.
**a.** Calculate a two-sided 95% confidence interval on the proportion of such tears that will heal.
**b.** Calculate a 95% lower confidence bound on the proportion of such tears that will heal.

**8.4.3** WP An article in the *Journal of the American Statistical Association* ["Illustration of Bayesian Inference in Normal Data Models Using Gibbs Sampling" (1990, Vol. 85, pp. 972–985)] measured the weight of 30 rats under experiment controls. Suppose that 12 were underweight rats.
**a.** Calculate a 95% two-sided confidence interval on the true proportion of rats that would show underweight from the experiment.
**b.** Using the point estimate of $p$ obtained from the preliminary sample, what sample size is needed to be 95% confident that the error in estimating the true value of $p$ is less than 0.02?
**c.** How large must the sample be if you wish to be at least 95% confident that the error in estimating $p$ is less than 0.02, regardless of the true value of $p$?

**8.4.4** WP SS The Arizona Department of Transportation wishes to survey state residents to determine what proportion of the population would like to increase statewide highway speed limits from 65 mph to 75 mph. How many residents does the department need to survey if it wants to be at least 99% confident that the sample proportion is within 0.05 of the true proportion?

**8.4.5** The U.S. Postal Service (USPS) has used optical character recognition (OCR) since the mid-1960s. In 1983, USPS began deploying the technology to major post offices throughout the country (*www.britannica.com*). Suppose that in a random sample of 500 handwritten zip code digits, 466 were read correctly.
**a.** Construct a 95% confidence interval for the true proportion of correct digits that can be automatically read.
**b.** What sample size is needed to reduce the margin of error to 1%?
**c.** How would the answer to part (b) change if you had to assume that the machine read only one-half of the digits correctly?

**8.4.6** Information on a packet of seeds claims that 93% of them will germinate. Of the 200 seeds that were planted, only 180 germinated.
**a.** Find a 95% confidence interval for the true proportion of seeds that germinate based on this sample.
**b.** Does this seem to provide evidence that the claim is wrong?

**8.4.7** Use the data from Exercise 8.4.5 to compute the two-sided Agresti-Coull CI on the proportion of digits read correctly. Compare and discuss the relationship of this interval to the one computed in Exercise 8.4.5.

**8.4.8** Use the data from Exercise 8.4.2 to compute the two-sided Agresti-Coull CI on the proportion of tears that heal. Compare and discuss the relationship of this interval to the one computed in Exercise 8.4.2.

## Exercises for Section 8.6

WP Problem available in WileyPLUS at instructor's discretion.
WP GO Tutorial Tutoring problem available in WileyPLUS at instructor's discretion
SS Student solution available in WileyPLUS.
VS Student video solution available in WileyPLUS.

**8.6.1** Consider the tire-testing data described in Exercise 8.2.3. Compute a 95% prediction interval on the life of the next tire of this type tested under conditions that are similar to those employed in the original test. Compare the length of the prediction interval with the length of the 95% CI on the population mean.

**8.6.2** Consider the natural frequency of beams described in Exercise 8.2.8. Compute a 90% prediction interval on the diameter of the natural frequency of the next beam of this type that will be tested. Compare the length of the prediction interval with the length of the 90% CI on the population mean.

**8.6.3** Consider the television tube brightness test described in Exercise 8.2.7. Compute a 99% prediction interval on the brightness of the next tube tested. Compare the length of the prediction interval with the length of the 99% CI on the population mean.